Activities Workbook

for
Active Calculus - Multivariable

Activities Workbook for Active Calculus - Multivariable

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Preface

This Activities Workbook for *Active Calculus - Multivariable* collects all the Preview Activities and Activities in a way that each starts on a new page. The Activities Workbook is designed to be used by students who wish to have a complete set of the activities to work through as they read the book in an electronic format and for instructors who wish to have a one activity per page format to make printing for distribution in class easier.

The design of this workbook is such that each Preview Activity and Activity starts on a right-hand page. As a result, most left-hand pages in this workbook are intentionally left blank as a place for student work associated with one of the adjacent activities.

Contents

9 Multivariable and Vector Functions

9.1 Functions of Several Variables and Three Dimensional Space

Preview Activity 9.1.1. Suppose you invest money in an account that pays 5% interest compounded continuously. If you invest P dollars in the account, the amount A of money in the account after t years is given by

$$A = Pe^{0.05t}.$$

The variables P and t are independent of each other, so using functional notation we write

$$A(P,t) = Pe^{0.05t}.$$

a. Find the amount of money in the account after 7 years if you originally invest 1000 dollars.

b. Evaluate $A(5000, 8)$. Explain in words what this calculation represents.

c. Now consider only the situation where the amount invested is fixed at 1000 dollars. Calculate the amount of money in the account after t years as indicated in Table 9.1.1. Round payments to the nearest penny.

Table 9.1.1 Amount of money in an account with an initial investment of 1000 dollars.

Duration (in years)	2	3	4	5	6
Amount (dollars)					

d. Now consider the situation where we want to know the amount of money in the account after 10 years given various initial investments. Calculate the amount of money in the account as indicated in Table 9.1.2. Round payments to the nearest penny.

Table 9.1.2 Amount of money in an account after 10 years.

Initial investment (dollars)	500	1000	5000	7500	10000
Amount (dollars)					

e. Describe as best you can the combinations of initial investments and time that result in an account containing \$10,000.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.1.2. Identify the domain of each of the following functions. Draw a picture of each domain in the xy-plane.

a. $f(x,y) = x^2 + y^2$

b. $f(x,y) = \sqrt{x^2 + y^2}$

c. $Q(x,y) = \dfrac{x+y}{x^2 - y^2}$

d. $s(x,y) = \dfrac{1}{\sqrt{1 - xy^2}}$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.1.3. Complete Table 9.1.7 by filling in the missing values of the function f. Round entries to the nearest tenth.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.1.4.

a Consider the set of points (x, y, z) that satisfy the equation $x = 2$. Describe this set as best you can.

b Consider the set of points (x, y, z) that satisfy the equation $y = -1$. Describe this set as best you can.

c Consider the set of points (x, y, z) that satisfy the equation $z = 0$. Describe this set as best you can.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.1.5.

Let $P = (x_0, y_0, z_0)$ and $Q = (x_1, y_1, z_1)$ be two points in $\mathbb{R}^3$. These two points form opposite vertices of a rectangular box whose sides are planes parallel to the coordinate planes as illustrated in Figure 9.1.12, and the distance between P and Q is the length of the blue diagonal shown in Figure 9.1.12.

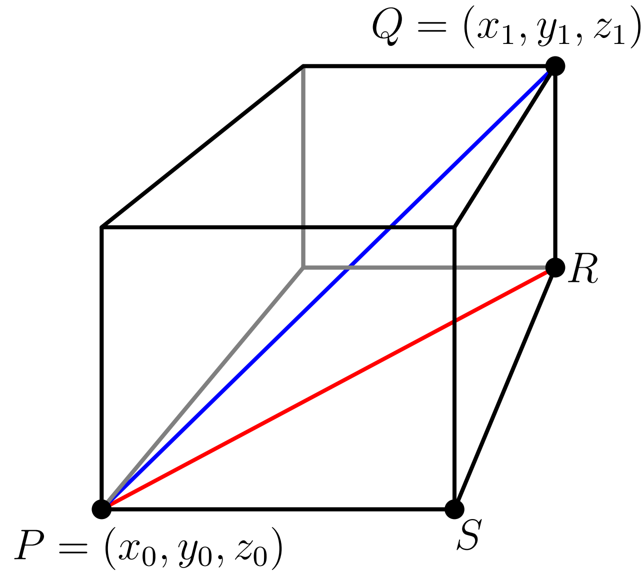


Figure 9.1.12 The distance formula in $\mathbb{R}^3$.

a. Consider the right triangle PRS in the base of the box whose hypotenuse is shown as the red line in Figure 9.1.12. What are the coordinates of the vertices of this triangle? Since this right triangle lies in a plane, we can use the Pythagorean Theorem to find a formula for the length of the hypotenuse of this triangle. Find such a formula, which will be in terms of x_0, y_0, x_1, and y_1.

b. Now notice that the triangle PRQ whose hypotenuse is the blue segment connecting the points P and Q with a leg as the hypotenuse PR of the triangle found in part (a) lies entirely in a plane, so we can again use the Pythagorean Theorem to find the length of its hypotenuse. Explain why the length of this hypotenuse, which is the distance between the points P and Q, is

$$\sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2}.$$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.1.6. In the following questions, we investigate the use of traces to better understand a function through both tables and graphs.

a. Identify the $y = 0.6$ trace for the distance function f defined by $f(x, y) = \frac{x^2 \sin(2y)}{g}$ by highlighting or circling the appropriate cells in Table 9.1.7. Write a sentence to describe the behavior of the function along this trace.

b. Identify the $x = 150$ trace for the distance function by highlighting or circling the appropriate cells in Table 9.1.7. Write a sentence to describe the behavior of the function along this trace.

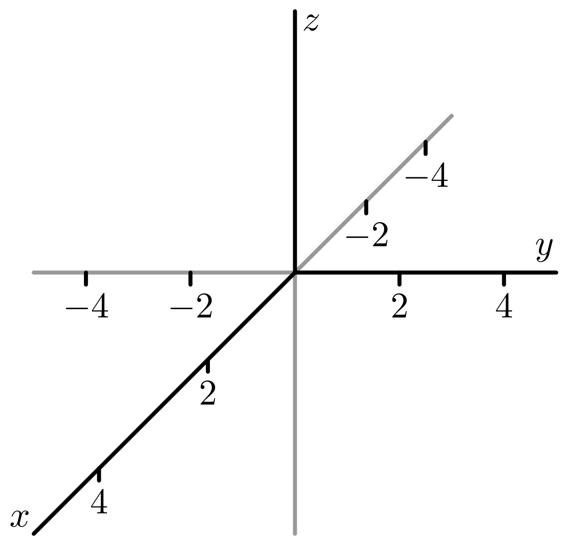


Figure 9.1.15 Coordinate axes to sketch traces.

c. For the function g defined by $g(x, y) = x^2 + y^2 + 1$, explain the type of function that each trace in the x direction will be (keeping y constant). Plot the $y = -4$, $y = -2$, $y = 0$, $y = 2$, and $y = 4$ traces in 3-dimensional coordinate system provided in Figure 9.1.15.

d. For the function g defined by $g(x, y) = x^2 + y^2 + 1$, explain the type of function that each trace in the y direction will be (keeping x constant). Plot the $x = -4$, $x = -2$, $x = 0$, $x = 2$, and $x = 4$ traces in 3-dimensional coordinate system in Figure 9.1.15.

e. Describe the surface generated by the function g.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.1.7. On the topographical map of the Porcupine Mountains in Figure 9.1.16,

a. identify the highest and lowest points you can find;

b. from a point of your choice, determine a path of steepest ascent that leads to the highest point;

c. from that same initial point, determine the least steep path that leads to the highest point.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.1.8.

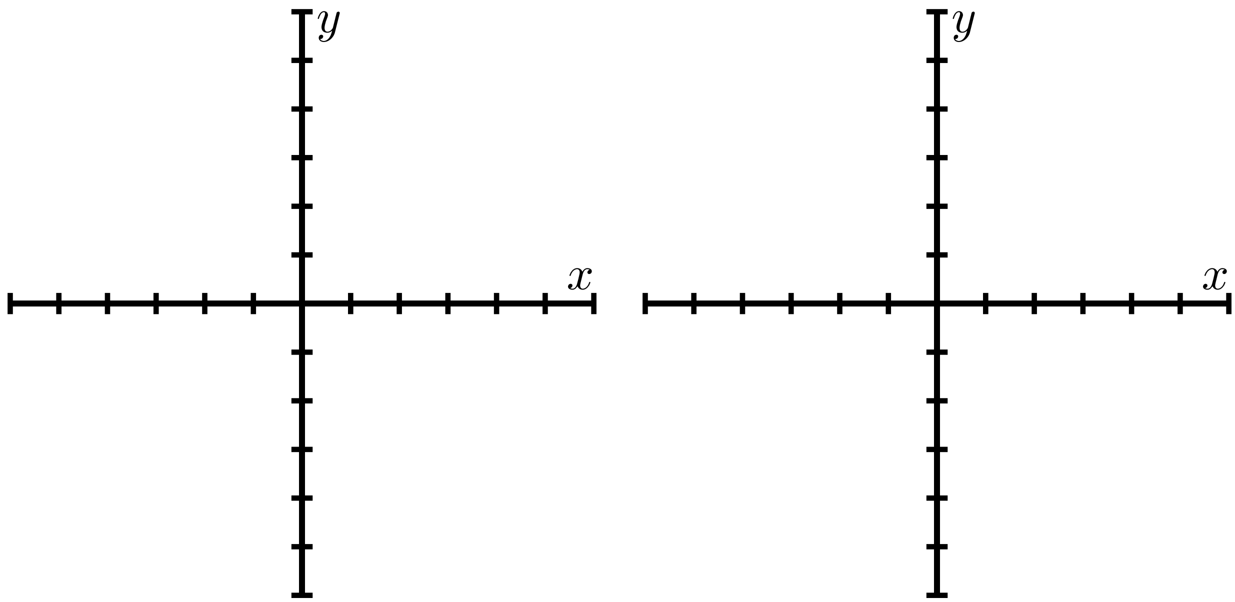


Figure 9.1.19 Left: Level curves for $f(x, y) = x^2 + y^2$. Right: Level curves for $g(x, y) = \sqrt{x^2 + y^2}$.

a. Let $f(x, y) = x^2 + y^2$. Draw the level curves $f(x, y) = k$ for $k = 1$, $k = 2$, $k = 3$, and $k = 4$ on the left set of axes given in Figure 9.1.19. (You decide on the scale of the axes.) Explain what the surface defined by f looks like.

b. Let $g(x, y) = \sqrt{x^2 + y^2}$. Draw the level curves $g(x, y) = k$ for $k = 1$, $k = 2$, $k = 3$, and $k = 4$ on the right set of axes given in Figure 9.1.19. (You decide on the scale of the axes.) Explain what the surface defined by g looks like.

c. Compare and contrast the graphs of f and g. How are they alike? How are they different? Use traces for each function to help answer these questions.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

9.2 Vectors

Preview Activity 9.2.1. Postscript is a programming language whose primary purpose is to describe the appearance of text or graphics. A simple set of Postscript commands that produces the triangle in the plane with vertices $(0,0)$, $(1,1)$, and $(1,-1)$ is the following:

```
(0,0) moveto
 (1,1) lineto stroke
 (1,-1) lineto stroke
 (0,0) lineto stroke
```

The key idea in these commands is that we start at the origin, then tell Postscript that we want to start at the point $(0,0)$, draw a line from the point $(0,0)$ to the point $(1,1)$ (this is what the lineto and stroke commands do), then draw lines from $(1,1)$ to $(1,-1)$ and $(1,-1)$ back to the origin. Each of these commands encodes two important pieces of information: a direction in which to move and a distance to move. Mathematically, we can capture this information succinctly in a vector. To do so, we record the movement on the map in a pair $\langle x, y\rangle$ (this pair $\langle x, y\rangle$ is a **vector**), where x is the horizontal displacement and y the vertical displacement from one point to another. So, for example, the vector from the origin to the point $(1,1)$ is represented by the vector $\langle 1, 1\rangle$.

a. What is the vector $\mathbf{v}_1 = \langle x, y\rangle$ that describes the displacement from the point $(1,1)$ to the point $(1,-1)$? How can we use this vector to determine the distance from the point $(1,1)$ to the point $(1,-1)$?

b. Suppose we want to draw the triangle with vertices $A = (2,3)$, $B = (-3,1)$, and $C = (4,-2)$. As a shorthand notation, we will denote the vector from the point A to the point B as $\overrightarrow{AB}$

   i. Determine the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$, and $\overrightarrow{AC}$.

   ii. What relationship do you see among the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$, and $\overrightarrow{AC}$? Explain why this relationship should hold.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.2.2. An article by C.Kenneth Tanner of the University of Georgia argues that, due to the concept of social distance, a secondary school classroom for 20 students should have 1344 square feet of floor space. Suppose a classroom is 32 feet by 42 feet by 8 feet. Set the origin O of the classroom to be its center. In this classroom, a student is sitting on a chair whose seat is at location $A = (9, -6, -1.5)$, an overhead projector is located at position $B = (0, 1, 3)$, and the teacher is standing at point $C = (-2, 20, -4)$, all distances measured in feet. Determine the components of the indicated vectors and explain in context what each represents.

a. $\overrightarrow{OA}$ b. $\overrightarrow{OB}$ c. $\overrightarrow{OC}$ d. $\overrightarrow{AB}$ e. $\overrightarrow{AC}$ f. $\overrightarrow{BC}$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.2.3. Let $\mathbf{u} = \langle 2, 3\rangle$, $\mathbf{v} = \langle -1, 4\rangle$.

a. Using the two specific vectors above, what is the natural way to define the vector sum $\mathbf{u} + \mathbf{v}$?

b. In general, how do you think the vector sum $\mathbf{a} + \mathbf{b}$ of vectors $\mathbf{a} = \langle a_1, a_2\rangle$ and $\mathbf{b} = \langle b_1, b_2\rangle$ in $\mathbb{R}^2$ should be defined? Write a formal definition of a vector sum based on your intuition.

c. In general, how do you think the vector sum $\mathbf{a} + \mathbf{b}$ of vectors $\mathbf{a} = \langle a_1, a_2, a_3\rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3\rangle$ in $\mathbb{R}^3$ should be defined? Write a formal definition of a vector sum based on your intuition.

d. Returning to the specific vector $\mathbf{v} = \langle -1, 4\rangle$ given above, what is the natural way to define the scalar multiple $\frac{1}{2}\mathbf{v}$?

e. In general, how do you think a scalar multiple of a vector $\mathbf{a} = \langle a_1, a_2\rangle$ in $\mathbb{R}^2$ by a scalar c should be defined? how about for a scalar multiple of a vector $\mathbf{a} = \langle a_1, a_2, a_3\rangle$ in $\mathbb{R}^3$ by a scalar c? Write a formal definition of a scalar multiple of a vector based on your intuition.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.2.4.

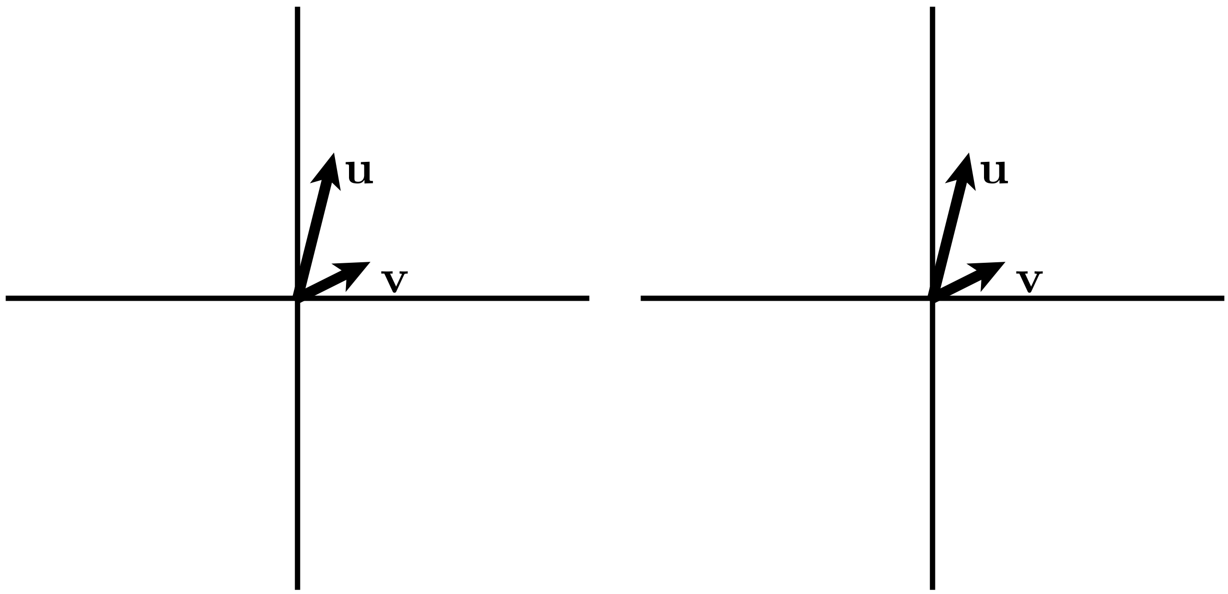


Figure 9.2.8 Left: Sketch sums. Right: Sketch multiples.

Suppose that $\mathbf{u}$ and $\mathbf{v}$ are the vectors shown in Figure 9.2.8.

a. On the axes at left in Figure 9.2.8, sketch the vectors $\mathbf{u}+\mathbf{v}$, $\mathbf{v}-\mathbf{u}$, $2\mathbf{u}$, $-2\mathbf{u}$, and $-3\mathbf{v}$.

b. What is $0\mathbf{v}$?

c. On the axes at right in Figure 9.2.8, sketch the vectors $-3\mathbf{v}$, $-2\mathbf{v}$, $-1\mathbf{v}$, $2\mathbf{v}$, and $3\mathbf{v}$.

d. Give a geometric description of the set of terminal points of the vectors $t\mathbf{v}$ where t is any scalar.

e. On the set of axes at right in Figure 9.2.8, sketch the vectors $\mathbf{u}-3\mathbf{v}$, $\mathbf{u}-2\mathbf{v}$, $\mathbf{u}-\mathbf{v}$, $\mathbf{u}+\mathbf{v}$, and $\mathbf{u}+2\mathbf{v}$.

f. Give a geometric description of the set of terminal points of the vectors $\mathbf{u}+t\mathbf{v}$ where t is any scalar.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.2.5.

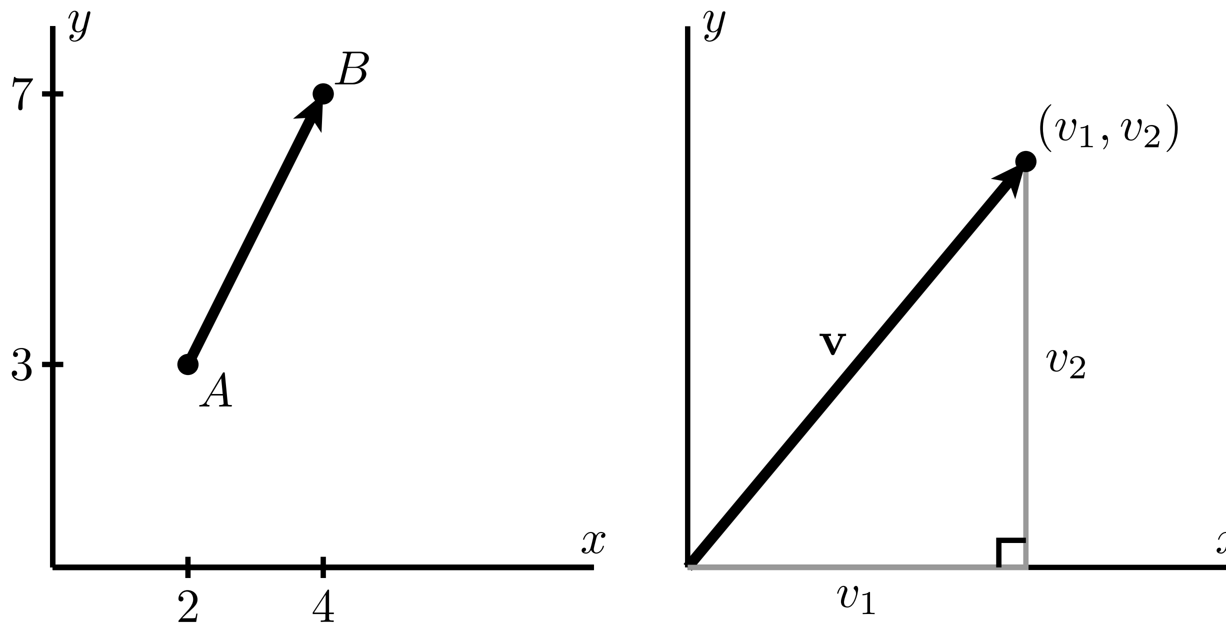


Figure 9.2.9 Left: $\overrightarrow{AB}$. Right: An arbitrary vector, $\mathbf{v}$.

a. Let $A = (2, 3)$ and $B = (4, 7)$, as shown at left in Figure 9.2.9. Compute $|\overrightarrow{AB}|$.

b. Let $\mathbf{v} = \langle v_1, v_2 \rangle$ be the vector in $\mathbb{R}^2$ with components v_1 and v_2 as shown at right in Figure 9.2.9. Use the distance formula to find a general formula for $|\mathbf{v}|$.

c. Let $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ be a vector in $\mathbb{R}^3$. Use the distance formula to find a general formula for $|\mathbf{v}|$.

d. Suppose that $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle -1, 2 \rangle$. Find $|\mathbf{u}|$, $|\mathbf{v}|$, and $|\mathbf{u} + \mathbf{v}|$. Is it true that $|\mathbf{u} + \mathbf{v}| = |\mathbf{u}| + |\mathbf{v}|$?

e. Under what conditions will $|\mathbf{u} + \mathbf{v}| = |\mathbf{u}| + |\mathbf{v}|$? (Hint: Think about how $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} + \mathbf{v}$ form the sides of a triangle.)

f. With the vector $\mathbf{u} = \langle 2, 3 \rangle$, find the lengths of $2\mathbf{u}$, $3\mathbf{u}$, and $-2\mathbf{u}$, respectively, and use proper notation to label your results.

g. If t is any scalar, how is $|t\mathbf{u}|$ related to $|\mathbf{u}|$?

h. A *unit vector* is a vector whose magnitude is 1. Of the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{i} + \mathbf{j}$, which are unit vectors?

i. Find a unit vector $\mathbf{v}$ whose direction is the same as $\mathbf{u} = \langle 2, 3 \rangle$. (Hint: Consider the result of part (g).)

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

9.3 The Dot Product

Preview Activity 9.3.1. For two-dimensional vectors $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$, the dot product is simply the scalar obtained by

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

a. If $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{v} = \langle -2, 1 \rangle$, find the dot product $\mathbf{u} \cdot \mathbf{v}$.

b. Find $\mathbf{i} \cdot \mathbf{i}$ and $\mathbf{i} \cdot \mathbf{j}$.

c. If $\mathbf{u} = \langle 3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{u}$. How is this related to $|\mathbf{u}|$?

d. On the axes in Figure 9.3.1, plot the vectors $\mathbf{u} = \langle 1, 3 \rangle$ and $\mathbf{v} = \langle -3, 1 \rangle$. Then, find $\mathbf{u} \cdot \mathbf{v}$. What is the angle between these vectors?

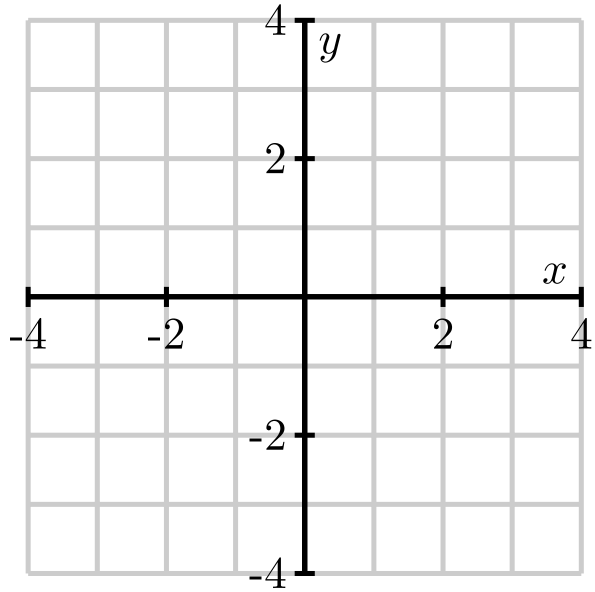


Figure 9.3.1 For part (d)

e. On the axes in Figure 9.3.2, plot the vector $\mathbf{u} = \langle 1, 3 \rangle$.

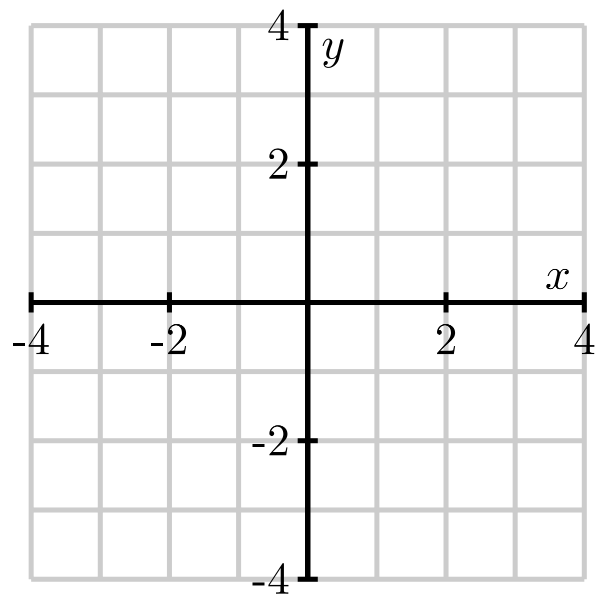


Figure 9.3.2 For part (e)

For each of the following vectors **v**, plot the vector on Figure 9.3.2 and then compute the dot product $\mathbf{u} \cdot \mathbf{v}$.

- $\mathbf{v} = \langle 3, 2\rangle$.
- $\mathbf{v} = \langle 3, 0\rangle$.
- $\mathbf{v} = \langle 3, -1\rangle$.
- $\mathbf{v} = \langle 3, -2\rangle$.
- $\mathbf{v} = \langle 3, -4\rangle$.

f. Based upon the previous part of this activity, what do you think is the sign of the dot product in the following three cases shown in Figure 9.3.3?

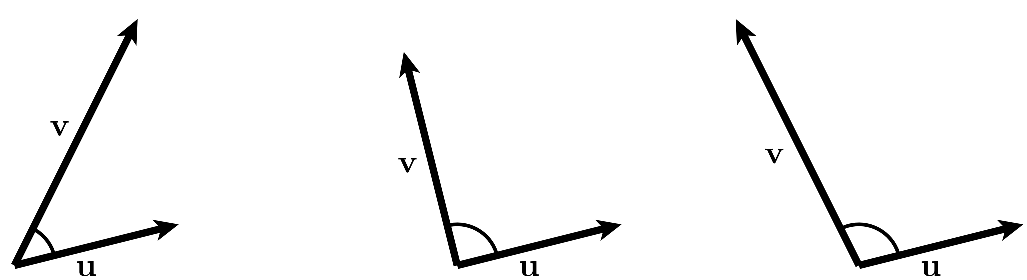


Figure 9.3.3 For part (f)

Activity 9.3.2. Determine each of the following.

a. $\langle 1, 2, -3\rangle \cdot \langle 4, -2, 0\rangle$.

b. $\langle 0, 3, -2, 1\rangle \cdot \langle 5, -6, 0, 4\rangle$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.3.3. Determine each of the following.

a. The length of the vector $\mathbf{u} = \langle 1, 2, -3 \rangle$ using the dot product.

b. The angle between the vectors $\mathbf{u} = \langle 1, 2 \rangle$ and $\mathbf{v} = \langle 4, -1 \rangle$ to the nearest tenth of a degree.

c. The angle between the vectors $\mathbf{y} = \langle 1, 2, -3 \rangle$ and $\mathbf{z} = \langle -2, 1, 1 \rangle$ to the nearest tenth of a degree.

d. If the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$ is a right angle, what does the expression $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos(\theta)$ say about their dot product?

e. If the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$ is acute—that is, less than $\pi/2$—what does the expression $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos(\theta)$ say about their dot product?

f. If the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$ is obtuse—that is, greater than $\pi/2$—what does the expression $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos(\theta)$ say about their dot product?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.3.4. Determine the work done by a 25 pound force acting at a 30° angle to the direction of the object's motion, if the object is pulled 10 feet. In addition, is more work or less work done if the angle to the direction of the object's motion is 60°?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.3.5. Let $\mathbf{u} = \langle 2, 6\rangle$.

a. Let $\mathbf{v} = \langle 4, -8\rangle$. Find $\text{comp}_{\mathbf{v}}\mathbf{u}$, $\text{proj}_{\mathbf{v}}\mathbf{u}$ and $\text{proj}_{\perp\mathbf{v}}\mathbf{u}$, and draw a picture to illustrate. Finally, express $\mathbf{u}$ as the sum of two vectors where one is parallel to $\mathbf{v}$ and the other is perpendicular to $\mathbf{v}$.

b. Now let $\mathbf{v} = \langle -2, 4\rangle$. Without doing any calculations, find $\text{proj}_{\mathbf{v}}\mathbf{u}$. Explain your reasoning. (Hint: Refer to the picture you drew in part (a).)

c. Find a vector $\mathbf{w}$ not parallel to $\mathbf{z} = \langle 3, 4\rangle$ such that $\text{proj}_{\mathbf{z}}\mathbf{w}$ has length 10. Note that there are infinitely many different answers.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

9.4 The Cross Product

Preview Activity 9.4.1. The cross product of two vectors, $\mathbf{u}$ and $\mathbf{v}$, will itself be a **vector** denoted $\mathbf{u} \times \mathbf{v}$. The direction of $\mathbf{u} \times \mathbf{v}$ is determined by the right-hand rule: if we point the index finger of our right hand in the direction of $\mathbf{u}$ and our middle finger in the direction of $\mathbf{v}$, then our thumb points in the direction of $\mathbf{u} \times \mathbf{v}$.

a. We begin by defining the cross products using the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$. Referring to Figure 9.4.1, explain why $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ in that order form a right-hand system. We then define $\mathbf{i} \times \mathbf{j}$ to be $\mathbf{k}$ — that is $\mathbf{i} \times \mathbf{j} = \mathbf{k}$.

b. Now explain why $\mathbf{i}$, $\mathbf{k}$, and $-\mathbf{j}$ in that order form a right-hand system. We then define $\mathbf{i} \times \mathbf{k}$ to be $-\mathbf{j}$ — that is $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$.

c. Continuing in this way, complete the missing entries in Table 9.4.2.

   Table 9.4.2 Table of cross products involving i, j, and k.

$\mathbf{i} \times \mathbf{j} = \mathbf{k}$	$\mathbf{i} \times \mathbf{k} = -\mathbf{j}$	$\mathbf{j} \times \mathbf{k} =$
$\mathbf{j} \times \mathbf{i} =$	$\mathbf{k} \times \mathbf{i} =$	$\mathbf{k} \times \mathbf{j} =$

d. Up to this point, the products you have seen, such as the product of real numbers and the dot product of vectors, have been commutative, meaning that the product does not depend on the order of the terms. For instance, $2 \cdot 5 = 5 \cdot 2$. The table above suggests, however, that the cross product is *anti-commutative*: for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$. If we consider the case when $\mathbf{u} = \mathbf{v}$, this shows that $\mathbf{v} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{v})$. What does this tell us about $\mathbf{v} \times \mathbf{v}$; in particular, what vector is unchanged by scalar multiplication by -1?

e. It is not difficult to show that the cross product interacts with scalar multiplication and vector addition as one would expect: that is

$$(c\mathbf{u}) \times \mathbf{v} = c(\mathbf{u} \times \mathbf{v})$$
$$(\mathbf{u} + \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \times \mathbf{w}) + (\mathbf{v} \times \mathbf{w})$$

   We can combine these properties to make cross product calculations a bit easier. For example,

$$\begin{aligned}(2\mathbf{i} + \mathbf{j}) \times \mathbf{k} &= (2\mathbf{i} \times \mathbf{k}) + (\mathbf{j} \times \mathbf{k}) \\ &= 2(\mathbf{i} \times \mathbf{k}) + (\mathbf{j} \times \mathbf{k}) \\ &= -2\mathbf{j} + \mathbf{i}.\end{aligned}$$

   Using these properties along with Table 9.4.2, find the cross product $\mathbf{u} \times \mathbf{v}$ if $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{v} = -\mathbf{i} + \mathbf{k}$.

f. Verify that the cross product $\mathbf{u} \times \mathbf{v}$ you just found in part (e) is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

g. Consider the vectors $\mathbf{u}$ and $\mathbf{v}$ in the xy-plane as shown below in Figure 9.4.3.

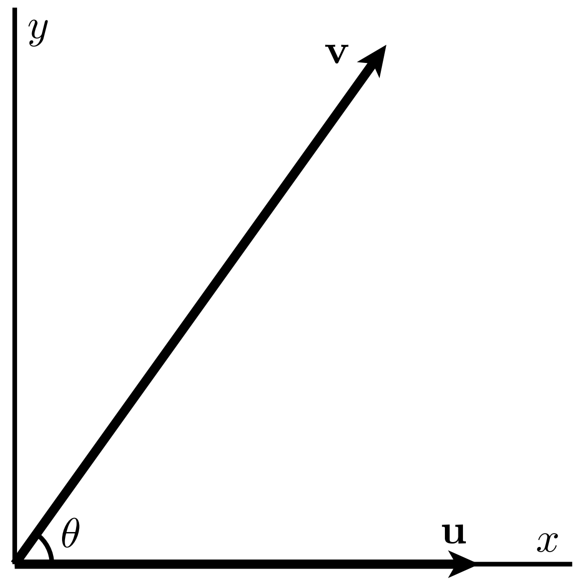


Figure 9.4.3 Two vectors in the xy-plane

Explain why $\mathbf{u} = |\mathbf{u}|\mathbf{i}$ and $\mathbf{v} = |\mathbf{v}|\cos(\theta)\mathbf{i} + |\mathbf{v}|\sin(\theta)\mathbf{j}$. Then compute the length of $|\mathbf{u} \times \mathbf{v}|$.

Activity 9.4.2. Suppose $\mathbf{u} = \langle 0, 1, 3\rangle$ and $\mathbf{v} = \langle 2, -1, 0\rangle$. Use the formula (9.4.1) for the following.

a. Find the cross product $\mathbf{u} \times \mathbf{v}$.

b. Evaluate the dot products $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v})$ and $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v})$. What does this tell you about the geometric relationship among $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$?

c. Find the cross product $\mathbf{v} \times \mathbf{i}$.

d. Multiplication of real numbers is *associative*, which means, for instance, that $(2 \cdot 5) \cdot 3 = 2 \cdot (5 \cdot 3)$. Is it true that the cross product of vectors is associative? For instance, is it true that $(\mathbf{u} \times \mathbf{v}) \times \mathbf{i} = \mathbf{u} \times (\mathbf{v} \times \mathbf{i})$?

e. Find the cross product $\mathbf{u} \times \mathbf{u}$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.4.3.

a. Find the area of the parallelogram formed by the vectors $\mathbf{u} = \langle 1, 3, -2 \rangle$ and $\mathbf{v} = \langle 3, 0, 1 \rangle$.

b. Find the area of the parallelogram in $\mathbb{R}^3$ whose vertices are $(1, 0, 1)$, $(0, 0, 1)$, $(2, 1, 0)$, and $(1, 1, 0)$. (Hint: It might be helpful to draw a picture to see how the vertices are arranged so you can determine which vectors you might use.)

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.4.4. Suppose $\mathbf{u} = \langle 3, 5, -1 \rangle$ and $\mathbf{v} = \langle 2, -2, 1 \rangle$.

a. Find two unit vectors orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

b. Find the volume of the parallelepiped formed by the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w} = \langle 3, 3, 1 \rangle$.

c. Find a vector orthogonal to the plane containing the points $(0, 1, 2)$, $(4, 1, 0)$, and $(-2, 2, 2)$.

d. Given the vectors $\mathbf{u}$ and $\mathbf{v}$ shown below in Figure 9.4.7, sketch the cross products $\mathbf{u} \times \mathbf{v}$ and $\mathbf{v} \times \mathbf{u}$.

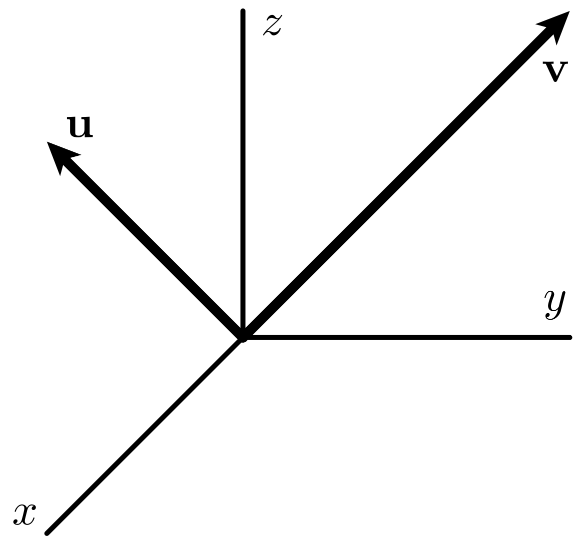


Figure 9.4.7 Vectors $\mathbf{u}$ and $\mathbf{v}$

e. Do the vectors $\mathbf{a} = \langle 1, 3, -2 \rangle$,$\mathbf{b} = \langle 2, 1, -4 \rangle$, and $\mathbf{c} = \langle 0, 1, 0 \rangle$ in standard position lie in the same plane? Use the concepts from this section to explain.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

9.5 Lines and Planes in Space

Preview Activity 9.5.1. We are familiar with equations of lines in the plane in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the y-intercept. In this activity, we explore a more flexible way of representing lines that we can use not only in the plane, but in higher dimensions as well.
To begin, consider the line through the point $(2, -1)$ with slope $\frac{2}{3}$ as shown in Figure 9.5.1.

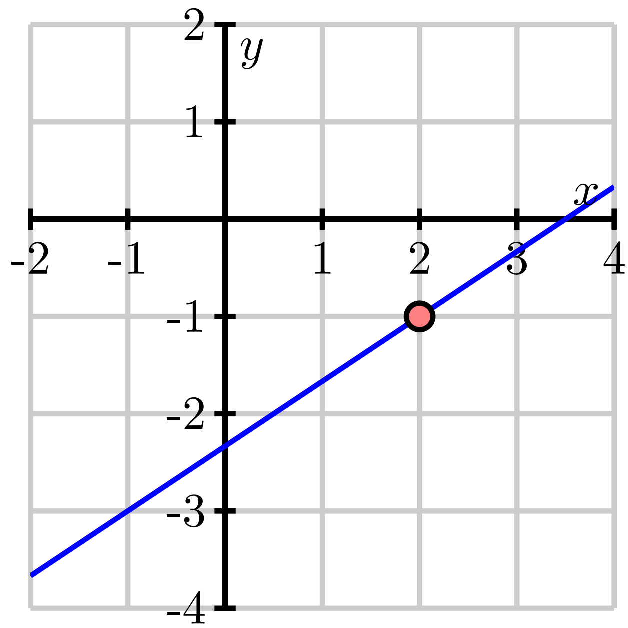


Figure 9.5.1 The line through $(2, -1)$ with slope $\frac{2}{3}$.

a. Suppose we increase x by 1 from the point $(2, -1)$. How does the y-value change? What is the point on the line with x-coordinate 3?

b. Suppose we decrease x by 3.25 from the point $(2, -1)$. How does the y-value change? What is the point on the line with x-coordinate -1.25?

c. Now, suppose we increase x by some arbitrary value $3t$ from the point $(2, -1)$. How does the y-value change? What is the point on the line with x-coordinate $2 + 3t$?

d. Observe that the slope of the line is related to any vector whose y-component divided by the x-component is the slope of the line. For the line in this exercise, we might use the vector $\langle 3, 2\rangle$, which describes the direction of the line. Explain why the terminal points of the vectors $\mathbf{r}(t)$, where

$$\mathbf{r}(t) = \langle 2, -1\rangle + \langle 3, 2\rangle t,$$

trace out the graph of the line through the point $(2, -1)$ with slope $\frac{2}{3}$.

e. Now we extend this vector approach to $\mathbb{R}^3$ and consider a second example. Let $\mathcal{L}$ be the line in $\mathbb{R}^3$ through the point $(1, 0, 2)$ in the direction of the vector $\langle 2, -1, 4\rangle$. Find the coordinates of three distinct points on line $\mathcal{L}$. Explain your thinking.

f. Find a vector in the form

$$\mathbf{r}(t) = \langle x_0, y_0, z_0\rangle + \langle a, b, c\rangle t$$

whose terminal points trace out the line $\mathcal{L}$ that is described in (e). That is, you should be able to locate any point on the line by determining a corresponding value of t.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.5.2. Let $P_1 = (1, 2, -1)$ and $P_2 = (-2, 1, -2)$. Let $\mathcal{L}$ be the line in $\mathbb{R}^3$ through P_1 and P_2, and note that three snapshots of this line are shown in Figure 9.5.5.

a. Find a direction vector for the line $\mathcal{L}$.

b. Find a vector equation of $\mathcal{L}$ in the form $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$.

c. Consider the vector equation $\mathbf{s}(t) = \langle -5, 0, -3 \rangle + t\langle 6, 2, 2 \rangle$. What is the direction of the line given by $\mathbf{s}(t)$? Is this new line parallel to line $\mathcal{L}$?

d. Do $\mathbf{r}(t)$ and $\mathbf{s}(t)$ represent the same line, $\mathcal{L}$? Explain.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.5.3. Let $P_1 = (1, 2, -1)$ and $P_2 = (-2, 1, -2)$, and let $\mathcal{L}$ be the line in $\mathbb{R}^3$ through P_1 and P_2, which is the same line as in Activity 9.5.2.

a. Find parametric equations of the line $\mathcal{L}$.

b. Does the point $(1, 2, 1)$ lie on $\mathcal{L}$? If so, what value of t results in this point?

c. Consider another line, $\mathcal{K}$, whose parametric equations are

$$x(s) = 11 + 4s, \quad y(s) = 1 - 3s, \quad z(s) = 3 + 2s.$$

What is the direction of the line $\mathcal{K}$?

d. Do the lines $\mathcal{L}$ and $\mathcal{K}$ intersect? If so, provide the point of intersection and the t and s values, respectively, that result in the point. If not, explain why.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.5.4.

a. Write a scalar equation of the plane p_1 passing through the point $(0, 2, 4)$ and perpendicular to the vector $\mathbf{n} = \langle 2, -1, 1 \rangle$.

b. Is the point $(2, 0, 2)$ on the plane p_1?

c. Write a scalar equation of the plane p_2 that is parallel to p_1 and passing through the point $(3, 0, 4)$. (Hint: Compare normal vectors of the planes.)

d. Write a parametric description of the line l passing through the point $(2, 0, 2)$ and perpendicular to the plane p_3 described by the equation $x + 2y - 2z = 7$.

e. Find the point at which l intersects the plane p_3.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.5.5. Let $P_0 = (1, 2, -1)$, $P_1 = (1, 0, -1)$, and $P_2 = (0, 1, 3)$ and let p be the plane containing P_0, P_1, and P_2.

a. Determine the components of the vectors $\overrightarrow{P_0P_1}$ and $\overrightarrow{P_0P_2}$.

b. Find a normal vector $\mathbf{n}$ to the plane p.

c. Find a scalar equation of the plane p.

d. Consider a second plane, q, with scalar equation $-3(x - 1) + 4(y + 3) + 2(z - 5) = 0$. Find two different points on plane q, as well as a vector $\mathbf{m}$ that is normal to q.

e. The angle between two planes is the acute angle between their respective normal vectors. What is the angle between planes p and q?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

9.6 Vector-Valued Functions

Preview Activity 9.6.1. In this activity we consider how we might use vectors to define a curve in space.

a. On a single set of axes in $\mathbb{R}^2$, draw the vectors

- $\langle \cos(0), \sin(0) \rangle$,
- $\langle \cos\left(\frac{\pi}{2}\right), \sin\left(\frac{\pi}{2}\right) \rangle$,
- $\langle \cos(\pi), \sin(\pi) \rangle$, and
- $\left\langle \cos\left(\dfrac{3\pi}{2}\right), \sin\left(\dfrac{3\pi}{2}\right) \right\rangle$

with their initial points at the origin.

b. On the same set of axes, draw the vectors

- $\langle \cos\left(\frac{\pi}{4}\right), \sin\left(\frac{\pi}{4}\right) \rangle$,
- $\langle \cos\left(\frac{3\pi}{4}\right), \sin\left(\frac{3\pi}{4}\right) \rangle$,
- $\langle \cos\left(\frac{5\pi}{4}\right), \sin\left(\frac{5\pi}{4}\right) \rangle$, and
- $\left\langle \cos\left(\dfrac{7\pi}{4}\right), \sin\left(\dfrac{7\pi}{4}\right) \right\rangle$

with their initial points at the origin.

c. Based on the pictures from parts (a) and (b), sketch the set of *terminal* points of all of the vectors of the form $\langle \cos(t), \sin(t) \rangle$, where t assumes values from 0 to 2π. What is the resulting figure? Why?

d. Suppose we sketched the terminal points of all vectors of the form $\langle \cos(t), \sin(t) \rangle$, where t assumes values from 0 to π. How does the resulting picture differ from the one in part (c)? What about for t from 0 to 4π?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.6.2. The same curve can be represented with different parameterizations. Use appropriate technology to plot the curves generated by the following vector-valued functions for values of t from 0 to 2π. Compare and contrast the graphs — explain how they are alike and how they are different.

a. $\mathbf{r}(t) = \langle \sin(t), \cos(t) \rangle$

b. $\mathbf{r}(t) = \langle \sin(2t), \cos(2t) \rangle$

c. $\mathbf{r}(t) = \langle \cos(t+\pi), \sin(t+\pi) \rangle$

d. $\mathbf{r}(t) = \langle \cos(t^2), \sin(t^2) \rangle$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.6.3. Vector-valued functions can be used to generate many interesting curves. Graph each of the following using an appropriate technological tool, and then write one sentence for each function to describe the behavior of the resulting curve.

a. $\mathbf{r}(t) = \langle t\cos(t), t\sin(t)\rangle$

b. $\mathbf{r}(t) = \langle \sin(t)\cos(t), t\sin(t)\rangle$

c. $\mathbf{r}(t) = \langle \sin(5t), \sin(4t)\rangle$

d. $\mathbf{r}(t) = \langle t^2\sin(t)\cos(t), 0.9t\cos(t^2), \sin(t)\rangle$ (Note that this defines a curve in 3-space.)

e. Experiment with different formulas for $x(t)$ and $y(t)$ and ranges for t to see what other interesting curves you can generate. Share your best results with peers.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.6.4. Consider the paraboloid defined by $f(x, y) = x^2 + y^2$.

a. Find a parameterization for the $x = 2$ trace of f. What type of curve does this trace describe?

b. Find a parameterization for the $y = -1$ trace of f. What type of curve does this trace describe?

c. Find a parameterization for the level curve $f(x, y) = 25$. What type of curve does this trace describe?

d. How do your responses change to all three of the preceding questions if you instead consider the function g defined by $g(x, y) = x^2 - y^2$? (Hint for generating one of the parameterizations: $\sec^2(t) - \tan^2(t) = 1$.)

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

9.7 Derivatives and Integrals of Vector-Valued Functions

Preview Activity 9.7.1. Let $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(2t)\mathbf{j}$ describe the path traveled by an object at time t.

a. Use appropriate technology to help you sketch the graph of the vector-valued function $\mathbf{r}$, and then locate and label the point on the graph when $t = \pi$.

b. Recall that for functions of a single variable, the derivative of a sum is the sum of the derivatives; that is, $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$. With this idea in mind and viewing $\mathbf{i}$ and $\mathbf{j}$ as constant vectors, what do you expect the derivative of $\mathbf{r}$ to be? Write a proposed formula for $\mathbf{r}'(t)$.

c. Use your result from part (b) to compute $\mathbf{r}'(\pi)$. Sketch this vector $\mathbf{r}'(\pi)$ as emanating from the point on the graph of $\mathbf{r}$ when $t = \pi$, and explain what you think $\mathbf{r}'(\pi)$ tells us about the object's motion.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.7.2. Let's investigate how we can interpret the derivative $\mathbf{r}'(t)$. Let $\mathbf{r}$ be the vector-valued function whose graph is shown in Figure 9.7.2, and let h be a scalar that represents a small change in time. The vector $\mathbf{r}(t)$ is the blue vector in Figure 9.7.2 and $\mathbf{r}(t+h)$ is the green vector.

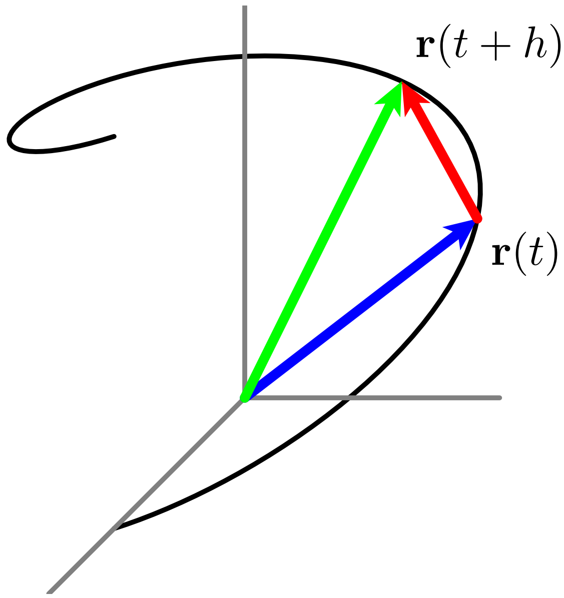


Figure 9.7.2 A single difference quotient.

a. Is the quantity $\mathbf{r}(t+h) - \mathbf{r}(t)$ a vector or a scalar? Identify this object in Figure 9.7.2.

b. Is $\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$ a vector or a scalar? Sketch a representative vector $\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$ with $h < 1$ in Figure 9.7.2.

c. Think of $\mathbf{r}(t)$ as providing the position of an object moving along the curve these vectors trace out. What do you think that the vector $\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$ measures? Why? (Hint: You might think analogously about difference quotients such as $\frac{f(x+h)-f(x)}{h}$ or $\frac{s(t+h)-s(t)}{h}$ from calculus I.)

d. Figure 9.7.3 presents three snapshots of the vectors $\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$ as we let $h \to 0$. Write 2-3 sentences to describe key attributes of the vector

$$\lim_{h\to 0} \frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}.$$

(Hint: Compare to limits such as $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ or $\lim_{h\to 0} \frac{s(t+h)-s(t)}{h}$ from calculus I, keeping in mind that in three dimensions there is no general concept of slope.)

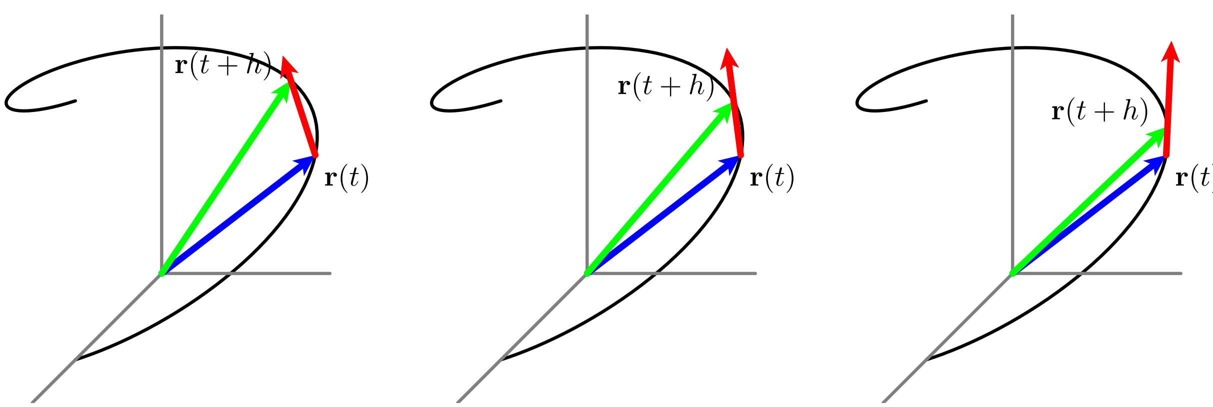


Figure 9.7.3 Snapshots of several difference quotients.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.7.3. For each of the following vector-valued functions, find $\mathbf{r}'(t)$.

a. $\mathbf{r}(t) = \langle \cos(t), t\sin(t), \ln(t)\rangle$.

b. $\mathbf{r}(t) = \langle t^2 + 3t, e^{-2t}, \frac{t}{t^2+1}\rangle$.

c. $\mathbf{r}(t) = \langle \tan(t), \cos(t^2), te^{-t}\rangle$.

d. $\mathbf{r}(t) = \langle \sqrt{t^4 + 4}, \sin(3t), \cos(4t)\rangle$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.7.4. The left side of Figure 9.7.4 shows the curve described by the vector-valued function $\mathbf{r}$ defined by

$$\mathbf{r}(t) = \left\langle 2t - \frac{1}{2}t^2 + 1, t - 1 \right\rangle.$$

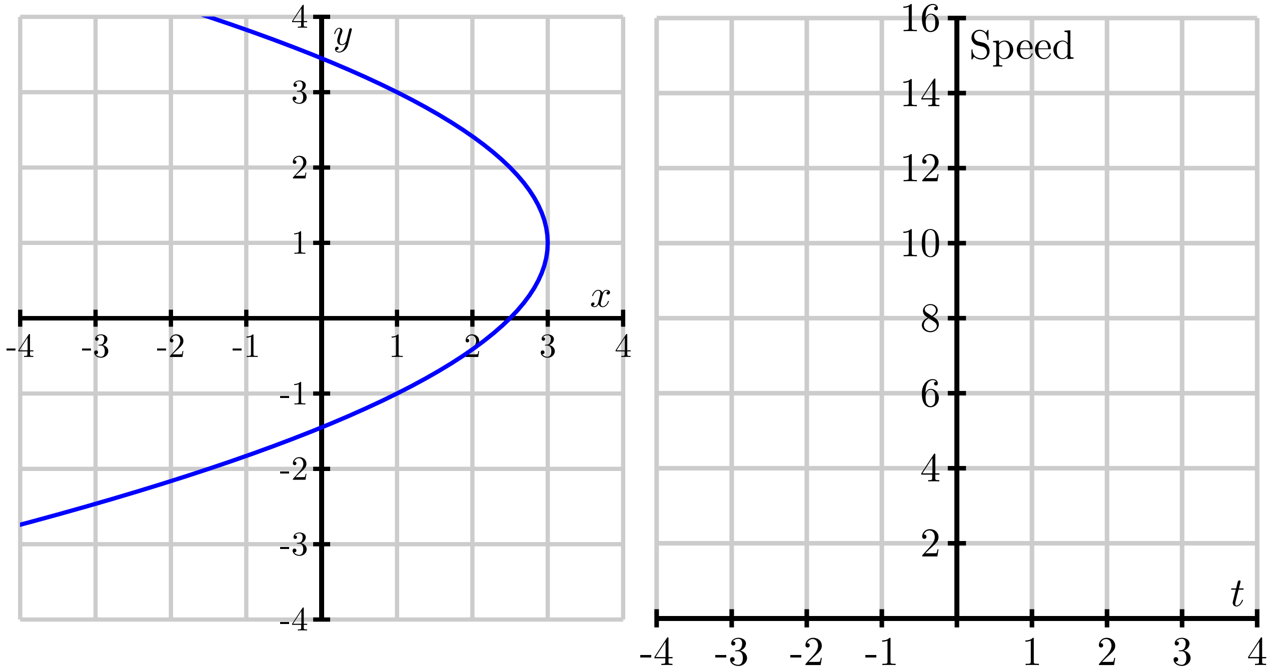


Figure 9.7.4 The curve $\mathbf{r}(t) = \langle 2t - \frac{1}{2}t^2 + 1, t - 1\rangle$ and its speed.

a. Find the object's velocity $\mathbf{v}(t)$.

b. Find the object's acceleration $\mathbf{a}(t)$.

c. Indicate on the left of Figure 9.7.4 the object's position, velocity and acceleration at the times $t = 0, 2, 4$. Draw the velocity and acceleration vectors with their tails placed at the object's position.

d. Recall that the speed is $|\mathbf{v}| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$. Find the object's speed and graph it as a function of time t on the right of Figure 9.7.4. When is the object's speed the slowest? When is the speed increasing? When is it decreasing?

e. What seems to be true about the angle between $\mathbf{v}$ and $\mathbf{a}$ when the speed is at a minimum? What is the angle between $\mathbf{v}$ and $\mathbf{a}$ when the speed is increasing? when the speed is decreasing?

f. Since the square root is an increasing function, we see that the speed increases precisely when $\mathbf{v} \cdot \mathbf{v}$ is increasing. Use the product rule for the dot product to express $\frac{d}{dt}(\mathbf{v} \cdot \mathbf{v})$ in terms of the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$. Use this to explain why the speed is increasing when $\mathbf{v} \cdot \mathbf{a} > 0$ and decreasing when $\mathbf{v} \cdot \mathbf{a} < 0$. Compare this to part (d).

g. Show that the speed's rate of change is

$$\frac{d}{dt}|\mathbf{v}(t)| = \operatorname{comp}_{\mathbf{v}}\mathbf{a}.$$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.7.5. Let

$$\mathbf{r}(t) = \cos(t)\mathbf{i} - \sin(t)\mathbf{j} + t\mathbf{k}.$$

Sketch the curve using some appropriate tool.

a. Determine the coordinates of the point on the curve traced out by $\mathbf{r}(t)$ when $t = \pi$.

b. Find a direction vector for the line tangent to the graph of $\mathbf{r}$ at the point where $t = \pi$.

c. Find the parametric equations of the line tangent to the graph of $\mathbf{r}$ when $t = \pi$.

d. Sketch a plot of the curve $\mathbf{r}(t)$ and its tangent line near the point where $t = \pi$. In addition, include a sketch of $\mathbf{r}'(\pi)$. What is the important role of $\mathbf{r}'(\pi)$ in this activity?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.7.6. Suppose a moving object in space has its velocity given by

$$\mathbf{v}(t) = (-2\sin(2t))\mathbf{i} + (2\cos(t))\mathbf{j} + \left(1 - \frac{1}{1+t}\right)\mathbf{k}.$$

A graph of the position of the object for times t in $[-0.5, 3]$ is shown in Figure 9.7.6. Suppose further that the object is at the point $(1.5, -1, 0)$ at time $t = 0$.

a. Determine $\mathbf{a}(t)$, the acceleration of the object at time t.

b. Determine $\mathbf{r}(t)$, position of the object at time t.

c. Compute and sketch the position, velocity, and acceleration vectors of the object at time $t = 1$, using Figure 9.7.6.

d. Finally, determine the vector equation for the tangent line, $\mathbf{L}(t)$, that is tangent to the position curve at $t = 1$.

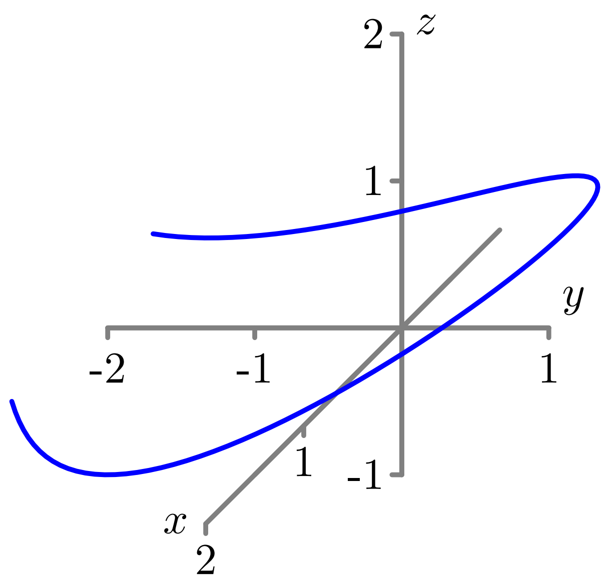


Figure 9.7.6 The position graph for the function in Activity 9.7.6.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

9.8 Arc Length and Curvature

Preview Activity 9.8.1. In earlier investigations, we have used integration to calculate quantities such as area, volume, mass, and work. We are now interested in determining the length of a space curve. Consider the smooth curve in 3-space defined by the vector-valued function $\mathbf{r}$, where

$$\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle = \langle \cos(t), \sin(t), t\rangle$$

for t in the interval $[0, 2\pi]$. Pictures of the graph of $\mathbf{r}$ are shown in Figure 9.8.1. We will use the integration process to calculate the length of this curve. In this situation we partition the interval $[0, 2\pi]$ into n subintervals of equal length and let $0 = t_0 < t_1 < t_2 < \cdots < t_n = b$ be the endpoints of the subintervals. We then approximate the length of the curve on each subinterval with some related quantity that we can compute. In this case, we approximate the length of the curve on each subinterval with the length of the segment connecting the endpoints. Figure 9.8.1 illustrates the process in three different instances using increasing values of n.

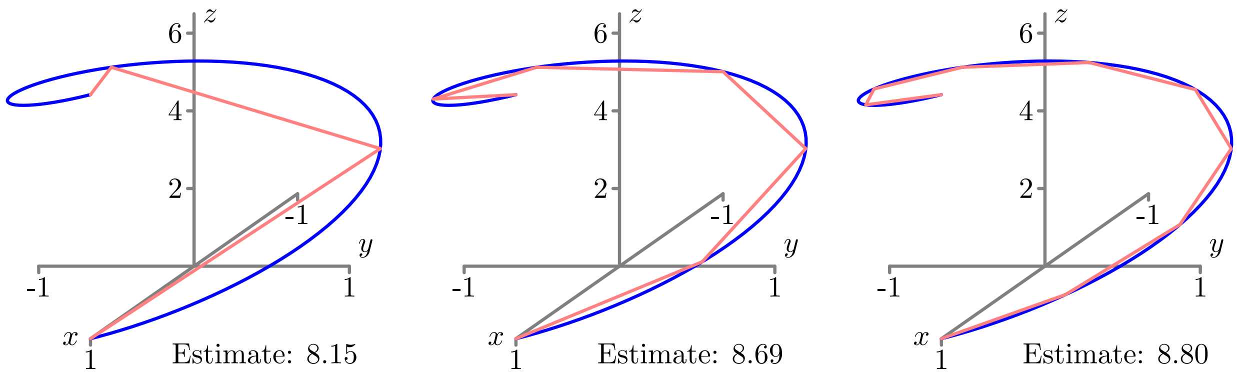


Figure 9.8.1 Approximating the length of the curve with $n = 3$, $n = 6$, and $n = 9$.

a. Write a formula for the length of the line segment that connects the endpoints of the curve on the ith subinterval $[t_{i-1}, t_i]$. (This length is our approximation of the length of the curve on this interval.)

b. Use your formula in part (a) to write a sum that adds all of the approximations to the lengths on each subinterval.

c. What do we need to do with the sum in part (b) in order to obtain the exact value of the length of the graph of $\mathbf{r}(t)$ on the interval $[0, 2\pi]$?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.8.2. Here we calculate the arc length of two familiar curves.

a. Use Equation (9.8.1) to calculate the circumference of a circle of radius r.

b. Find the exact length of the spiral defined by $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ on the interval $[0, 2\pi]$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.8.3. Let $y = f(x)$ define a smooth curve in 2-space. Parameterize this curve and use Equation (9.8.1) to show that the length of the curve defined by f on an interval $[a, b]$ is

$$\int_a^b \sqrt{1 + [f'(t)]^2}\, dt.$$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.8.4. In this activity we parameterize a line in 2-space in terms of arc length. Consider the line with parametric equations

$$x(t) = x_0 + at \qquad \text{and} \qquad y(t) = y_0 + bt.$$

a. To write t in terms of s, evaluate the integral

$$s = L(t) = \int_0^t \sqrt{(x'(w))^2 + (y'(w))^2}\, dw$$

to determine the length of the line from time 0 to time t.

b. Use the formula from (a) for s in terms of t to write t in terms of s. Then explain why a parameterization of the line in terms of arc length is

$$x(s) = x_0 + \frac{a}{\sqrt{a^2+b^2}}s \qquad \text{and} \qquad y(s) = y_0 + \frac{b}{\sqrt{a^2+b^2}}s. \tag{9.8.4}$$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.8.5.

a. We should expect that the curvature of a line is 0 everywhere. To show that our definition of curvature measures this correctly in 2-space, recall that (9.8.4) gives us the arc length parameterization

$$x(s) = x_0 + \frac{a}{\sqrt{a^2+b^2}}s \qquad \text{and} \qquad y(s) = y_0 + \frac{b}{\sqrt{a^2+b^2}}s$$

of a line. Use this information to explain why the curvature of a line is 0 everywhere.

b. Recall that an arc length parameterization of a circle in 2-space of radius a centered at the origin is, from (9.8.3),

$$\mathbf{r}(s) = \left\langle a\cos\left(\frac{s}{a}\right),\ a\sin\left(\frac{s}{a}\right)\right\rangle.$$

Show that the curvature of this circle is the constant $\frac{1}{a}$. What can you say about the relationship between the size of the radius of a circle and the value of its curvature? Why does this make sense?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 9.8.6. Use one of the two formulas for κ in terms of t to help you answer the following questions.

a. The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has parameterization

$$\mathbf{r}(t) = \langle a\cos(t), b\sin(t)\rangle.$$

Find the curvature of the ellipse. Assuming $0 < b < a$, at what points is the curvature the greatest and at what points is the curvature the smallest? Does this agree with your intuition?

b. The standard helix has parameterization $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$. Find the curvature of the helix. Does the result agree with your intuition?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

10 Derivatives of Multivariable Functions

10.1 Limits

Preview Activity 10.1.1. We investigate the limits of several different functions by working with tables and graphs.

a. Consider the function f defined by

$$f(x) = 3 - x.$$

Complete Table 10.1.1.

Table 10.1.1 Values of $f(x) = 3 - x$.

x	-0.2	-0.1	0.0	0.1	0.2
$f(x)$					

What does the table suggest regarding $\lim_{x\to 0} f(x)$?

b. Explain how your results in (a) are reflected in Figure 10.1.2.

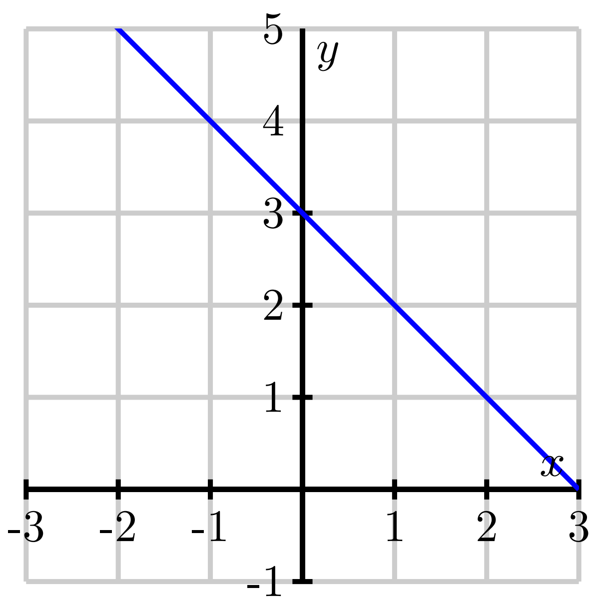


Figure 10.1.2 The graph of $f(x) = 3 - x$.

c. Next, consider

$$g(x) = \frac{x}{|x|}.$$

Complete Table 10.1.3 with values near $x = 0$, the point at which g is not defined.

Table 10.1.3 Values of $g(x) = \frac{x}{|x|}$.

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$g(x)$						

What does this suggest about $\lim_{x\to 0} g(x)$?

d. Explain how your results in (c) are reflected in Figure 10.1.4.

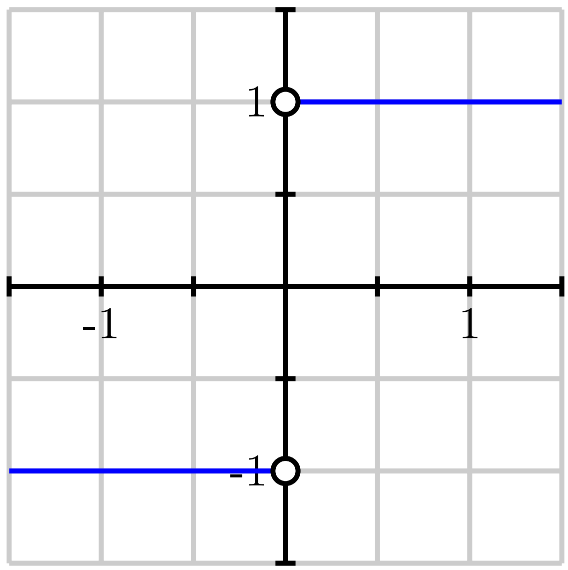


Figure 10.1.4 The graph of $g(x) = \frac{x}{|x|}$.

e. Now, let's examine a function of two variables. Let

$$f(x, y) = 3 - x - 2y.$$

Complete Table 10.1.5.

Table 10.1.5 Values of $f(x, y) = 3 - x - 2y$.

$x \backslash y$	-1.0	-0.1	0.0	0.1	1.0
-1.0		4.2			
-0.1					1.1
0.0				2.8	
0.1	4.9				
1.0			2.0		

What does the table suggest about $\lim_{(x,y)\to(0,0)} f(x, y)$?

f. Explain how your results in (e) are reflected in Figure 10.1.6. Compare this limit to the limit in part (a). How are the limits similar and how are they different?

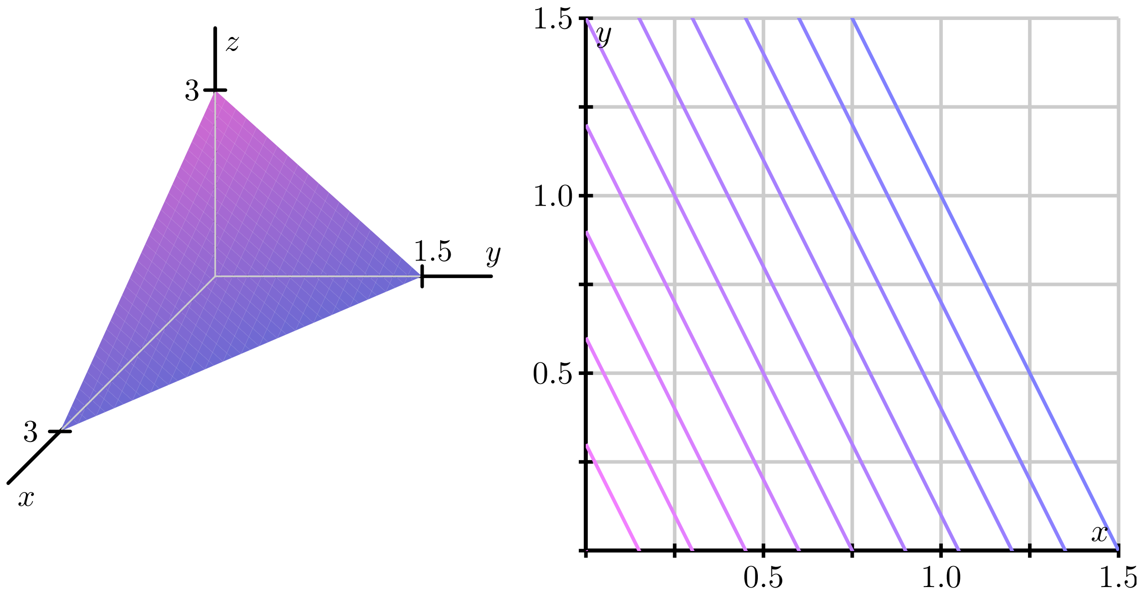


Figure 10.1.6 Left: The graph of $f(x, y) = 3 - x - 2y$. Right: A contour plot.

g. Finally, consider

$$g(x, y) = \frac{2xy}{x^2 + y^2},$$

which is not defined at $(0, 0)$. Complete Table 10.1.7. Round to three decimal places.

Table 10.1.7 Values of $g(x, y) = \frac{2xy}{x^2+y^2}$**.**

$x \backslash y$	-1.0	-0.1	0.0	0.1	1.0
-1.0		0.198			
-0.1					-0.198
0.0			—	0.000	
0.1	-0.198				
1.0			0.000		

What does this suggest about $\lim_{(x,y)\to(0,0)} g(x, y)$?

h. Explain how your results are reflected in Figure 10.1.8. Compare this limit to the limit in part (c). How are the results similar and how are they different?

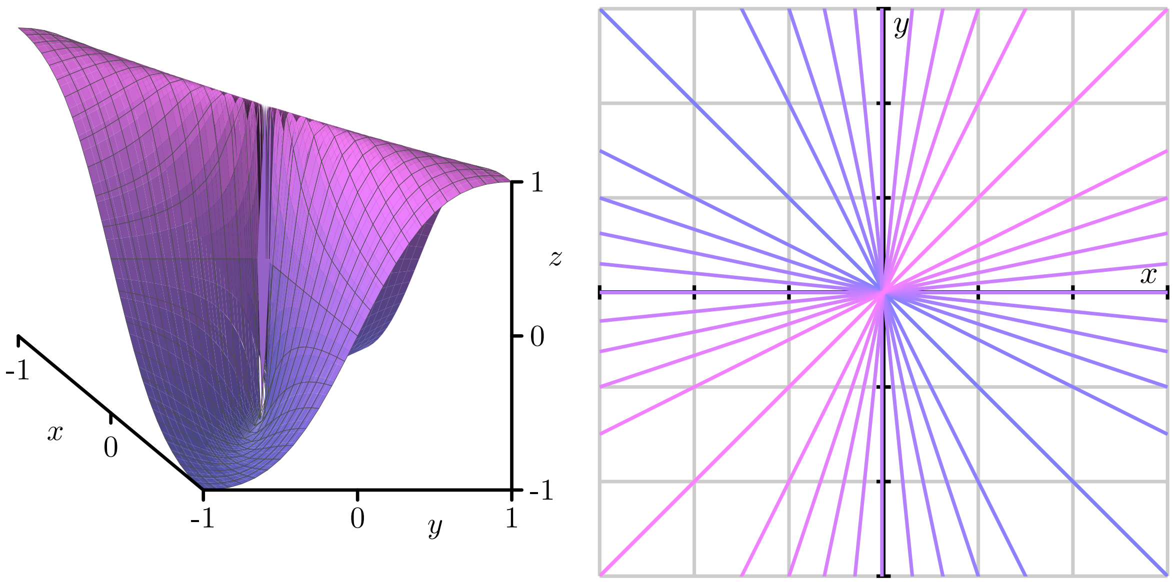


Figure 10.1.8 Left: The graph of $g(x,y) = \frac{2xy}{x^2+y^2}$. Right: A contour plot.

Activity 10.1.2. Consider the function f, defined by

$$f(x,y) = \frac{y}{\sqrt{x^2+y^2}},$$

whose graph is shown below in Figure 10.1.18

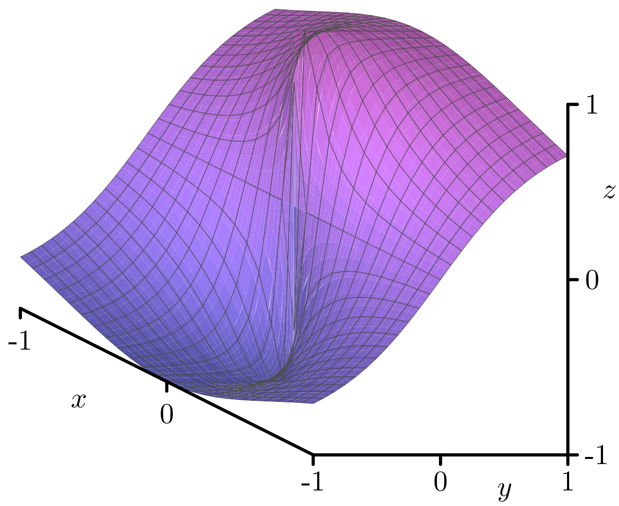


Figure 10.1.18 The graph of $f(x,y) = \frac{y}{\sqrt{x^2+y^2}}$.

a. Is f defined at the point $(0,0)$? What, if anything, does this say about whether f has a limit at the point $(0,0)$?

b. Values of f (to three decimal places) at several points close to $(0,0)$ are shown in Table 10.1.19.

Table 10.1.19 Values of a function f.

$x\backslash y$	-1.000	-0.100	0.000	0.100	1.000
-1.000	-0.707	—	0.000	—	0.707
-0.100	—	-0.707	0.000	0.707	—
0.000	-1.000	-1.000	—	1.000	1.000
0.100	—	-0.707	0.000	0.707	—
1.000	-0.707	—	0.000	—	0.707

Based on these calculations, state whether f has a limit at $(0,0)$ and give an argument supporting your statement. (Hint: The blank spaces in the table are there to help you see the patterns.)

c. Now we formalize the conjecture from the previous part by considering what happens if we restrict our attention to different paths. First, we look at f for points in the domain along the x-axis; that is, we consider what happens when $y = 0$. What is the behavior of $f(x,0)$ as $x \to 0$? If we approach $(0,0)$ by moving along the x-axis, what value do we find as the limit?

d. What is the behavior of f along the line $y = x$ when $x > 0$; that is, what is the value of $f(x,x)$ when $x > 0$? If we approach $(0,0)$ by moving along the line $y = x$ in the first quadrant (thus considering $f(x,x)$ as $x \to 0^+$), what value do we find as the limit?

e. In general, if $\lim_{(x,y)\to(0,0)} f(x,y) = L$, then $f(x,y)$ approaches L as (x,y) approaches $(0,0)$, regardless of the path we take in letting $(x,y) \to (0,0)$. Explain what the last two parts of this activity imply about the existence of $\lim_{(x,y)\to(0,0)} f(x,y)$.

f. Shown below in Figure 10.1.20 is a set of contour lines of the function f. What is the behavior of $f(x,y)$ as (x,y) approaches $(0,0)$ along any straight line? How does this observation reinforce your conclusion about the existence of $\lim_{(x,y)\to(0,0)} f(x, y)$ from the previous part of this activity? (Hint: Use the fact that a non-vertical line has equation $y = mx$ for some constant m.)

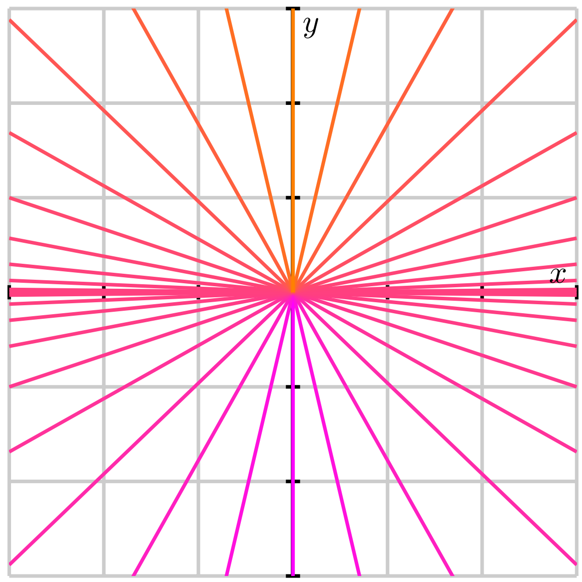


Figure 10.1.20 Contour lines of $f(x, y) = \frac{y}{\sqrt{x^2+y^2}}$.

Activity 10.1.3. Let's consider the function g defined by

$$g(x, y) = \frac{x^2 y}{x^4 + y^2}$$

and investigate the limit $\lim_{(x,y)\to(0,0)} g(x, y)$.

a. What is the behavior of g on the x-axis? That is, what is $g(x, 0)$ and what is the limit of g as (x, y) approaches $(0, 0)$ along the x-axis?

b. What is the behavior of g on the y-axis? That is, what is $g(0, y)$ and what is the limit of g as (x, y) approaches $(0, 0)$ along the y-axis?

c. What is the behavior of g on the line $y = mx$? That is, what is $g(x, mx)$ and what is the limit of g as (x, y) approaches $(0, 0)$ along the line $y = mx$?

d. Based on what you have seen so far, do you think $\lim_{(x,y)\to(0,0)} g(x, y)$ exists? If so, what do you think its value is?

e. Now consider the behavior of g on the parabola $y = x^2$? What is $g(x, x^2)$ and what is the limit of g as (x, y) approaches $(0, 0)$ along this parabola?

f. State whether the limit $\lim_{(x,y)\to(0,0)} g(x, y)$ exists or not and provide a justification of your statement.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

10.2 First-Order Partial Derivatives

Preview Activity 10.2.1. Suppose we take out an \$18,000 car loan at interest rate r and we agree to pay off the loan in t years. The monthly payment, in dollars, is

$$M(r,t) = \frac{1500r}{1 - \left(1 + \frac{r}{12}\right)^{-12t}}.$$

a. What is the monthly payment if the interest rate is 3% so that $r = 0.03$, and we pay the loan off in $t = 4$ years?

b. Suppose the interest rate is fixed at 3%. Express M as a function f of t alone using $r = 0.03$. That is, let $f(t) = M(0.03, t)$. Sketch the graph of f on the left of Figure 10.2.1. Explain the meaning of the function f.

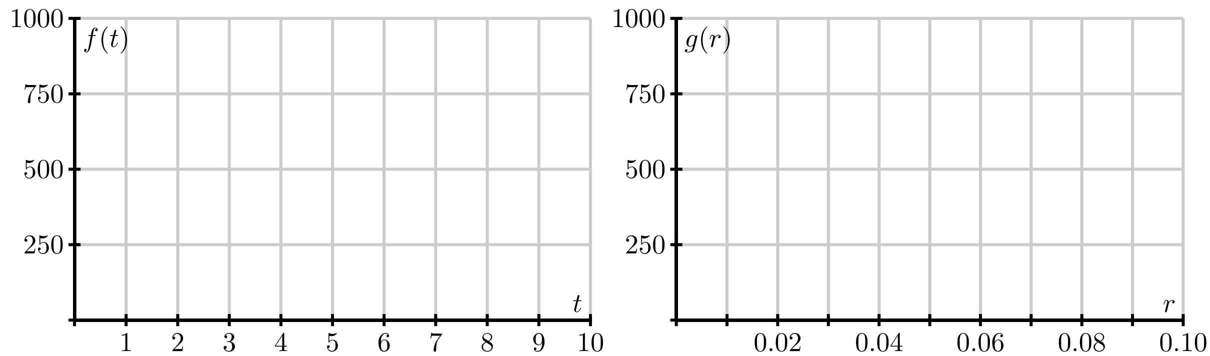


Figure 10.2.1 Left: Graphs $f(t) = M(0.03, t)$. Right: Graph $g(r) = M(r, 4)$.

c. Find the instantaneous rate of change $f'(4)$ and state the units on this quantity. What information does $f'(4)$ tell us about our car loan? What information does $f'(4)$ tell us about the graph you sketched in (b)?

d. Express M as a function of r alone, using a fixed time of $t = 4$. That is, let $g(r) = M(r, 4)$. Sketch the graph of g on the right of Figure 10.2.1. Explain the meaning of the function g.

e. Find the instantaneous rate of change $g'(0.03)$ and state the units on this quantity. What information does $g'(0.03)$ tell us about our car loan? What information does $g'(0.03)$ tell us about the graph you sketched in (d)?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.2.2. Consider the function f defined by

$$f(x,y) = \frac{xy^2}{x+1}$$

at the point $(1,2)$.

a. Write the trace $f(x,2)$ at the fixed value $y=2$. On the left side of Figure 10.2.5, draw the graph of the trace with $y=2$ around the point where $x=1$, indicating the scale and labels on the axes. Also, sketch the tangent line at the point $x=1$.

Figure 10.2.5 Traces of $f(x,y) = \frac{xy^2}{x+1}$.

b. Find the partial derivative $f_x(1,2)$ and relate its value to the sketch you just made.

c. Write the trace $f(1,y)$ at the fixed value $x=1$. On the right side of Figure 10.2.5, draw the graph of the trace with $x=1$ indicating the scale and labels on the axes. Also, sketch the tangent line at the point $y=2$.

d. Find the partial derivative $f_y(1,2)$ and relate its value to the sketch you just made.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.2.3.

a. If $f(x, y) = 3x^3 - 2x^2y^5$, find the partial derivatives f_x and f_y.

b. If $f(x, y) = \dfrac{xy^2}{x+1}$, find the partial derivatives f_x and f_y.

c. If $g(r, s) = rs\cos(r)$, find the partial derivatives g_r and g_s.

d. Assuming $f(w, x, y) = (6w+1)\cos(3x^2 + 4xy^3 + y)$, find the partial derivatives f_w, f_x, and f_y.

e. Find all possible first-order partial derivatives of $q(x, t, z) = \dfrac{x2^tz^3}{1+x^2}$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.2.4. The speed of sound C traveling through ocean water is a function of temperature, salinity and depth. It may be modeled by the function

$$C = 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3 + (1.34 - 0.01T)(S - 35) + 0.016D.$$

Here C is the speed of sound in meters/second, T is the temperature in degrees Celsius, S is the salinity in grams/liter of water, and D is the depth below the ocean surface in meters.

a. State the units in which each of the partial derivatives, C_T, C_S and C_D, are expressed and explain the physical meaning of each.

b. Find the partial derivatives C_T, C_S and C_D.

c. Evaluate each of the three partial derivatives at the point where $T = 10$, $S = 35$ and $D = 100$. What does the sign of each partial derivatives tell us about the behavior of the function C at the point $(10, 35, 100)$?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.2.5. The wind chill, as frequently reported, is a measure of how cold it feels outside when the wind is blowing. In Table 10.2.7, the wind chill w, measured in degrees Fahrenheit, is a function of the wind speed v, measured in miles per hour, and the ambient air temperature T, also measured in degrees Fahrenheit. We thus view w as being of the form $w = w(v, T)$.

Table 10.2.7 Wind chill as a function of wind speed and temperature.

$v\backslash T$	-30	-25	-20	-15	-10	-5	0	5	10	15	20
5	-46	-40	-34	-28	-22	-16	-11	-5	1	7	13
10	-53	-47	-41	-35	-28	-22	-16	-10	-4	3	9
15	-58	-51	-45	-39	-32	-26	-19	-13	-7	0	6
20	-61	-55	-48	-42	-35	-29	-22	-15	-9	-2	4
25	-64	-58	-51	-44	-37	-31	-24	-17	-11	-4	3
30	-67	-60	-53	-46	-39	-33	-26	-19	-12	-5	1
35	-69	-62	-55	-48	-41	-34	-27	-21	-14	-7	0
40	-71	-64	-57	-50	-43	-36	-29	-22	-15	-8	-1

a. Estimate the partial derivative $w_v(20, -10)$. What are the units on this quantity and what does it mean? (Recall that we can estimate a partial derivative of a single variable function f using the symmetric difference quotient $\frac{f(x+h)-f(x-h)}{2h}$ for small values of h. A partial derivative is a derivative of an appropriate trace.)

b. Estimate the partial derivative $w_T(20, -10)$. What are the units on this quantity and what does it mean?

c. Use your results to estimate the wind chill $w(18, -10)$. (Recall from single variable calculus that for a function f of x, $f(x + h) \approx f(x) + hf'(x)$.)

d. Use your results to estimate the wind chill $w(20, -12)$.

e. Consider how you might combine your previous results to estimate the wind chill $w(18, -12)$. Explain your process.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.2.6. Shown below in Figure 10.2.8 is a contour plot of a function f. The values of the function on a few of the contours are indicated to the left of the figure.

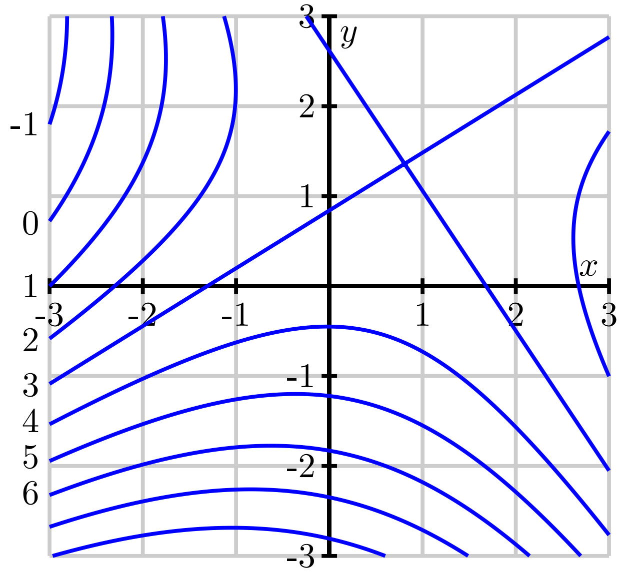


Figure 10.2.8 A contour plot of f.

a. Estimate the partial derivative $f_x(-2, -1)$. (Hint: How can you find values of f that are of the form $f(-2 + h)$ and $f(-2 - h)$ so that you can use a symmetric difference quotient?)

b. Estimate the partial derivative $f_y(-2, -1)$.

c. Estimate the partial derivatives $f_x(-1, 2)$ and $f_y(-1, 2)$.

d. Locate, if possible, one point (x, y) where $f_x(x, y) = 0$.

e. Locate, if possible, one point (x, y) where $f_x(x, y) < 0$.

f. Locate, if possible, one point (x, y) where $f_y(x, y) > 0$.

g. Suppose you have a different function g, and you know that $g(2, 2) = 4$, $g_x(2, 2) > 0$, and $g_y(2, 2) > 0$. Using this information, sketch a possibility for the contour $g(x, y) = 4$ passing through $(2, 2)$ on the left side of Figure 10.2.9. Then include possible contours $g(x, y) = 3$ and $g(x, y) = 5$.

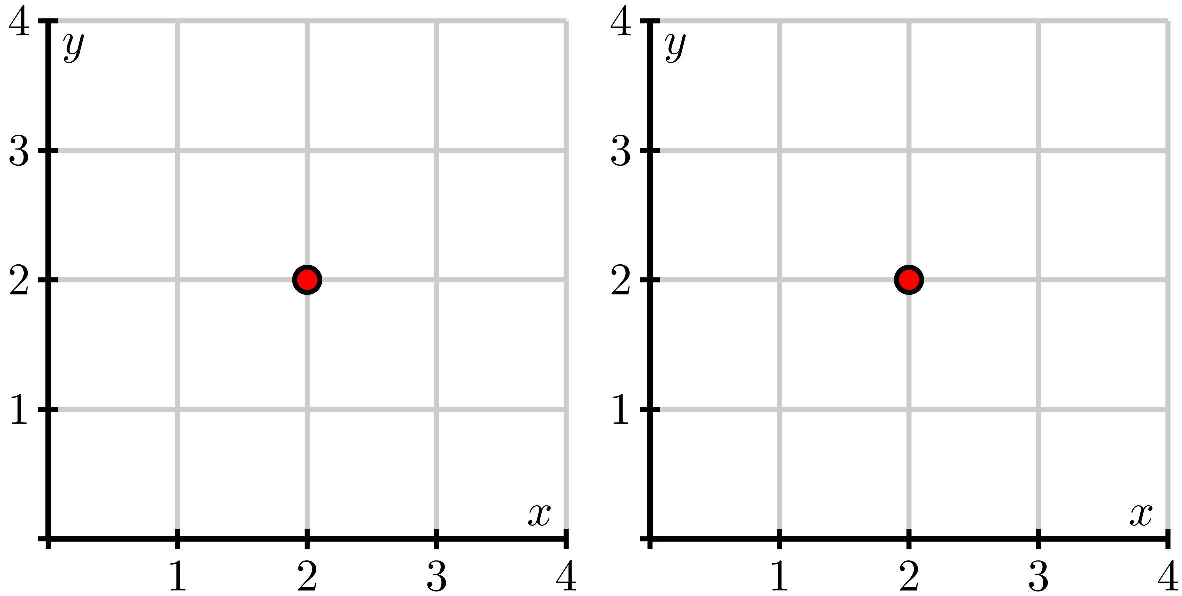


Figure 10.2.9 Plots for contours of g and h.

h. Suppose you have yet another function h, and you know that $h(2,2) = 4$, $h_x(2,2) < 0$, and $h_y(2,2) > 0$. Using this information, sketch a possible contour $h(x,y) = 4$ passing through $(2,2)$ on the right side of Figure 10.2.9. Then include possible contours $h(x,y) = 3$ and $h(x,y) = 5$.

10.3 Second-Order Partial Derivatives

Preview Activity 10.3.1. Once again, let's consider the function f defined by $f(x,y) = \frac{x^2 \sin(2y)}{32}$ that measures a projectile's range as a function of its initial speed x and launch angle y. The graph of this function, including traces with $x = 150$ and $y = 0.6$, is shown in Figure 10.3.1.

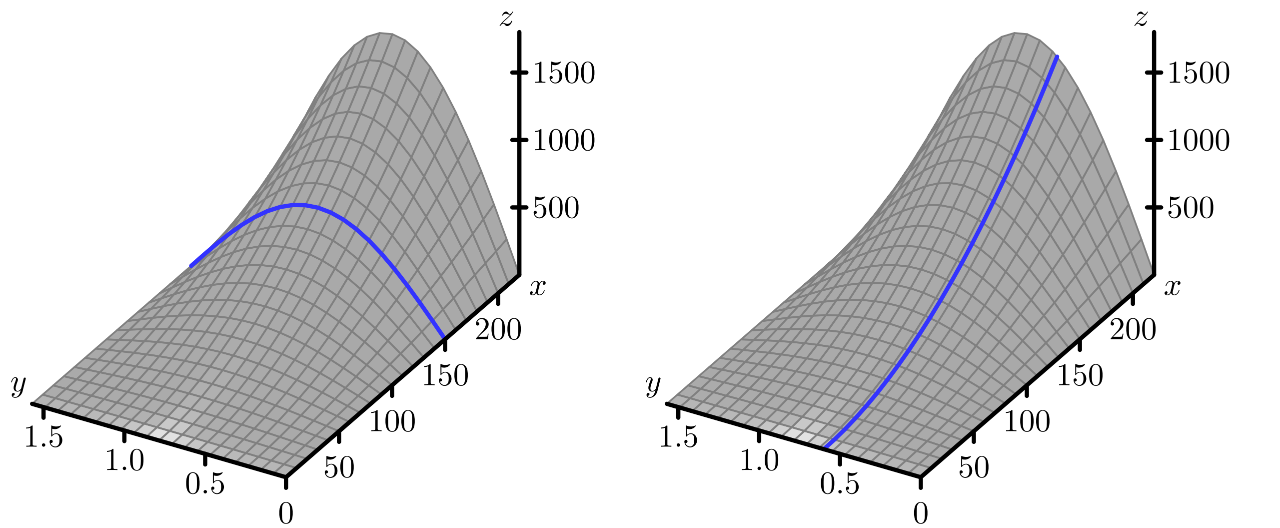


Figure 10.3.1 The distance function with traces $x = 150$ and $y = 0.6$.

a. Compute the partial derivative f_x. Notice that f_x itself is a new function of x and y, so we may now compute the partial derivatives of f_x. Find the partial derivative $f_{xx} = (f_x)_x$ and show that $f_{xx}(150, 0.6) \approx 0.058$.

b. Figure 10.3.2 shows the trace of f with $y = 0.6$ with three tangent lines included. Explain how your result from part (a) of this preview activity is reflected in this figure.

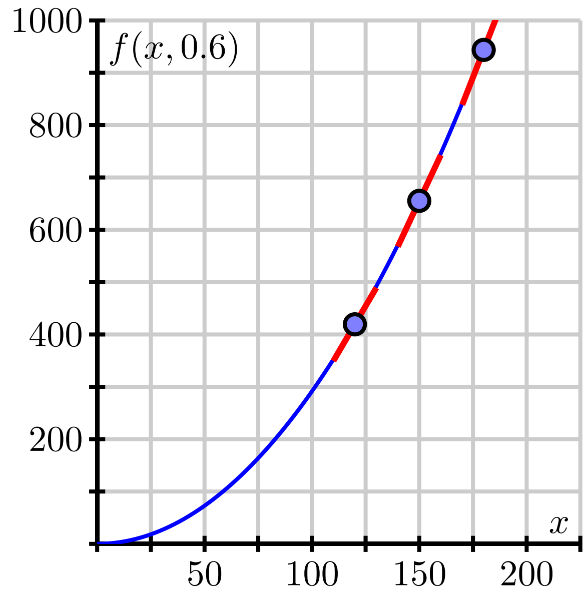


Figure 10.3.2 The trace with $y = 0.6$.

c. Determine the partial derivative f_y, and then find the partial derivative $f_{yy} = (f_y)_y$. Evaluate $f_{yy}(150, 0.6)$.

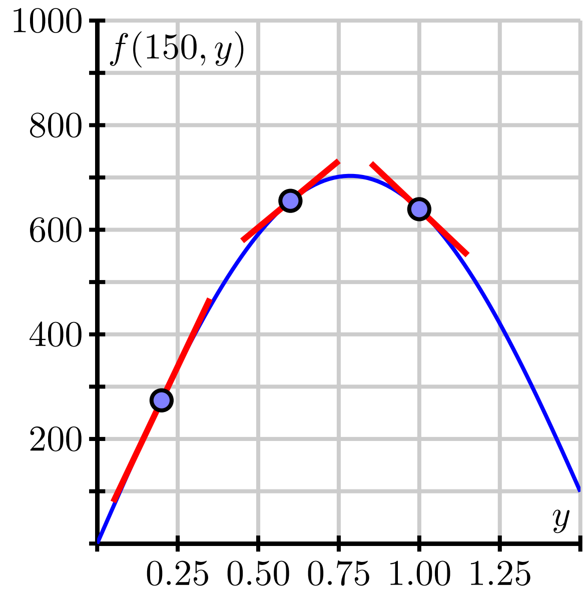


Figure 10.3.3 More traces of the range function.

d. Figure 10.3.3 shows the trace $f(150, y)$ and includes three tangent lines. Explain how the value of $f_{yy}(150, 0.6)$ is reflected in this figure.

e. Because f_x and f_y are each functions of both x and y, they each have two partial derivatives. Not only can we compute $f_{xx} = (f_x)_x$, but also $f_{xy} = (f_x)_y$; likewise, in addition to $f_{yy} = (f_y)_y$, but also $f_{yx} = (f_y)_x$. For the range function $f(x, y) = \frac{x^2 \sin(2y)}{32}$, use your earlier computations of f_x and f_y to now determine f_{xy} and f_{yx}. Write one sentence to explain how you calculated these "mixed" partial derivatives.

Activity 10.3.2. Find all second order partial derivatives of the following functions. For each partial derivative you calculate, state explicitly which variable is being held constant.

a. $f(x, y) = x^2 y^3$

b. $f(x, y) = y\cos(x)$

c. $g(s, t) = st^3 + s^4$

d. How many second order partial derivatives does the function h defined by $h(x, y, z) = 9x^9 z - xyz^9 + 9$ have? Find h_{xz} and h_{zx} (you do not need to find the other second order partial derivatives).

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.3.3. We continue to consider the function f defined by $f(x, y) = \sin(x)e^{-y}$.

a. In Figure 10.3.5, we see the trace of $f(x, y) = \sin(x)e^{-y}$ that has x held constant with $x = 1.75$. We also see three different lines that are tangent to the trace of f in the y direction at values of y that are increasing from left to right in the figure. Write a couple of sentences that describe whether the slope of the tangent lines to this curve increase or decrease as y increases, and, after computing $f_{yy}(x, y)$, explain how this observation is related to the value of $f_{yy}(1.75, y)$. Be sure to address the notion of concavity in your response.(You need to be careful about the directions in which x and y are increasing.)

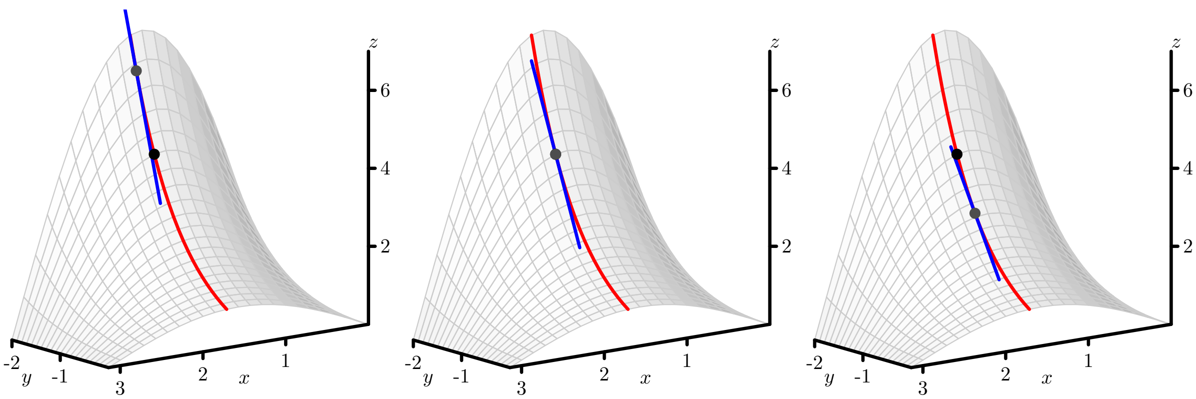


Figure 10.3.5 The tangent lines to a trace with increasing y.

b. In Figure 10.3.6, we start to think about the mixed partial derivative, f_{xy}. Here, we first hold y constant to generate the first-order partial derivative f_x, and then we hold x constant to compute f_{xy}. This leads to first thinking about a trace with x being constant, followed by slopes of tangent lines in the x-direction that slide along the original trace. You might think of sliding your pencil down the trace with x constant in a way that its slope indicates $(f_x)_y$ in order to further animate the three snapshots shown in the figure.

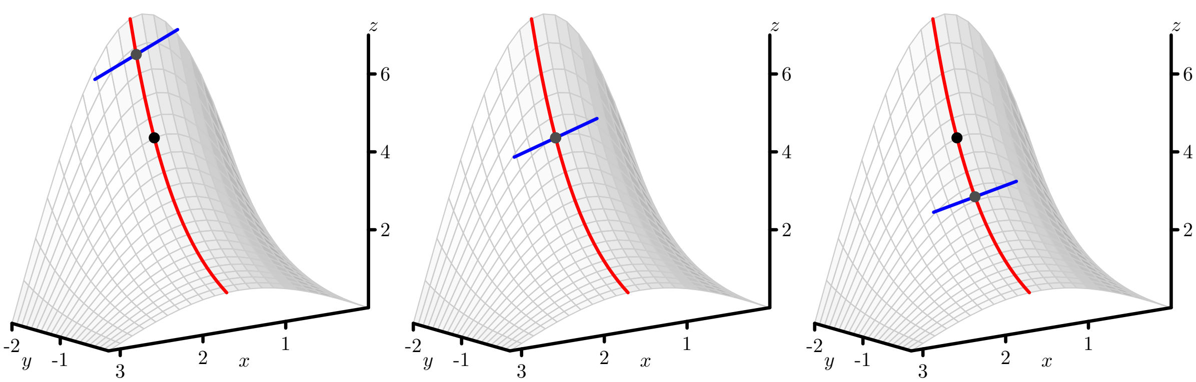


Figure 10.3.6 The trace of $z = f(x, y) = \sin(x)e^{-y}$ with $x = 1.75$, along with tangent lines in the y-direction at three different points.

Based on Figure 10.3.6, is $f_{xy}(1.75, -1.5)$ positive or negative? Why?

c. Determine the formula for $f_{xy}(x, y)$, and hence evaluate $f_{xy}(1.75, -1.5)$. How does this value compare with your observations in (b)?

d. We know that $f_{xx}(1.75, -1.5)$ measures the concavity of the $y = -1.5$ trace, and that $f_{yy}(1.75, -1.5)$ measures the concavity of the $x = 1.75$ trace. What do you think the quantity $f_{xy}(1.75, -1.5)$ measures?

e. On Figure 10.3.6, sketch the trace with $y = -1.5$, and sketch three tangent lines whose slopes correspond to the value of $f_{yx}(x, -1.5)$ for three different values of x, the middle of which is $x = -1.5$. Is $f_{yx}(1.75, -1.5)$ positive or negative? Why? What does $f_{yx}(1.75, -1.5)$ measure?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.3.4. As we saw in Activity 10.2.5, the wind chill $w(v, T)$, in degrees Fahrenheit, is a function of the wind speed, in miles per hour, and the air temperature, in degrees Fahrenheit. Some values of the wind chill are recorded in Table 10.3.7.

Table 10.3.7 Wind chill as a function of wind speed and temperature.

$v\backslash T$	-30	-25	-20	-15	-10	-5	0	5	10	15	20
5	-46	-40	-34	-28	-22	-16	-11	-5	1	7	13
10	-53	-47	-41	-35	-28	-22	-16	-10	-4	3	9
15	-58	-51	-45	-39	-32	-26	-19	-13	-7	0	6
20	-61	-55	-48	-42	-35	-29	-22	-15	-9	-2	4
25	-64	-58	-51	-44	-37	-31	-24	-17	-11	-4	3
30	-67	-60	-53	-46	-39	-33	-26	-19	-12	-5	1
35	-69	-62	-55	-48	-41	-34	-27	-21	-14	-7	0
40	-71	-64	-57	-50	-43	-36	-29	-22	-15	-8	-1

a. Estimate the partial derivatives $w_T(20, -15)$, $w_T(20, -10)$, and $w_T(20, -5)$. Use these results to estimate the second-order partial $w_{TT}(20, -10)$.

b. In a similar way, estimate the second-order partial $w_{vv}(20, -10)$.

c. Estimate the partial derivatives $w_T(20, -10)$, $w_T(25, -10)$, and $w_T(15, -10)$, and use your results to estimate the partial $w_{Tv}(20, -10)$.

d. In a similar way, estimate the partial derivative $w_{vT}(20, -10)$.

e. Write several sentences that explain what the values $w_{TT}(20, -10)$, $w_{vv}(20, -10)$, and $w_{Tv}(20, -10)$ indicate regarding the behavior of $w(v, T)$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

10.4 Linearization: Tangent Planes and Differentials

Preview Activity 10.4.1. Let $f(x,y) = 6 - \frac{x^2}{2} - y^2$, and let $(x_0, y_0) = (1,1)$.

a. Evaluate $f(x,y) = 6 - \frac{x^2}{2} - y^2$ and its partial derivatives at (x_0, y_0); that is, find $f(1,1)$, $f_x(1,1)$, and $f_y(1,1)$.

b. We know one point on the tangent plane; namely, the z-value of the tangent plane agrees with the z-value on the graph of $f(x,y) = 6 - \frac{x^2}{2} - y^2$ at the point (x_0, y_0). In other words, both the tangent plane and the graph of the function f contain the point (x_0, y_0, z_0). Use this observation to determine z_0 in the expression $z = z_0 + a(x - x_0) + b(y - y_0)$.

c. Sketch the traces of $f(x,y) = 6 - \frac{x^2}{2} - y^2$ for $y = y_0 = 1$ and $x = x_0 = 1$ below in Figure 10.4.4.

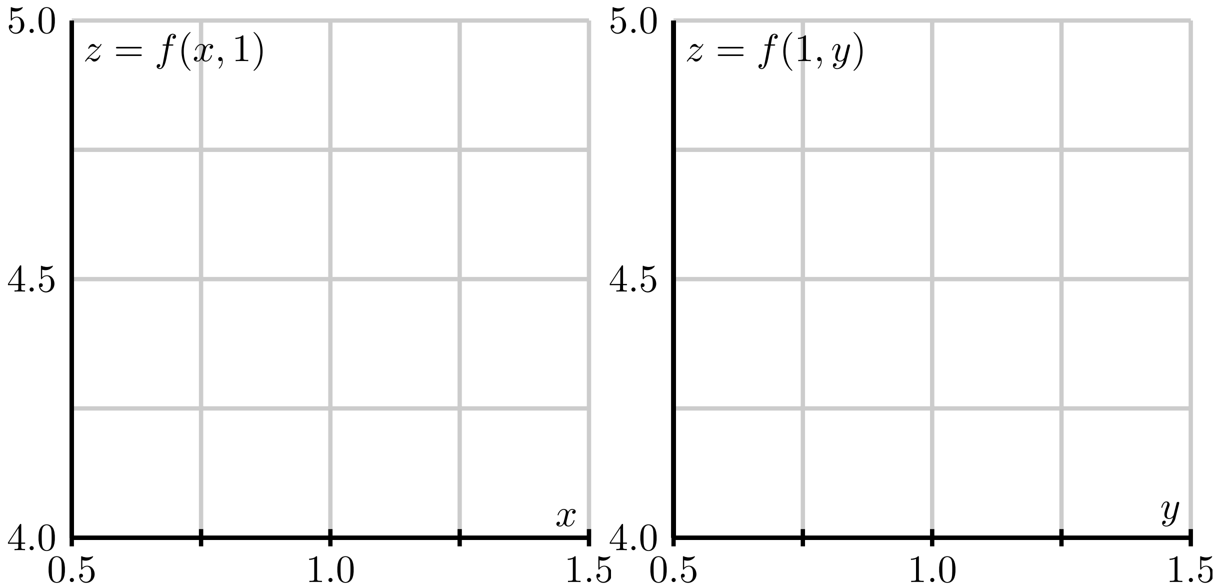


Figure 10.4.4 The traces of $f(x,y)$ with $y = y_0 = 1$ and $x = x_0 = 1$.

d. Determine the equation of the tangent line of the trace that you sketched in the previous part with $y = 1$ (in the x direction) at the point $x_0 = 1$.

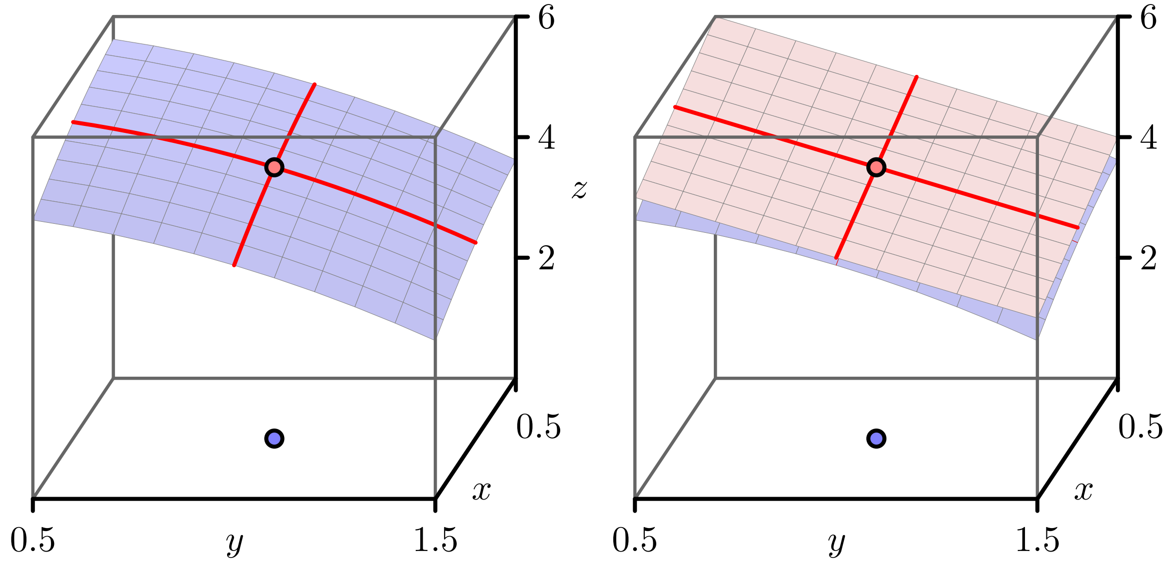


Figure 10.4.5 The traces of $f(x, y)$ and the tangent plane.

e. Figure 10.4.5 shows the traces of the function and the traces of the tangent plane. Explain how the tangent line of the trace of f, whose equation you found in the last part of this activity, is related to the tangent plane. How does this observation help you determine the constant a in the equation for the tangent plane $z = z_0 + a(x - x_0) + b(y - y_0)$? (Hint: How do you think $f_x(x_0, y_0)$ should be related to $z_x(x_0, y_0)$?)

f. In a similar way to what you did in (d), determine the equation of the tangent line of the trace with $x = 1$ at the point $y_0 = 1$. Explain how this tangent line is related to the tangent plane, and use this observation to determine the constant b in the equation for the tangent plane $z = z_0 + a(x - x_0) + b(y - y_0)$. (Hint: How do you think $f_y(x_0, y_0)$ should be related to $z_y(x_0, y_0)$?)

g. Finally, write the equation $z = z_0 + a(x - x_0) + b(y - y_0)$ of the tangent plane to the graph of $f(x, y) = 6 - x^2/2 - y^2$ at the point $(x_0, y_0) = (1, 1)$.

Activity 10.4.2.

a. Find the equation of the tangent plane to $f(x, y) = 2 + 4x - 3y$ at the point $(1, 2)$. Simplify as much as possible. Does the result surprise you? Explain.

b. Find the equation of the tangent plane to $f(x, y) = x^2y$ at the point $(1, 2)$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.4.3. In what follows, we find the linearization of several different functions that are given in algebraic, tabular, or graphical form.

a. Find the linearization $L(x, y)$ for the function g defined by

$$g(x, y) = \frac{x}{x^2 + y^2}$$

at the point $(1, 2)$. Then use the linearization to estimate the value of $g(0.8, 2.3)$.

b. Table 10.4.8 provides a collection of values of the wind chill $w(v, T)$, in degrees Fahrenheit, as a function of wind speed, in miles per hour, and temperature, also in degrees Fahrenheit.

Table 10.4.8 Wind chill as a function of wind speed and temperature.

$v\backslash T$	-30	-25	-20	-15	-10	-5	0	5	10	15	20
5	-46	-40	-34	-28	-22	-16	-11	-5	1	7	13
10	-53	-47	-41	-35	-28	-22	-16	-10	-4	3	9
15	-58	-51	-45	-39	-32	-26	-19	-13	-7	0	6
20	-61	-55	-48	-42	-35	-29	-22	-15	-9	-2	4
25	-64	-58	-51	-44	-37	-31	-24	-17	-11	-4	3
30	-67	-60	-53	-46	-39	-33	-26	-19	-12	-5	1
35	-69	-62	-55	-48	-41	-34	-27	-21	-14	-7	0
40	-71	-64	-57	-50	-43	-36	-29	-22	-15	-8	-1

Use the data to first estimate the appropriate partial derivatives, and then find the linearization $L(v, T)$ at the point $(20, -10)$. Finally, use the linearization to estimate $w(10, -10)$, $w(20, -12)$, and $w(18, -12)$. Compare your results to what you obtained in Activity 10.2.5

c. Figure 10.4.9 gives a contour plot of a continuously differentiable function f.

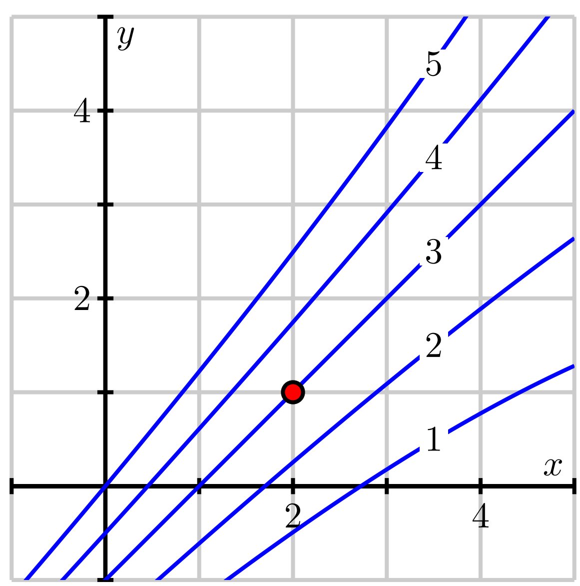


Figure 10.4.9 A contour plot of $f(x, y)$.

After estimating appropriate partial derivatives, determine the linearization $L(x, y)$ at the point $(2, 1)$, and use it to estimate $f(2.2, 1)$, $f(2, 0.8)$, and $f(2.2, 0.8)$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.4.4. The questions in this activity explore the differential in several different contexts.

a. Suppose that the elevation of a landscape is given by the function h, where we additionally know that $h(3,1) = 4.35$, $h_x(3,1) = 0.27$, and $h_y(3,1) = -0.19$. Assume that x and y are measured in miles in the easterly and northerly directions, respectively, from some base point $(0,0)$. Your GPS device says that you are currently at the point $(3,1)$. However, you know that the coordinates are only accurate to within 0.2 units; that is, $dx = \Delta x = 0.2$ and $dy = \Delta y = 0.2$. Estimate the uncertainty in your elevation using differentials.

b. The pressure, volume, and temperature of an ideal gas are related by the equation

$$P = P(T,V) = 8.31T/V,$$

where P is measured in kilopascals, V in liters, and T in kelvin. Find the pressure when the volume is 12 liters and the temperature is 310 K. Use differentials to estimate the change in the pressure when the volume increases to 12.3 liters and the temperature decreases to 305 K.

c. Refer to Table 10.4.8, the table of values of the wind chill $w(v,T)$, in degrees Fahrenheit, as a function of temperature, also in degrees Fahrenheit, and wind speed, in miles per hour. Suppose your anemometer says the wind is blowing at 25 miles per hour and your thermometer shows a reading of $-15°$ degrees. However, you know your thermometer is only accurate to within $2°$ degrees and your anemometer is only accurate to within 3 miles per hour. What is the wind chill based on your measurements? Estimate the uncertainty in your measurement of the wind chill.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

10.5 The Chain Rule

Preview Activity 10.5.1. Suppose you are driving around in the xy-plane in such a way that your position $\mathbf{r}(t)$ at time t is given by function

$$\mathbf{r}(t) = \langle x(t), y(t)\rangle = \langle 2 - t^2, t^3 + 1\rangle.$$

The path taken is shown on the left of Figure 10.5.1.

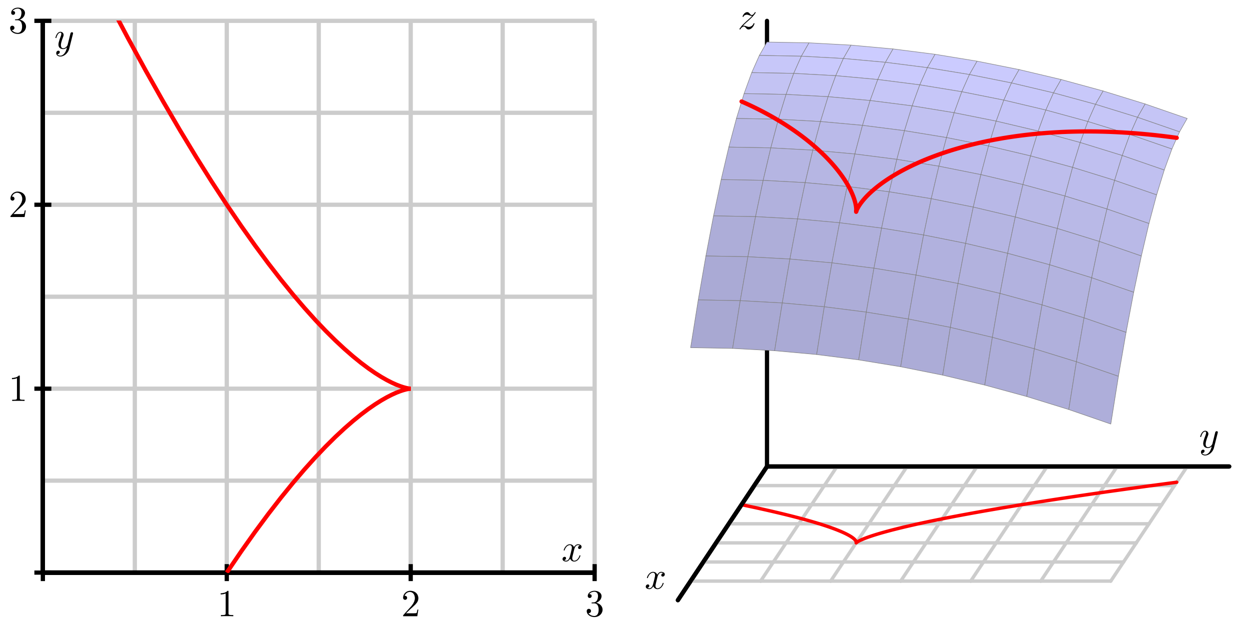


Figure 10.5.1 Left: Your position in the plane. Right: The corresponding temperature.

Suppose, furthermore, that the temperature at a point in the plane is given by

$$T(x, y) = 10 - \frac{1}{2}x^2 - \frac{1}{5}y^2,$$

and note that the surface generated by T is shown on the right of Figure 10.5.1. Therefore, as time passes, your position $(x(t), y(t))$ changes, and, as your position changes, the temperature $T(x, y)$ also changes.

a. The position function $\mathbf{r}$ provides a parameterization $x = x(t)$ and $y = y(t)$ of the position at time t. By substituting $x(t)$ for x and $y(t)$ for y in the formula for T, we can write $T = T(x(t), y(t))$ as a function of t. Make these substitutions to write T as a function of t and then use the Chain Rule from single variable calculus to find $\frac{dT}{dt}$. (Do not do any algebra to simplify the derivative, either before taking the derivative, nor after.)

b. Now we want to understand how the result from part (a) can be obtained from T as a multivariable function. Recall from the previous section that small changes in x and y produce a change in T that is approximated by

$$\Delta T \approx T_x \Delta x + T_y \Delta y.$$

The Chain Rule tells us about the instantaneous rate of change of T, and this can be found as

$$\lim_{\Delta t \to 0} \frac{\Delta T}{\Delta t} = \lim_{\Delta t \to 0} \frac{T_x \Delta x + T_y \Delta y}{\Delta t}. \tag{10.5.1}$$

Use equation (10.5.1) to explain why the instantaneous rate of change of T that results from a change in t is

$$\frac{dT}{dt} = \frac{\partial T}{\partial x}\frac{dx}{dt} + \frac{\partial T}{\partial y}\frac{dy}{dt}. \tag{10.5.2}$$

c. Using the original formulas for T, x, and y in the problem statement, calculate all of the derivatives in Equation (10.5.2) (with T_x and T_y in terms of x and y, and x' and y' in terms of t), and hence write the right-hand side of Equation (10.5.2) in terms of x, y, and t.

d. Compare the results of parts (a) and (c). Write a couple of sentences that identify specifically how each term in (c) relates to a corresponding terms in (a). This connection between parts (a) and (c) provides a multivariable version of the Chain Rule.

Activity 10.5.2. In the following questions, we apply the Chain Rule in several different contexts.

a. Suppose that we have a function z defined by $z(x, y) = x^2 + xy^3$. In addition, suppose that x and y are restricted to points that move around the plane by following a circle of radius 2 centered at the origin that is parameterized by

$$x(t) = 2\cos(t), \text{ and } y(t) = 2\sin(t).$$

   i. Use the Chain Rule to find the resulting instantaneous rate of change $\frac{dz}{dt}$.

   ii. Substitute $x(t)$ for x and $y(t)$ for y in the rule for z to write z in terms of t and calculate $\frac{dz}{dt}$ directly. Compare to the result of part (i.).

b. Suppose that the temperature on a metal plate is given by the function T with

$$T(x, y) = 100 - (x^2 + 4y^2),$$

where the temperature is measured in degrees Fahrenheit and x and y are each measured in feet.

   i. Find T_x and T_y. What are the units on these partial derivatives?

   ii. Suppose an ant is walking along the x-axis at the rate of 2 feet per minute toward the origin. When the ant is at the point $(2, 0)$, what is the instantaneous rate of change in the temperature dT/dt that the ant experiences. Include units on your response.

   iii. Suppose instead that the ant walks along an ellipse with $x = 6\cos(t)$ and $y = 3\sin(t)$, where t is measured in minutes. Find $\frac{dT}{dt}$ at $t = \pi/6$, $t = \pi/4$, and $t = \pi/3$. What does this seem to tell you about the path along which the ant is walking?

c. Suppose that you are walking along a surface whose elevation is given by a function f. Furthermore, suppose that if you consider how your location corresponds to points in the xy-plane, you know that when you pass the point $(2, 1)$, your velocity vector is $\mathbf{v} = \langle -1, 2 \rangle$. If some contours of f are as shown in Figure 10.5.2, estimate the rate of change df/dt when you pass through $(2, 1)$.

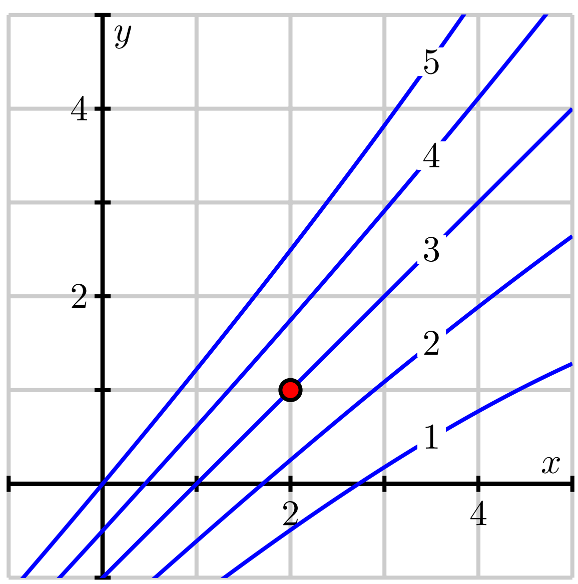


Figure 10.5.2 Some contours of f.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.5.3.

a. Figure 10.5.4 shows the tree diagram we construct when (a) z depends on w, x, and y, (b) w, x, and y each depend on u and v, and (c) u and v depend on t.

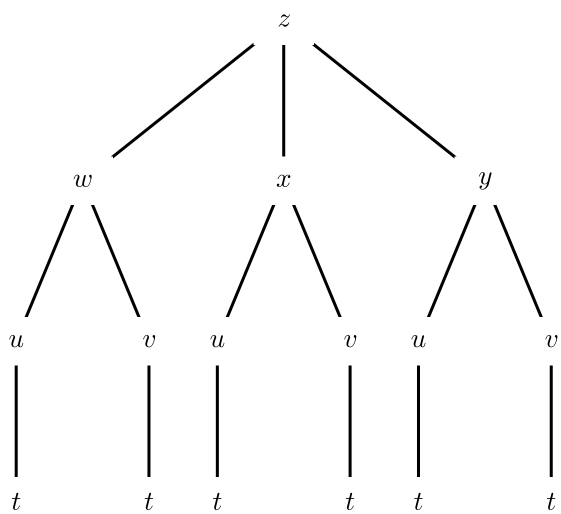


Figure 10.5.4 Three levels of dependencies

   i. Label the edges with the appropriate derivatives.

   ii. Use the Chain Rule to write $\frac{dz}{dt}$.

b. Suppose that $z = x^2 - 2xy^2$ and that

$$\begin{aligned} x &= r\cos(\theta) \\ y &= r\sin(\theta). \end{aligned}$$

   i. Construct a tree diagram representing the dependencies of z on x and y and x and y on r and θ.

   ii. Use the tree diagram to find $\frac{\partial z}{\partial r}$.

   iii. Now suppose that $r = 3$ and $\theta = \pi/6$. Find the values of x and y that correspond to these given values of r and θ, and then use the Chain Rule to find the value of the partial derivative $\frac{\partial z}{\partial \theta}\big|_{(3,\frac{\pi}{6})}$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

10.6 Directional Derivatives and the Gradient

Preview Activity 10.6.1. Let's consider the function f defined by

$$f(x,y) = 30 - x^2 - \frac{1}{2}y^2,$$

and suppose that f measures the temperature, in degrees Celsius, at a given point in the plane, where x and y are measured in feet. Assume that the positive x-axis points due east, while the positive y-axis points due north. A contour plot of f is shown in Figure 10.6.1

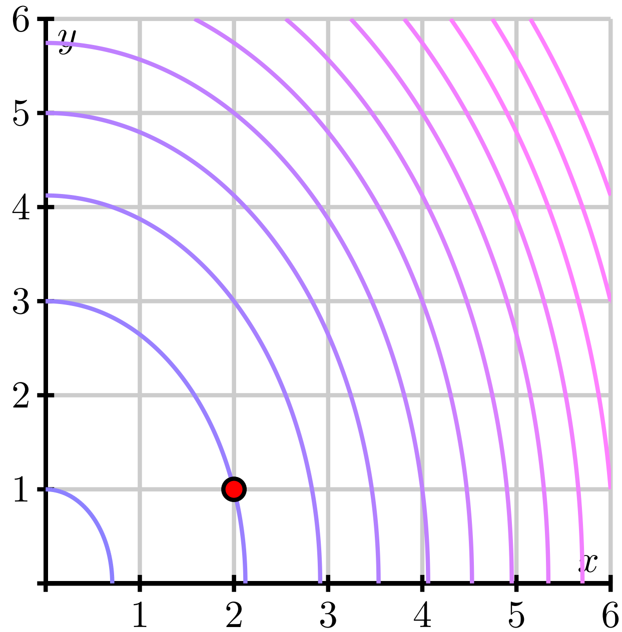


Figure 10.6.1 A contour plot of $f(x,y) = 30 - x^2 - \frac{1}{2}y^2$.

a. Suppose that a person is walking due east, and thus parallel to the x-axis. At what instantaneous rate is the temperature changing with respect to x at the moment the walker passes the point $(2,1)$? What are the units on this rate of change?

b. Next, determine the instantaneous rate of change of temperature with respect to distance at the point $(2,1)$ if the person is instead walking due north. Again, include units on your result.

c. Now, rather than walking due east or due north, let's suppose that the person is walking with velocity given by the vector $\mathbf{v} = \langle 3,4\rangle$, where time is measured in seconds. Note that the person's speed is thus $|\mathbf{v}| = 5$ feet per second. Find parametric equations for the person's path; that is, parameterize the line through $(2,1)$ using the direction vector $\mathbf{v} = \langle 3,4\rangle$. Let $x(t)$ denote the x-coordinate of the line, and $y(t)$ its y-coordinate. Make sure your parameterization places the walker at the point $(2,1)$ when $t = 0$.

d. With the parameterization in (c), we can now view the temperature f as not only a function of x and y, but also of time, t. Hence, use the chain rule to determine the value of $\frac{df}{dt}\Big|_{t=0}$. What are the units on your answer? What is the practical meaning of this result?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.6.2. Let $f(x,y) = 3xy - x^2y^3$.

a. Determine $f_x(x,y)$ and $f_y(x,y)$.

b. Use Equation (10.6.3) to determine $D_{\mathbf{i}}f(x,y)$ and $D_{\mathbf{j}}f(x,y)$. What familiar function is $D_{\mathbf{i}}f$? What familiar function is $D_{\mathbf{j}}f$? (Recall that $\mathbf{i}$ is the unit vector in the positive x-direction and $\mathbf{j}$ is the unit vector in the positive y-direction.)

c. Use Equation (10.6.3) to find the derivative of f in the direction of the vector $\mathbf{v} = \langle 2, 3\rangle$ at the point $(1,-1)$. Remember that a unit direction vector is needed.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.6.3. Let's consider the function f defined by $f(x, y) = x^2 - y^2$. Some contours for this function are shown in Figure 10.6.3.

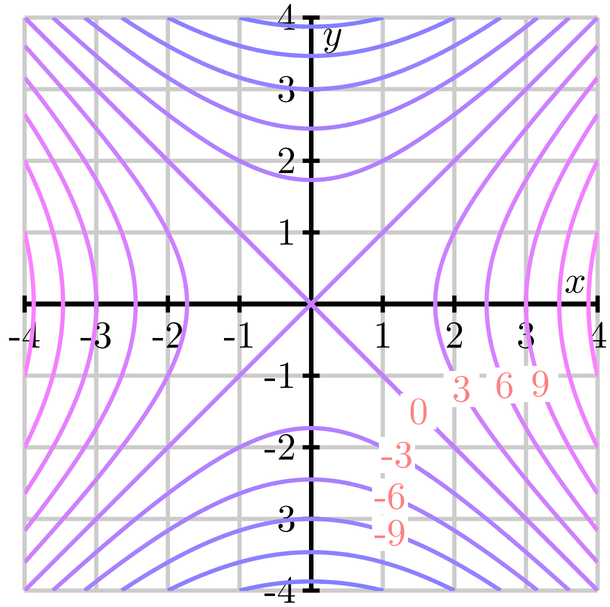


Figure 10.6.3 Contours of $f(x, y) = x^2 - y^2$.

a. Find the gradient $\nabla f(x, y)$.

b. For each of the following points (x_0, y_0), evaluate the gradient $\nabla f(x_0, y_0)$ and sketch the gradient vector with its tail at (x_0, y_0). Some of the vectors are too long to fit onto the plot, but we'd like to draw them to scale; to do so, scale each vector by a factor of $1/4$.

- $(x_0, y_0) = (2, 0)$
- $(x_0, y_0) = (0, 2)$
- $(x_0, y_0) = (2, 2)$
- $(x_0, y_0) = (2, 1)$
- $(x_0, y_0) = (-3, 2)$
- $(x_0, y_0) = (-2, -4)$
- $(x_0, y_0) = (0, 0)$

c. What do you notice about the relationship between the gradient at (x_0, y_0) and the contour passing through that point?

d. Does f increase or decrease in the direction of $\nabla f(x_0, y_0)$? Provide a justification for your response.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.6.4. In this activity we investigate how the gradient is related to the directions of greatest increase and decrease of a function. Let f be a differentiable function and $\mathbf{u}$ a unit vector.

a. Let θ be the angle between $\nabla f(x_0, y_0)$ and $\mathbf{u}$. Use the relationship between the dot product and the angle between two vectors to explain why

$$D_{\mathbf{u}} f(x_0, y_0) = |\langle f_x(x_0, y_0), f_y(x_0, y_0)\rangle| \cos(\theta). \tag{10.6.6}$$

b. At the point (x_0, y_0), the only quantity in Equation (10.6.6) that can change is θ (which determines the direction $\mathbf{u}$ of travel). Explain why $\theta = 0$ makes the quantity

$$|\langle f_x(x_0, y_0), f_y(x_0, y_0)\rangle| \cos(\theta)$$

as large as possible.

c. When $\theta = 0$, in what direction does the unit vector $\mathbf{u}$ point relative to $\nabla f(x_0, y_0)$? Why? What does this tell us about the direction of greatest increase of f at the point (x_0, y_0)?

d. In what direction, relative to $\nabla f(x_0, y_0)$, does f decrease most rapidly at the point (x_0, y_0)?

e. State the unit vectors $\mathbf{u}$ and $\mathbf{v}$ (in terms of $\nabla f(x_0, y_0)$) that provide the directions of greatest increase and decrease for the function f at the point (x_0, y_0). What important assumption must we make regarding $\nabla f(x_0, y_0)$ in order for these vectors to exist?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.6.5. Consider the function f defined by $f(x, y) = -x + 2xy - y$.

a. Find the gradient $\nabla f(1, 2)$ and sketch it on Figure 10.6.5.

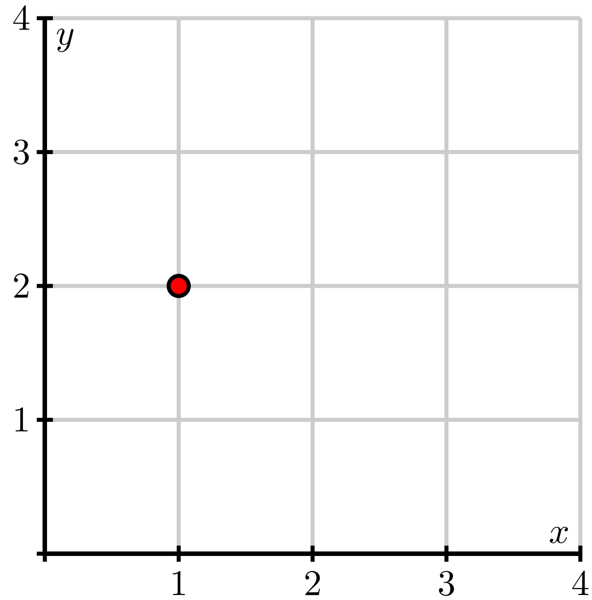


Figure 10.6.5 A plot for the gradient $\nabla f(1, 2)$.

b. Sketch the unit vector $\mathbf{z} = \left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle$ on Figure 10.6.5 with its tail at $(1, 2)$. Now find the directional derivative $D_{\mathbf{z}} f(1, 2)$.

c. What is the slope of the graph of f in the direction $\mathbf{z}$? What does the sign of the directional derivative tell you?

d. Consider the vector $\mathbf{v} = \langle 2, -1 \rangle$ and sketch $\mathbf{v}$ on Figure 10.6.5 with its tail at $(1, 2)$. Find a unit vector $\mathbf{w}$ pointing in the same direction of $\mathbf{v}$. Without computing $D_{\mathbf{w}} f(1, 2)$, what do you know about the sign of this directional derivative? Now verify your observation by computing $D_{\mathbf{w}} f(1, 2)$.

e. In which direction (that is, for what unit vector $\mathbf{u}$) is $D_{\mathbf{u}} f(1, 2)$ the greatest? What is the slope of the graph in this direction?

f. Corresponding, in which direction is $D_{\mathbf{u}} f(1, 2)$ least? What is the slope of the graph in this direction?

g. Sketch two unit vectors $\mathbf{u}$ for which $D_{\mathbf{u}} f(1, 2) = 0$ and then find component representations of these vectors.

h. Suppose you are standing at the point $(3, 3)$. In which direction should you move to cause f to increase as rapidly as possible? At what rate does f increase in this direction?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.6.6.

a. The temperature $T(x, y)$ has its maximum value at the fighter jet's location. State the fighter jet's location and explain how Figure 10.6.6 tells you this.

b. Determine ∇T at the fighter jet's location and give a justification for your response.

c. Suppose that a different function f has a local maximum value at (x_0, y_0). Sketch the behavior of some possible contours near this point. What is $\nabla f(x_0, y_0)$? (Hint: What is the direction of greatest increase in f at the local maximum?)

d. Suppose that a function g has a local minimum value at (x_0, y_0). Sketch the behavior of some possible contours near this point. What is $\nabla g(x_0, y_0)$?

e. If a function g has a local minimum at (x_0, y_0), what is the direction of greatest increase of g at (x_0, y_0)?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

10.7 Optimization

Preview Activity 10.7.1. Let $z = f(x, y)$ be a differentiable function, and suppose that at the point (x_0, y_0), f achieves a local maximum. That is, the value of $f(x_0, y_0)$ is greater than the value of $f(x, y)$ for all (x, y) nearby (x_0, y_0). You might find it helpful to sketch a rough picture of a possible function f that has this property.

a. If we consider the trace given by holding $y = y_0$ constant, then the single-variable function defined by $f(x, y_0)$ must have a local maximum at x_0. What does this say about the value of the partial derivative $f_x(x_0, y_0)$?

b. In the same way, the trace given by holding $x = x_0$ constant has a local maximum at $y = y_0$. What does this say about the value of the partial derivative $f_y(x_0, y_0)$?

c. What may we now conclude about the gradient $\nabla f(x_0, y_0)$ at the local maximum? How is this consistent with the statement "f increases most rapidly in the direction $\nabla f(x_0, y_0)$?"

d. How will the tangent plane to the surface $z = f(x, y)$ appear at the point $(x_0, y_0, f(x_0, y_0))$?

e. By first computing the partial derivatives, find any points at which $f(x, y) = 2x - x^2 - (y + 2)^2$ may have a local maximum.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.7.2. Find the critical points of each of the following functions. Then, using appropriate technology, plot the graphs of the surfaces near each critical point and compare the graph to your work.

a. $f(x, y) = 2 + x^2 + y^2$

b. $f(x, y) = 2 + x^2 - y^2$

c. $f(x, y) = 2x - x^2 - \frac{1}{4}y^2$

d. $f(x, y) = |x| + |y|$

e. $f(x, y) = 2xy - 4x + 2y - 3$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.7.3. Recall that the *Second Derivative Test* for single-variable functions states that if x_0 is a critical point of a function f so that $f'(x_0) = 0$ and if $f''(x_0)$ exists, then

- if $f''(x_0) < 0$, x_0 is a local maximum,
- if $f''(x_0) > 0$, x_0 is a local minimum, and
- if $f''(x_0) = 0$, this test yields no information.

Our goal in this activity is to understand a similar test for classifying extreme values of functions of two variables. Consider the following three functions:

$$f_1(x, y) = 4 - x^2 - y^2, \quad f_2(x, y) = x^2 + y^2, \quad f_3(x, y) = x^2 - y^2.$$

You can verify that each function has a critical point at the origin $(0, 0)$. You should check this.

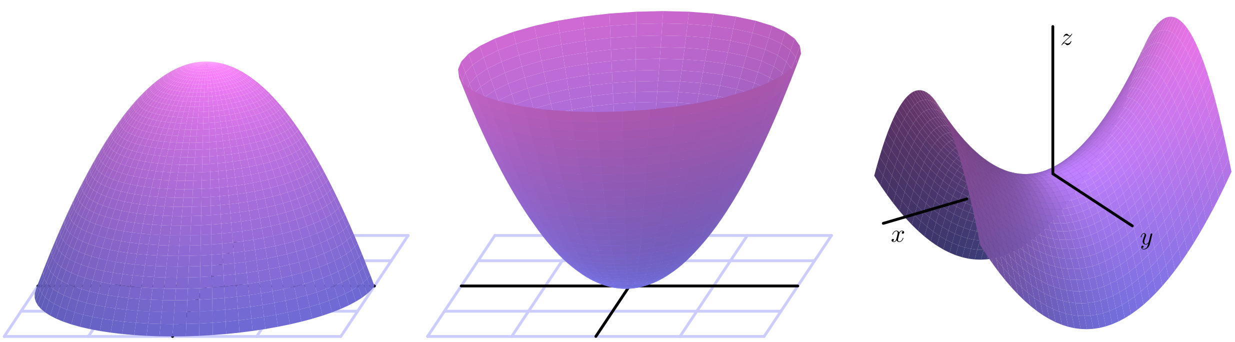


Figure 10.7.6 Three surfaces.

a. The graphs of these three functions are shown in Figure 10.7.6, with $z = 4 - x^2 - y^2$ at left, $z = x^2 + y^2$ in the middle, and $z = x^2 - y^2$ at right. Use the graphs to decide if a function has a local maximum, local minimum, saddle point, or none of the above at the origin.

b. There is no single second derivative of a function of two variables, so we consider a quantity that combines the second order partial derivatives. Let $D = f_{xx}f_{yy} - f_{xy}^2$. Calculate D at the origin for each of the functions f_1, f_2, and f_3. What difference do you notice between the values of D when a function has a maximum or minimum value at the origin versus when a function has a saddle point at the origin?

c. Now consider the cases where $D > 0$. It is in these cases that a function has a local maximum or minimum at a point. What is necessary in these cases is to find a condition that will distinguish between a maximum and a minimum. In the cases where $D > 0$ at the origin, evaluate $f_{xx}(0, 0)$. What value does $f_{xx}(0, 0)$ have when f has a local maximum value at the origin? When f has a local minimum value at the origin? Explain why. (Hint: This should look very similar to the Second Derivative Test for functions of a single variable.) What would happen if we considered the values of $f_{yy}(0, 0)$ instead?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.7.4. Find the critical points of the following functions and use the Second Derivative Test to classify the critical points.

a. $f(x, y) = 3x^3 + y^2 - 9x + 4y$

b. $f(x, y) = xy + \dfrac{2}{x} + \dfrac{4}{y}$

c. $f(x, y) = x^3 + y^3 - 3xy$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.7.5. While the quantity of a product demanded by consumers is often a function of the price of the product, the demand for a product may also depend on the price of other products. For instance, the demand for blue jeans at Old Navy may be affected not only by the price of the jeans themselves, but also by the price of khakis.

Suppose we have two goods whose respective prices are p_1 and p_2. The demand for these goods, q_1 and q_2, depend on the prices as

$$q_1 = 150 - 2p_1 - p_2 \tag{10.7.1}$$
$$q_2 = 200 - p_1 - 3p_2. \tag{10.7.2}$$

The seller would like to set the prices p_1 and p_2 in order to maximize revenue. We will assume that the seller meets the full demand for each product. Thus, if we let R be the revenue obtained by selling q_1 items of the first good at price p_1 per item and q_2 items of the second good at price p_2 per item, we have

$$R = p_1q_1 + p_2q_2.$$

We can then write the revenue as a function of just the two variables p_1 and p_2 by using Equations (10.7.1) and (10.7.2), giving us

$$\begin{aligned} R(p_1, p_2) &= p_1(150 - 2p_1 - p_2) + p_2(200 - p_1 - 3p_2) \\ &= 150p_1 + 200p_2 - 2p_1p_2 - 2p_1^2 - 3p_2^2. \end{aligned}$$

A graph of R as a function of p_1 and p_2 is shown in Figure 10.7.7.

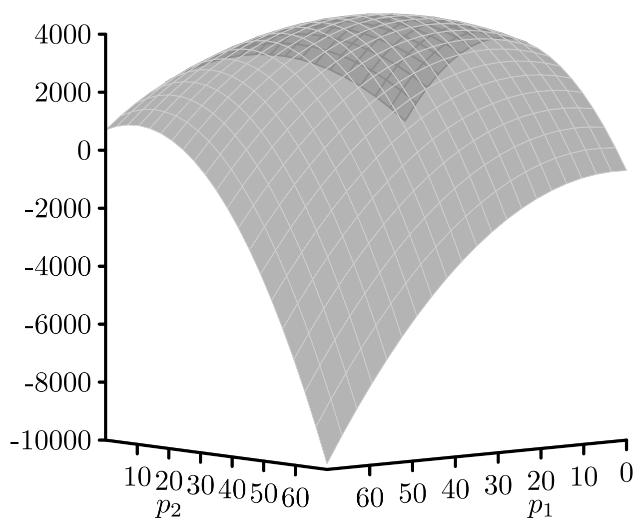


Figure 10.7.7 A revenue function.

a. Find all critical points of the revenue function, R. (Hint: You should obtain a system of two equations in two unknowns which can be solved by elimination or substitution.)

b. Apply the Second Derivative Test to determine the type of any critical point(s).

c. Where should the seller set the prices p_1 and p_2 to maximize the revenue?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.7.6. Let $f(x, y) = x^2 - 3y^2 - 4x + 6y$ with triangular domain R whose vertices are at $(0, 0)$, $(4, 0)$, and $(0, 4)$. The domain R and a graph of f on the domain appear in Figure 10.7.10.

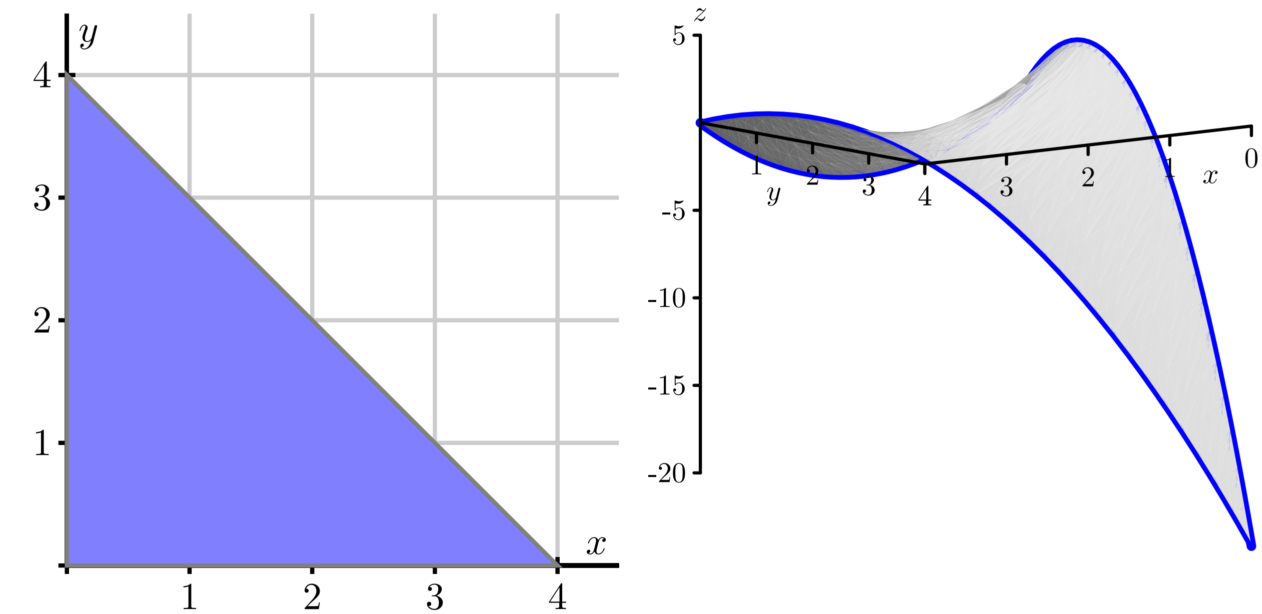


Figure 10.7.10 The domain of $f(x, y) = x^2 - 3y^2 - 4x + 6y$ and its graph.

a. Find all of the critical points of f in R.

b. Parameterize the horizontal leg of the triangular domain, and find the critical points of f on that leg. (Hint: You may need to consider endpoints.)

c. Parameterize the vertical leg of the triangular domain, and find the critical points of f on that leg. (Hint: You may need to consider endpoints.)

d. Parameterize the hypotenuse of the triangular domain, and find the critical points of f on the hypotenuse. (Hint: You may need to consider endpoints.)

e. Find the absolute maximum and absolute minimum values of f on R.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

10.8 Constrained Optimization: Lagrange Multipliers

Preview Activity 10.8.1. According to U.S. postal regulations, the girth plus the length of a parcel sent by mail may not exceed 108 inches, where by "girth" we mean the perimeter of the smallest end. Our goal is to find the largest possible volume of a rectangular parcel with a square end that can be sent by mail. (We solved this applied optimization problem in single variable *Active Calculus*, so it may look familiar. We take a different approach in this section, and this approach allows us to view most applied optimization problems from single variable calculus as constrained optimization problems, as well as provide us tools to solve a greater variety of optimization problems.) If we let x be the length of the side of one square end of the package and y the length of the package, then we want to maximize the volume $f(x, y) = x^2 y$ of the box subject to the constraint that the girth $(4x)$ plus the length (y) is as large as possible, or $4x + y = 108$. The equation $4x + y = 108$ is thus an external constraint on the variables.

a. The constraint equation involves the function g that is given by

$$g(x, y) = 4x + y.$$

Explain why the constraint is a contour of g, and is therefore a two-dimensional curve.

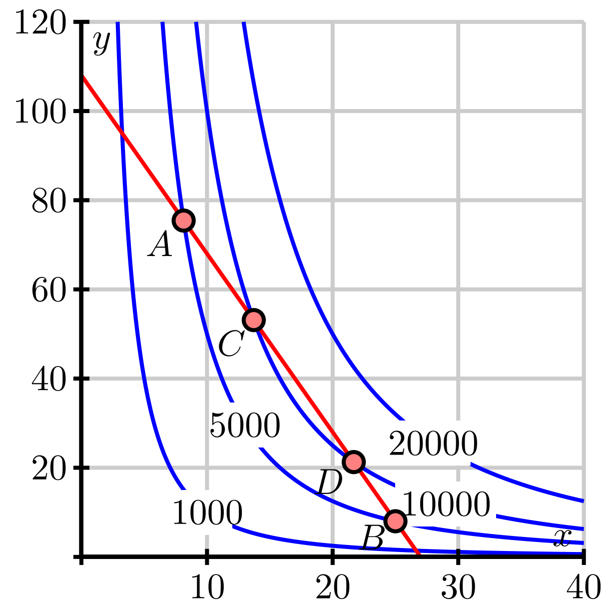


Figure 10.8.1 Contours of f and the constraint equation $g(x, y) = 108$.

b. Figure 10.8.1 shows the graph of the constraint equation $g(x, y) = 108$ along with a few contours of the volume function f. Since our goal is to find the maximum value of f subject to the constraint $g(x, y) = 108$, we want to find the point on our constraint curve that intersects the contours of f at which f has its largest value.

i. Points A and B in Figure 10.8.1 lie on a contour of f and on the constraint equation $g(x, y) = 108$. Explain why neither A nor B provides a maximum value of f that satisfies the constraint.

ii. Points C and D in Figure 10.8.1 lie on a contour of f and on the constraint equation $g(x, y) = 108$. Explain why neither C nor D provides a maximum value of f that satisfies the constraint.

iii. Based on your responses to parts i. and ii., draw the contour of f on which you believe f will achieve a maximum value subject to the constraint $g(x, y) = 108$. Explain why you drew the contour you did.

c. Recall that $g(x, y) = 108$ is a contour of the function g, and that the gradient of a function is always orthogonal to its contours. With this in mind, how should ∇f and ∇g be related at the optimal point? Explain.

Activity 10.8.2. A cylindrical soda can holds about 355 cc of liquid. In this activity, we want to find the dimensions of such a can that will minimize the surface area. For the sake of simplicity, assume the can is a perfect cylinder.

a. What are the variables in this problem? Based on the context, what restriction(s), if any, are there on these variables?

b. What quantity do we want to optimize in this problem? What equation describes the constraint? (You need to decide which of these functions plays the role of f and which plays the role of g in our discussion of Lagrange multipliers.)

c. Find λ and the values of your variables that satisfy Equation (10.8.1) in the context of this problem.

d. Determine the dimensions of the pop can that give the desired solution to this constrained optimization problem.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 10.8.3. Use the method of Lagrange multipliers to find the dimensions of the least expensive packing crate with a volume of 240 cubic feet when the material for the top costs \$2 per square foot, the bottom is \$3 per square foot and the sides are \$1.50 per square foot.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11 Multiple Integrals

11.1 Double Riemann Sums and Double Integrals over Rectangles

Preview Activity 11.1.1. In this activity we introduce the concept of a double Riemann sum.

a. Review the concept of the Riemann sum from single-variable calculus. Then, explain how we define the definite integral $\int_a^b f(x)\,dx$ of a continuous function of a single variable x on an interval $[a, b]$. Include a sketch of a continuous function on an interval $[a, b]$ with appropriate labeling in order to illustrate your definition.

b. In our upcoming study of integral calculus for multivariable functions, we will first extend the idea of the single-variable definite integral to functions of two variables over rectangular domains. To do so, we will need to understand how to partition a rectangle into subrectangles. Let R be rectangular domain $R = \{(x, y) : 0 \le x \le 6, 2 \le y \le 4\}$ (we can also represent this domain with the notation $[0, 6] \times [2, 4]$), as pictured in Figure 11.1.1.

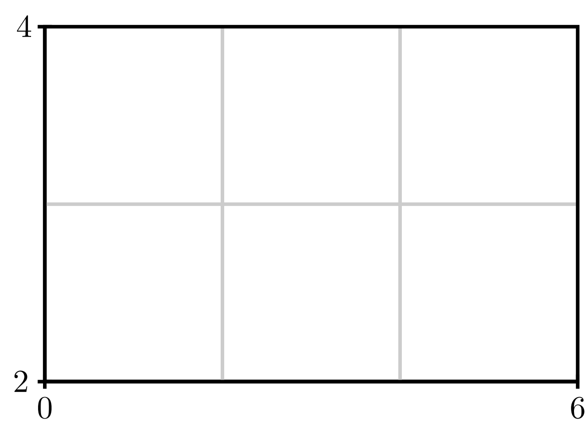


Figure 11.1.1 Rectangular domain R with subrectangles.

To form a partition of the full rectangular region, R, we will partition both intervals $[0, 6]$ and $[2, 4]$; in particular, we choose to partition the interval $[0, 6]$ into three uniformly sized subintervals and the interval $[2, 4]$ into two evenly sized subintervals as shown in Figure 11.1.1. In the following questions, we discuss how to identify the endpoints of each subinterval and the resulting subrectangles.

i. Let $0 = x_0 < x_1 < x_2 < x_3 = 6$ be the endpoints of the subintervals of $[0, 6]$ after partitioning. What is the length Δx of each subinterval $[x_{i-1}, x_i]$ for i from 1 to 3?

ii. Explicitly identify x_0, x_1, x_2, and x_3. On Figure 11.1.1 or your own version of the diagram, label these endpoints.

iii. Let $2 = y_0 < y_1 < y_2 = 4$ be the endpoints of the subintervals of $[2, 4]$ after partitioning. What is the length Δy of each subinterval $[y_{j-1}, y_j]$ for j from 1 to 2? Identify y_0, y_1, and y_2 and label these endpoints on Figure 11.1.1.

iv. Let R_{ij} denote the subrectangle $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$. Appropriately label each subrectangle in your drawing of Figure 11.1.1. How does the total number of subrectangles depend on the partitions of the intervals $[0, 6]$ and $[2, 4]$?

v. What is area ΔA of each subrectangle?

Activity 11.1.2. Let $f(x,y) = 100 - x^2 - y^2$ be defined on the rectangular domain $R = [a,b] \times [c,d]$. Partition the interval $[a,b]$ into four uniformly sized subintervals and the interval $[c,d]$ into three evenly sized subintervals as shown in Figure 11.1.2. As we did in Preview Activity 11.1.1, we will need a method for identifying the endpoints of each subinterval and the resulting subrectangles.

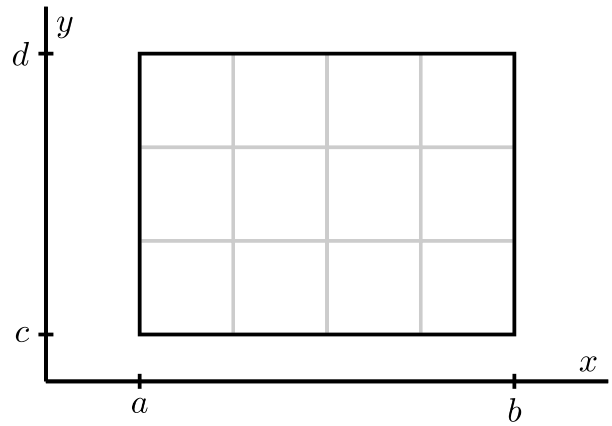


Figure 11.1.2 Rectangular domain with subrectangles.

a. Let $a = x_0 < x_1 < x_2 < x_3 < x_4 = b$ be the endpoints of the subintervals of $[a,b]$ after partitioning. Label these endpoints in Figure 11.1.2.

b. What is the length Δx of each subinterval $[x_{i-1}, x_i]$? Your answer should be in terms of a and b.

c. Let $c = y_0 < y_1 < y_2 < y_3 = d$ be the endpoints of the subintervals of $[c,d]$ after partitioning. Label these endpoints in Figure 11.1.2.

d. What is the length Δy of each subinterval $[y_{j-1}, y_j]$? Your answer should be in terms of c and d.

e. The partitions of the intervals $[a,b]$ and $[c,d]$ partition the rectangle R into subrectangles. How many subrectangles are there?

f. Let R_{ij} denote the subrectangle $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$. Label each subrectangle in Figure 11.1.2.

g. What is area ΔA of each subrectangle?

h. Now let $[a,b] = [0,8]$ and $[c,d] = [2,6]$. Let (x_{11}^*, y_{11}^*) be the point in the upper right corner of the subrectangle R_{11}. Identify and correctly label this point in Figure 11.1.2. Calculate the product

$$f(x_{11}^*, y_{11}^*)\Delta A.$$

Explain, geometrically, what this product represents.

i. For each i and j, choose a point (x_{ij}^*, y_{ij}^*) in the subrectangle $R_{i,j}$. Identify and correctly label these points in Figure 11.1.2. Explain what the product

$$f(x_{ij}^*, y_{ij}^*)\Delta A$$

represents.

j. If we were to add all the values $f(x_{ij}^*, y_{ij}^*)\Delta A$ for each i and j, what does the resulting number approximate about the surface defined by f on the domain R? (You don't actually need to add these values.)

k. Write a double sum using summation notation that expresses the arbitrary sum from part (j).

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.1.3. Let $f(x,y) = x + 2y$ and let $R = [0,2] \times [1,3]$.

a. Draw a picture of R. Partition $[0,2]$ into 2 subintervals of equal length and the interval $[1,3]$ into two subintervals of equal length. Draw these partitions on your picture of R and label the resulting subrectangles using the labeling scheme we established in the definition of a double Riemann sum.

b. For each i and j, let (x_{ij}^*, y_{ij}^*) be the midpoint of the rectangle R_{ij}. Identify the coordinates of each (x_{ij}^*, y_{ij}^*). Draw these points on your picture of R.

c. Calculate the Riemann sum

$$\sum_{j=1}^{n}\sum_{i=1}^{m} f(x_{ij}^*, y_{ij}^*) \cdot \Delta A$$

using the partitions we have described. If we let (x_{ij}^*, y_{ij}^*) be the midpoint of the rectangle R_{ij} for each i and j, then the resulting Riemann sum is called a *midpoint sum*.

d. Give two interpretations for the meaning of the sum you just calculated.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.1.4. Let $f(x, y) = \sqrt{4 - y^2}$ on the rectangular domain $R = [1, 7] \times [-2, 2]$. Partition $[1, 7]$ into 3 equal length subintervals and $[-2, 2]$ into 2 equal length subintervals. A table of values of f at some points in R is given in Table 11.1.8, and a graph of f with the indicated partitions is shown in Figure 11.1.9.

	-2	-1	0	1	2
1	0	$\sqrt{3}$	2	$\sqrt{3}$	0
2	0	$\sqrt{3}$	2	$\sqrt{3}$	0
3	0	$\sqrt{3}$	2	$\sqrt{3}$	0
4	0	$\sqrt{3}$	2	$\sqrt{3}$	0
5	0	$\sqrt{3}$	2	$\sqrt{3}$	0
6	0	$\sqrt{3}$	2	$\sqrt{3}$	0
7	0	$\sqrt{3}$	2	$\sqrt{3}$	0

Table 11.1.8 Table of values of $f(x, y) = \sqrt{4 - y^2}$**.**

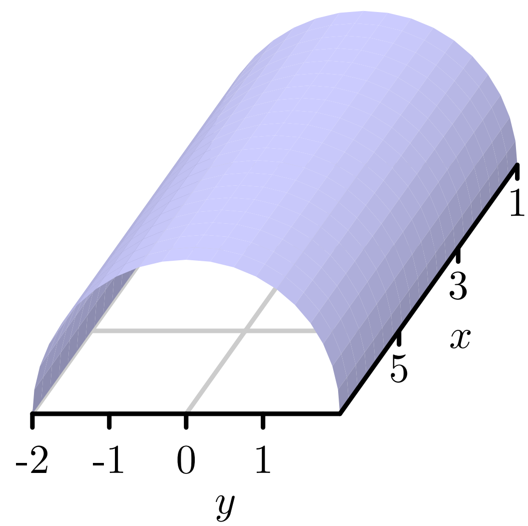


Figure 11.1.9 Graph of $f(x, y) = \sqrt{4 - y^2}$ on R.

a. Sketch the region R in the plane using the values in Table 11.1.8 as the partitions.

b. Calculate the double Riemann sum using the given partition of R and the values of f in the upper right corner of each subrectangle.

c. Use geometry to calculate the exact value of $\iint_R f(x, y)\, dA$ and compare it to your approximation. Describe one way we could obtain a better approximation using the given data.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.2 Iterated Integrals

Preview Activity 11.2.1. Let $f(x, y) = 25 - x^2 - y^2$ on the rectangular domain $R = [-3, 3] \times [-4, 4]$. As with partial derivatives, we may treat one of the variables in f as constant and think of the resulting function as a function of a single variable. Now we investigate what happens if we integrate instead of differentiate.

a. Choose a fixed value of x in the interior of $[-3, 3]$. Let

$$A(x) = \int_{-4}^{4} f(x, y)\, dy.$$

What is the geometric meaning of the value of $A(x)$ relative to the surface defined by f. (Hint: Think about the trace determined by the fixed value of x, and consider how $A(x)$ is related to the image at left in Figure 11.2.2.)

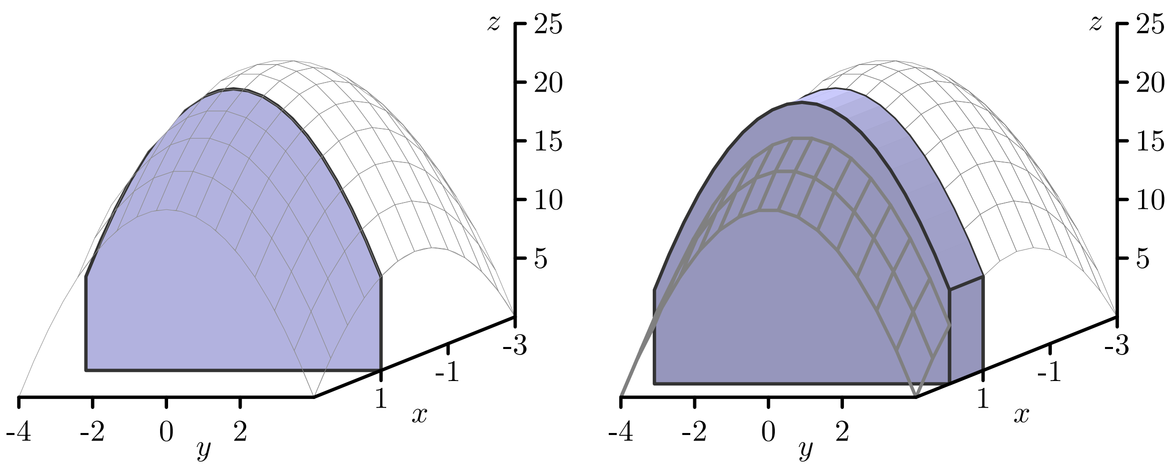


Figure 11.2.2 Left: A cross section with fixed x. Right: A cross section with fixed x and Δx.

b. For a fixed value of x, say x_i^*, what is the geometric meaning of $A(x_i^*)\,\Delta x$? (Hint: Consider how $A(x_i^*)\Delta x$ is related to the image at right in Figure 11.2.2.)

c. Since f is continuous on R, we can define the function $A = A(x)$ at every value of x in $[-3, 3]$. Now think about subdividing the x-interval $[-3, 3]$ into m subintervals, and choosing a value x_i^* in each of those subintervals. What will be the meaning of the sum $\sum_{i=1}^{m} A(x_i^*)\,\Delta x$?

d. Explain why $\int_{-3}^{3} A(x)\, dx$ will determine the exact value of the volume under the surface $z = f(x, y)$ over the rectangle R.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.2.2. Let $f(x, y) = 25 - x^2 - y^2$ on the rectangular domain $R = [-3, 3] \times [-4, 4]$.

a. Viewing x as a fixed constant, use the Fundamental Theorem of Calculus to evaluate the integral

$$A(x) = \int_{-4}^{4} f(x, y)\, dy.$$

Note that you will be integrating with respect to y, and holding x constant. Your result should be a function of x only.

b. Next, use your result from (a) along with the Fundamental Theorem of Calculus to determine the value of $\int_{-3}^{3} A(x)\, dx$.

c. What is the value of $\iint_R f(x, y)\, dA$? What are two different ways we may interpret the meaning of this value?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.2.3. Let $f(x, y) = x + y^2$ on the rectangle $R = [0, 2] \times [0, 3]$.

a. Evaluate $\iint_R f(x, y)\, dA$ using an iterated integral. Choose an order for integration by deciding whether you want to integrate first with respect to x or y.

b. Evaluate $\iint_R f(x, y)\, dA$ using the iterated integral whose order of integration is the opposite of the order you chose in (a).

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.3 Double Integrals over General Regions

Preview Activity 11.3.1. A tetrahedron is a three-dimensional figure with four faces, each of which is a triangle. A picture of the tetrahedron T with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ is shown at left in Figure 11.3.1. If we place one vertex at the origin and let vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ be determined by the edges of the tetrahedron that have one end at the origin, then a formula that tells us the volume V of the tetrahedron is

$$V = \frac{1}{6}|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|. \tag{11.3.1}$$

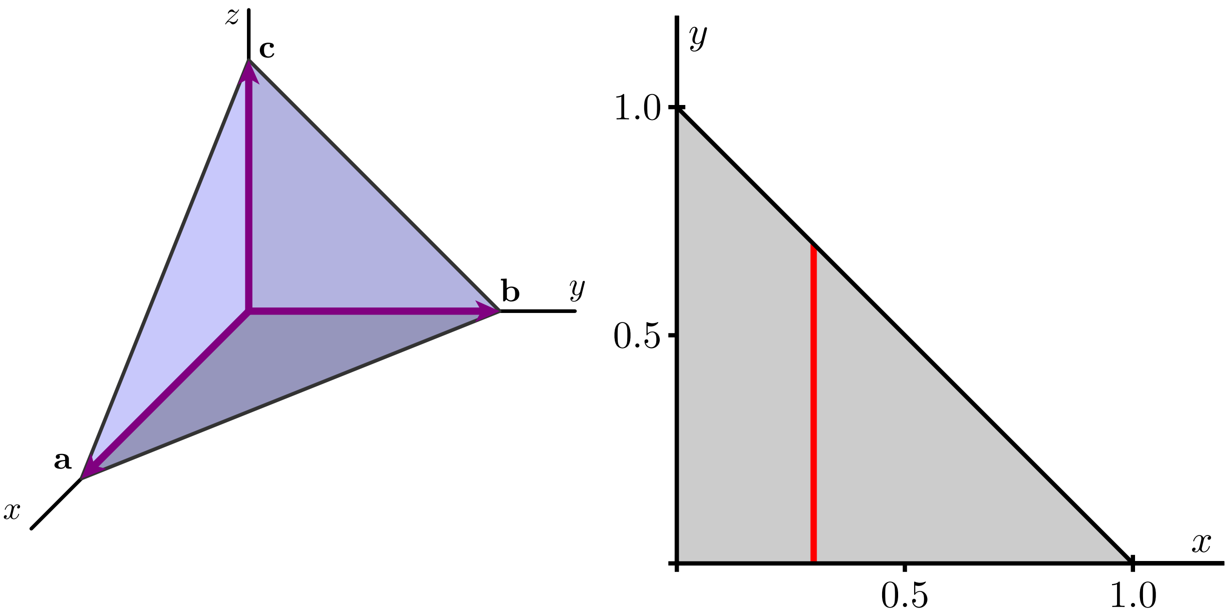


Figure 11.3.1 Left: The tetrahedron T. Right: Projecting T onto the xy-plane.

a. Use the formula (11.3.1) to find the volume of the tetrahedron T.

b. Instead of memorizing or looking up the formula for the volume of a tetrahedron, we can use a double integral to calculate the volume of the tetrahedron T. To see how, notice that the top face of the tetrahedron T is the plane whose equation is

$$z = 1 - (x + y).$$

Provided that we can use an iterated integral on a non-rectangular region, the volume of the tetrahedron will be given by an iterated integral of the form

$$\int_{x=?}^{x=?} \int_{y=?}^{y=?} 1 - (x + y)\, dy\, dx.$$

The issue that is new here is how we find the limits on the integrals; note that the outer integral's limits are in x, while the inner ones are in y, since we have chosen $dA = dy\, dx$. To see the domain over which we need to integrate, think of standing way above the tetrahedron looking straight down on it, which means we are projecting the entire tetrahedron onto the xy-plane. The resulting domain is the triangular region shown at right in Figure 11.3.1. Explain why we can represent the triangular region with the inequalities

$$0 \le y \le 1 - x \quad \text{and} \quad 0 \le x \le 1.$$

(Hint: Consider the cross sectional slice shown at right in Figure 11.3.1.)

c. Explain why it makes sense to now write the volume integral in the form

$$\int_{x=?}^{x=?} \int_{y=?}^{y=?} 1 - (x + y)\, dy\, dx = \int_{x=0}^{x=1} \int_{y=0}^{y=1-x} 1 - (x + y)\, dy\, dx.$$

d. Use the Fundamental Theorem of Calculus to evaluate the iterated integral

$$\int_{x=0}^{x=1} \int_{y=0}^{y=1-x} 1 - (x+y)\, dy\, dx$$

and compare to your result from part (a). (As with iterated integrals over rectangular regions, start with the inner integral.)

Activity 11.3.2. Consider the double integral $\iint_D (4 - x - 2y)\, dA$, where D is the triangular region with vertices (0,0), (4,0), and (0,2).

a. Write the given integral as an iterated integral of the form $\iint_D (4-x-2y)\, dy\, dx$. Draw a labeled picture of D with relevant cross sections.

b. Write the given integral as an iterated integral of the form $\iint_D (4-x-2y)\, dx\, dy$. Draw a labeled picture of D with relevant cross sections.

c. Evaluate the two iterated integrals from (a) and (b), and verify that they produce the same value. Give at least one interpretation of the meaning of your result.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.3.3. Consider the iterated integral $\int_{x=0}^{x=1}\int_{y=x}^{y=\sqrt{x}}(4x+10y)\,dy\,dx$.

a. Sketch the region of integration, D, for which

$$\iint_D (4x+10y)\,dA = \int_{x=0}^{x=1}\int_{y=x}^{y=\sqrt{x}}(4x+10y)\,dy\,dx.$$

b. Determine the equivalent iterated integral that results from integrating in the opposite order ($dx\,dy$, instead of $dy\,dx$). That is, determine the limits of integration for which

$$\iint_D (4x+10y)\,dA = \int_{y=?}^{y=?}\int_{x=?}^{x=?}(4x+10y)\,dx\,dy.$$

c. Evaluate one of the two iterated integrals above. Explain what the value you obtained tells you.

d. Set up and evaluate a single definite integral to determine the exact area of D, $A(D)$.

e. Determine the exact average value of $f(x,y)=4x+10y$ over D.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.3.4. Consider the iterated integral $\int_{x=0}^{x=4} \int_{y=x/2}^{y=2} e^{y^2}\, dy\, dx$.

a. Explain why we cannot find a simple antiderivative for e^{y^2} with respect to y, and thus are unable to evaluate $\int_{x=0}^{x=4} \int_{y=x/2}^{y=2} e^{y^2}\, dy\, dx$ in the indicated order using the Fundamental Theorem of Calculus.

b. Given that $\iint_D e^{y^2}\, dA = \int_{x=0}^{x=4} \int_{y=x/2}^{y=2} e^{y^2}\, dy\, dx$, sketch the region of integration, D.

c. Rewrite the given iterated integral in the opposite order, using $dA = dx\, dy$. (Hint: You may need more than one integral.)

d. Use the Fundamental Theorem of Calculus to evaluate the iterated integral you developed in (c). Write one sentence to explain the meaning of the value you found.

e. What is the important lesson this activity offers regarding the order in which we set up an iterated integral?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.4 Applications of Double Integrals

Preview Activity 11.4.1. Suppose that we have a flat, thin object (called a **lamina**) whose density varies across the object. We can think of the density on a lamina as a measure of mass per unit area. As an example, consider a circular plate D of radius 1 cm centered at the origin whose density δ varies depending on the distance from its center so that the density in grams per square centimeter at point (x, y) is

$$\delta(x, y) = 10 - 2(x^2 + y^2).$$

a. Suppose that we partition the plate into subrectangles R_{ij}, where $1 \leq i \leq m$ and $1 \leq j \leq n$, of equal area ΔA, and select a point (x_{ij}^*, y_{ij}^*) in R_{ij} for each i and j. What is the meaning of the quantity $\delta(x_{ij}^*, y_{ij}^*)\Delta A$?

b. State a double Riemann sum that provides an approximation of the mass of the plate.

c. Explain why the double integral

$$\iint_D \delta(x, y)\, dA$$

tells us the exact mass of the plate.

d. Determine an iterated integral which, if evaluated, would give the exact mass of the plate. Do not actually evaluate the integral. (This integral is considerably easier to evaluate in polar coordinates, which we will learn more about in Section 11.5.)

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.4.2. Let D be a half-disk lamina of radius 3 in quadrants IV and I, centered at the origin as shown in Figure 11.4.1. Assume the density at point (x, y) is given by $\delta(x, y) = x$. Find the exact mass of the lamina.

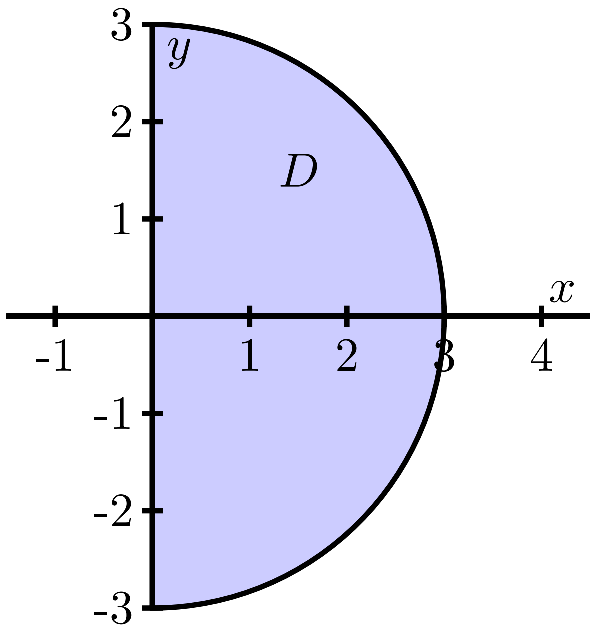


Figure 11.4.1 A half disk lamina.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.4.3. Suppose we want to find the area of the bounded region D between the curves

$$y = 1 - x^2 \qquad \text{and} \qquad y = x - 1.$$

A picture of this region is shown in Figure 11.4.2.

a. The volume of a solid with constant height is given by the area of the base times the height. Hence, we may interpret the area of the region D as the volume of a solid with base D and of uniform height 1. That is, the area of D is given by $\iint_D 1\, dA$. Write an iterated integral whose value is $\iint_D 1\, dA$. (Hint: Which order of integration might be more efficient? Why?)

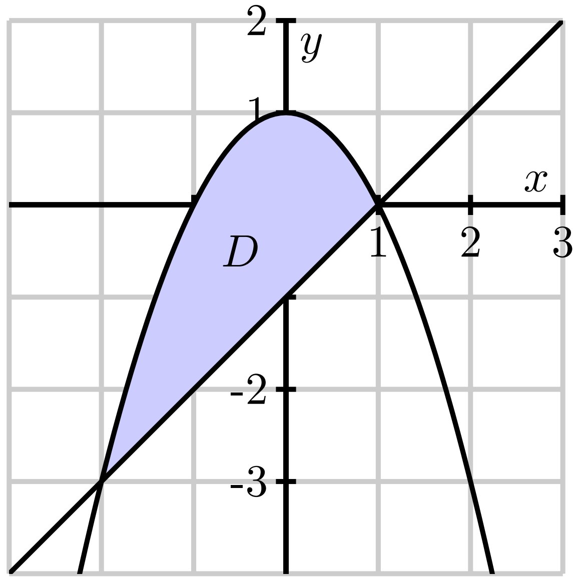


Figure 11.4.2 The graphs of $y = 1 - x^2$ and $y = x - 1$.

b. Evaluate the iterated integral from (a). What does the result tell you?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.4.4. In this activity we determine integrals that represent the center of mass of a lamina D described by the triangular region bounded by the x-axis and the lines $x = 1$ and $y = 2x$ in the first quadrant if the density at point (x, y) is $\delta(x, y) = 6x + 6y + 6$. A picture of the lamina is shown in Figure 11.4.4.

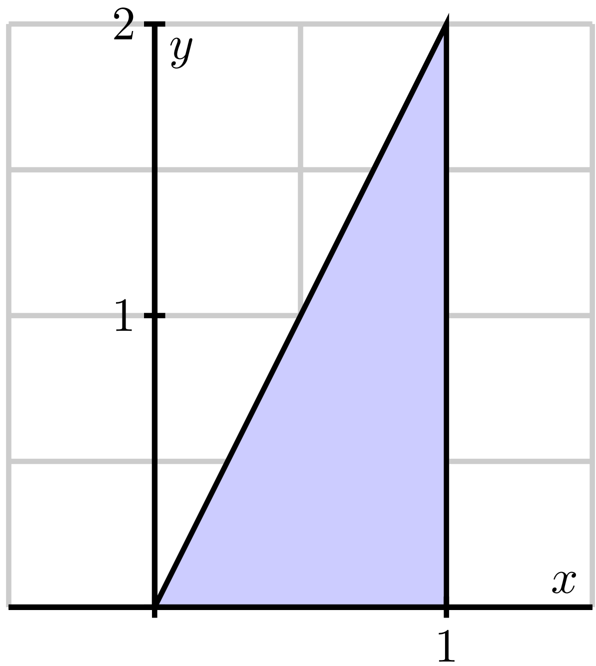


Figure 11.4.4 The lamina bounded by the x-axis and the lines $x = 1$ and $y = 2x$ in the first quadrant.

a. Set up an iterated integral that represents the mass of the lamina.

b. Assume the mass of the lamina is 14. Set up two iterated integrals that represent the coordinates of the center of mass of the lamina.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.4.5. A firm manufactures smoke detectors. Two components for the detectors come from different suppliers — one in Michigan and one in Ohio. The company studies these components for their reliability and their data suggests that if x is the life span (in years) of a randomly chosen component from the Michigan supplier and y the life span (in years) of a randomly chosen component from the Ohio supplier, then the joint probability density function f might be given by

$$f(x,y) = e^{-x}e^{-y}.$$

a. Theoretically, the components might last forever, so the domain D of the function f is the set D of all (x,y) such that $x \geq 0$ and $y \geq 0$. To show that f is a probability density function on D we need to demonstrate that

$$\int\int_D f(x,y)\,dA = 1,$$

or that

$$\int_0^\infty \int_0^\infty f(x,y)\,dy\,dx = 1.$$

Use your knowledge of improper integrals to verify that f is indeed a probability density function.

b. Assume that the smoke detector fails only if both of the supplied components fail. To determine the probability that a randomly selected detector will fail within one year, we will need to determine the probability that the life span of each component is between 0 and 1 years. Set up an appropriate iterated integral, and evaluate the integral to determine the probability.

c. What is the probability that a randomly chosen smoke detector will fail between years 3 and 7?

d. Suppose that the manufacturer determines that one of the components is more likely to fail than the other, and hence conjectures that the probability density function is instead $f(x,y) = Ke^{-x}e^{-2y}$. What is the value of K?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.5 Double Integrals in Polar Coordinates

Preview Activity 11.5.1. The coordinates of a point determine its location. In particular, the rectangular coordinates of a point P are given by an ordered pair (x, y), where x is the (signed) distance the point lies from the y-axis to P and y is the (signed) distance the point lies from the x-axis to P. In polar coordinates, we locate the point by considering the distance the point lies from the origin, $O = (0, 0)$, and the angle the line segment from the origin to P forms with the positive x-axis.

a. Determine the rectangular coordinates of the following points:

   (a) The point P that lies 1 unit from the origin on the positive x-axis.

   (b) The point Q that lies 2 units from the origin and such that $\overline{OQ}$ makes an angle of $\frac{\pi}{2}$ with the positive x-axis.

   (c) The point R that lies 3 units from the origin such that $\overline{OR}$ makes an angle of $\frac{2\pi}{3}$ with the positive x-axis.

b. Part (a) indicates that the two pieces of information completely determine the location of a point: either the traditional (x, y) coordinates, or alternately, the distance r from the point to the origin along with the angle θ that the line through the origin and the point makes with the positive x-axis. We write "(r, θ)" to denote the point's location in its polar coordinate representation. Find polar coordinates for the points with the given rectangular coordinates.

   i. $(0, -1)$ ii. $(-2, 0)$ iii. $(-1, 1)$

c. For each of the following points whose coordinates are given in polar form, determine the rectangular coordinates of the point.

   i. $(5, \frac{\pi}{4})$ ii. $(2, \frac{5\pi}{6})$ iii. $(\sqrt{3}, \frac{5\pi}{3})$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.5.2. Most polar graphing devices can plot curves in polar coordinates of the form $r = f(\theta)$. Use such a device to complete this activity.

a. Before plotting the polar curve $r = 1$ (where θ can have any value), think about what shape it should have, in light of how r is connected to x and y. Then use appropriate technology to draw the graph and test your intuition.

b. The equation $\theta = 1$ does not define r as a function of θ, so we can't graph this equation on many polar plotters. What do you think the graph of the polar curve $\theta = 1$ looks like? Why?

c. Before plotting the polar curve $r = \theta$, what do you think the graph looks like? Why? Use technology to plot the curve and compare your intuition.

d. What does the region defined by $1 \leq r \leq 3$ (where θ can have any value) look like? (Hint: Compare to your response from part (a).)

e. What does the region defined by $1 \leq r \leq 3$ and $\pi/4 \leq \theta \leq \pi/2$ look like?

f. Consider the curve $r = \sin(\theta)$. For some values of θ we will have $r < 0$. In these situations, we plot the point (r, θ) as $(|r|, \theta + \pi)$ (in other words, when $r < 0$, we reflect the point through the origin). With that in mind, what do you think the graph of $r = \sin(\theta)$ looks like? Plot this curve using technology and compare to your intuition.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.5.3. Consider a polar rectangle R, with r between r_i and r_{i+1} and θ between θ_j and θ_{j+1} as shown at left in Figure 11.5.2. Let $\Delta r = r_{i+1} - r_i$ and $\Delta\theta = \theta_{j+1} - \theta_j$. Let ΔA be the area of this region.

a. Explain why the area ΔA in polar coordinates is not $\Delta r\, \Delta\theta$.

b. Now find ΔA by the following steps:

   i. Find the area of the annulus (the washer-like region) between r_i and r_{i+1}, as shown at right in Figure 11.5.2. This area will be in terms of r_i and r_{i+1}.

   ii. Observe that the region R is only a portion of the annulus, so the area ΔA of R is only a fraction of the area of the annulus. For instance, if $\theta_{i+1} - \theta_i$ were $\frac{\pi}{4}$, then the resulting wedge would be

$$\frac{\frac{\pi}{4}}{2\pi} = \frac{1}{8}$$

   of the entire annulus. In this more general context, using the wedge between the two noted angles, what fraction of the area of the annulus is the area ΔA?

   iii. Write an expression for ΔA in terms of r_i, r_{i+1}, θ_j, and θ_{j+1}.

   iv. Finally, write the area ΔA in terms of r_i, r_{i+1}, Δr, and $\Delta\theta$, where each quantity appears only once in the expression. (Hint: Think about how to factor a difference of squares.)

c. As we take the limit as Δr and $\Delta\theta$ go to 0, Δr becomes dr, $\Delta\theta$ becomes $d\theta$, and ΔA becomes dA, the area element. Using your work in (iv), write dA in terms of r, dr, and $d\theta$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.5.4. Let $f(x, y) = x + y$ and $D = \{(x, y) : x^2 + y^2 \leq 4\}$.

a. Sketch the region D and then write the double integral of f over D as an iterated integral in rectangular coordinates.

b. Write the double integral of f over D as an iterated integral in polar coordinates.

c. Evaluate one of the iterated integrals. Why is the final value you found not surprising?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.5.5. Consider the circle given by $x^2 + (y-1)^2 = 1$ as shown in Figure 11.5.4.

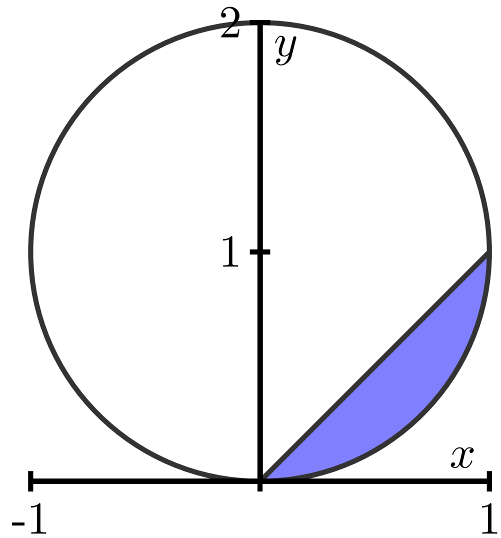


Figure 11.5.4 The graphs of $y = x$ and $x^2 + (y-1)^2 = 1$, for use in Activity 11.5.5.

a. Determine a polar curve in the form $r = f(\theta)$ that traces out the circle $x^2 + (y-1)^2 = 1$. (Hint: Recall that a circle centered at the origin of radius r can be described by the equations $x = r\cos(\theta)$ and $y = r\sin(\theta)$.)

b. Find the exact average value of $g(x, y) = \sqrt{x^2 + y^2}$ over the interior of the circle $x^2 + (y-1)^2 = 1$.

c. Find the volume under the surface $h(x, y) = x$ over the region D, where D is the region bounded above by the line $y = x$ and below by the circle (this is the shaded region in Figure 11.5.4).

d. Explain why in both (b) and (c) it is advantageous to use polar coordinates.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.6 Surfaces Defined Parametrically and Surface Area

Preview Activity 11.6.1. Recall the standard parameterization of the unit circle that is given by

$$x(t) = \cos(t) \quad \text{and} \quad y(t) = \sin(t),$$

where $0 \leq t \leq 2\pi$.

a. Determine a parameterization of the circle of radius 1 in $\mathbb{R}^3$ that has its center at $(0, 0, 1)$ and lies in the plane $z = 1$.

b. Determine a parameterization of the circle of radius 1 in 3-space that has its center at $(0, 0, -1)$ and lies in the plane $z = -1$.

c. Determine a parameterization of the circle of radius 1 in 3-space that has its center at $(0, 0, 5)$ and lies in the plane $z = 5$.

d. Taking into account your responses in (a), (b), and (c), describe the graph that results from the set of parametric equations

$$x(s, t) = \cos(t), \quad y(s, t) = \sin(t), \quad \text{and} \quad z(s, t) = s,$$

where $0 \leq t \leq 2\pi$ and $-5 \leq s \leq 5$. Explain your thinking.

e. Just as a cylinder can be viewed as a "stack" of circles of constant radius, a cone can be viewed as a stack of circles with varying radius. Modify the parametrizations of the circles above in order to construct the parameterization of a cone whose vertex lies at the origin, whose base radius is 4, and whose height is 3, where the base of the cone lies in the plane $z = 3$. Use appropriate technology to plot the parametric equations you develop. (Hint: The cross sections parallel to the xy-plane are circles, with the radii varying linearly as z increases.)

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.6.2. In this activity, we seek a parametrization of the sphere of radius R centered at the origin, as shown on the left in Figure 11.6.5. Notice that this sphere may be obtained by revolving a half-circle contained in the xz-plane about the z-axis, as shown on the right.

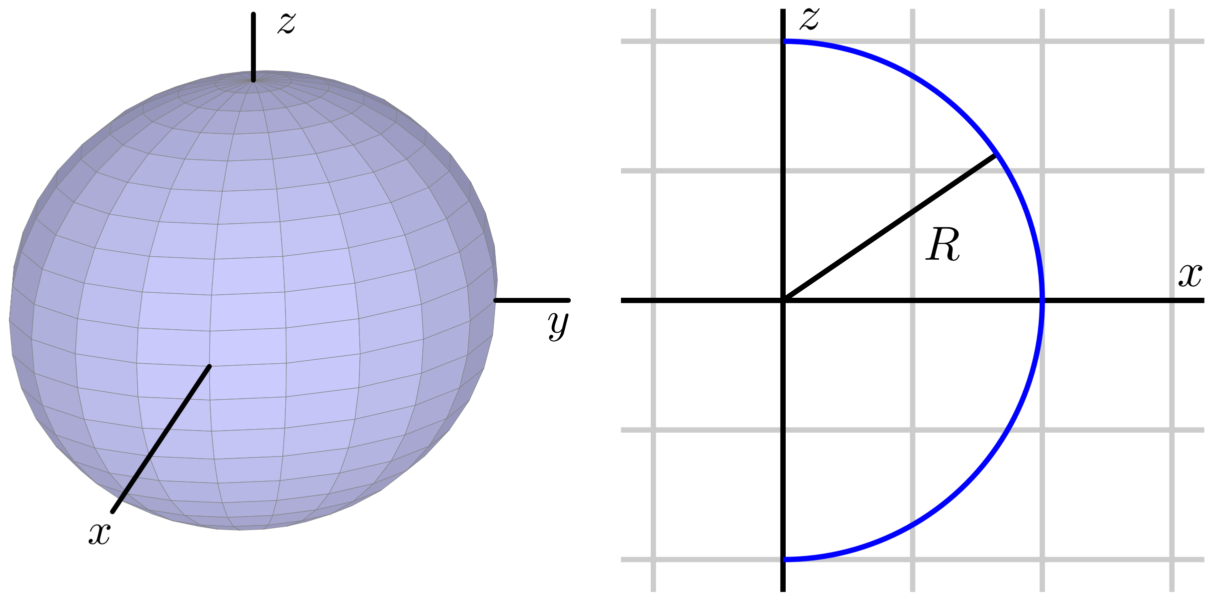


Figure 11.6.5 A sphere obtained by revolving a half-circle.

a. Begin by writing a parametrization of this half-circle using the parameter s:

$$x(s) = \dots \qquad z(s) = \dots .$$

Be sure to state the domain of the parameter s.

b. By revolving the points on this half-circle about the z-axis, obtain a parametrization $\mathbf{r}(s,t)$ of the points on the sphere of radius R. Be sure to include the domain of both parameters s and t. (Hint: What is the radius of the circle obtained when revolving a point on the half-circle around the z axis?)

c. Draw the surface defined by your parameterization with appropriate technology.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.6.3. Consider the cylinder with radius a and height h defined parametrically by

$$\mathbf{r}(s,t) = a\cos(s)\mathbf{i} + a\sin(s)\mathbf{j} + t\mathbf{k}$$

for $0 \le s \le 2\pi$ and $0 \le t \le h$, as shown in Figure 11.6.7.

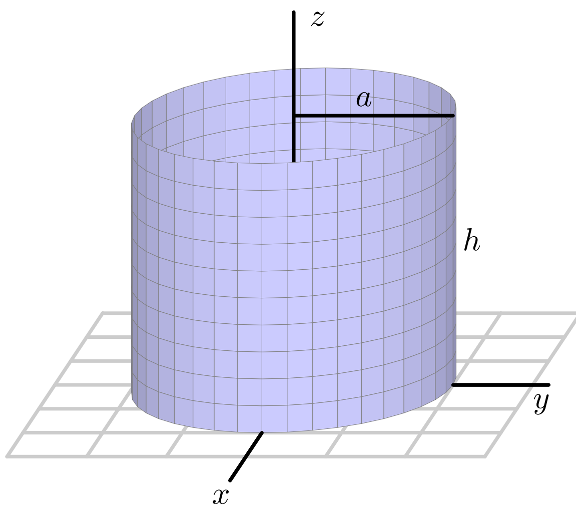


Figure 11.6.7 A cylinder.

a. Set up an iterated integral to determine the surface area of this cylinder.

b. Evaluate the iterated integral.

c. Recall that one way to think about the surface area of a cylinder is to cut the cylinder horizontally and find the perimeter of the resulting cross sectional circle, then multiply by the height. Calculate the surface area of the given cylinder using this alternate approach, and compare your work in (b).

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.6.4. Let $z = f(x, y)$ define a smooth surface, and consider the corresponding parameterization $\mathbf{r}(s,t) = \langle s, t, f(s,t)\rangle$.

a. Let D be a region in the domain of f. Using Equation (11.6.2), show that the area, S, of the surface defined by the graph of f over D is

$$S = \iint_D \sqrt{(f_x(x,y))^2 + (f_y(x,y))^2 + 1}\, dA.$$

b. Use the formula developed in (a) to calculate the area of the surface defined by $f(x,y) = \sqrt{4 - x^2}$ over the rectangle $D = [-2, 2] \times [0, 3]$.

c. Observe that the surface of the solid describe in (b) is half of a circular cylinder. Use the standard formula for the surface area of a cylinder to calculate the surface area in a different way, and compare your result from (b).

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.7 Triple Integrals

Preview Activity 11.7.1. Consider a solid piece of granite in the shape of a box $B = \{(x, y, z) : 0 \le x \le 4, 0 \le y \le 6, 0 \le z \le 8\}$, whose density varies from point to point. Let $\delta(x, y, z)$ represent the mass density of the piece of granite at point (x, y, z) in kilograms per cubic meter (so we are measuring x, y, and z in meters). Our goal is to find the mass of this solid.
Recall that if the density was constant, we could find the mass by multiplying the density and volume; since the density varies from point to point, we will use the approach we did with two-variable lamina problems, and slice the solid into small pieces on which the density is roughly constant.
Partition the interval $[0, 4]$ into 2 subintervals of equal length, the interval $[0, 6]$ into 3 subintervals of equal length, and the interval $[0, 8]$ into 2 subintervals of equal length. This partitions the box B into sub-boxes as shown in Figure 11.7.1.

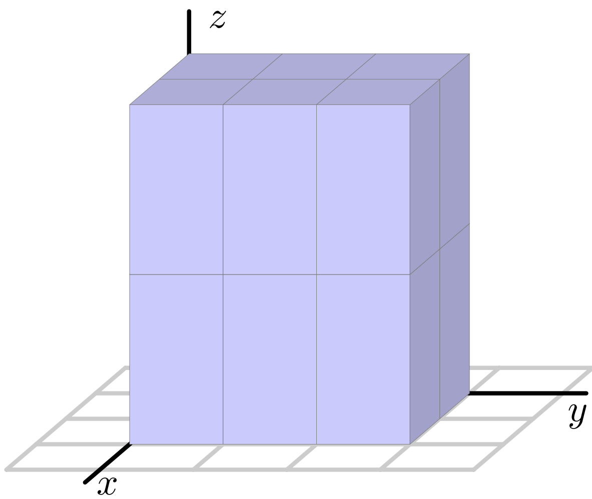


Figure 11.7.1 A partitioned three-dimensional domain.

a. Let $0 = x_0 < x_1 < x_2 = 4$ be the endpoints of the subintervals of $[0, 4]$ after partitioning. Draw a picture of Figure 11.7.1 and label these endpoints on your drawing. Do likewise with $0 = y_0 < y_1 < y_2 < y_3 = 6$ and $0 = z_0 < z_1 < z_2 = 8$ What is the length Δx of each subinterval $[x_{i-1}, x_i]$ for i from 1 to 2? the length of Δy? of Δz?

b. The partitions of the intervals $[0, 4]$, $[0, 6]$ and $[0, 8]$ partition the box B into sub-boxes. How many sub-boxes are there? What is volume ΔV of each sub-box?

c. Let B_{ijk} denote the sub-box $[x_{i-1}, x_i] \times [y_{j-1}, y_j] \times [z_{k-1}, z_k]$. Say that we choose a point $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$ in the i, j, kth sub-box for each possible combination of i, j, k. What is the meaning of $\delta(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$? What physical quantity will $\delta(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})\Delta V$ approximate?

d. What final step(s) would it take to determine the exact mass of the piece of granite?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.7.2.

a. Set up and evaluate the triple integral of $f(x,y,z) = x - y + 2z$ over the box $B = [-2,3] \times [1,4] \times [0,2]$.

b. Let S be the solid cone bounded by $z = \sqrt{x^2+y^2}$ and $z = 3$. A picture of S is shown at right in Figure 11.7.4. Our goal in what follows is to set up an iterated integral of the form

$$\int_{x=?}^{x=?} \int_{y=?}^{y=?} \int_{z=?}^{z=?} \delta(x,y,z)\, dz\, dy\, dx \tag{11.7.1}$$

to represent the mass of S in the setting where $\delta(x,y,z)$ tells us the density of S at the point (x,y,z). Our particular task is to find the limits on each of the three integrals.

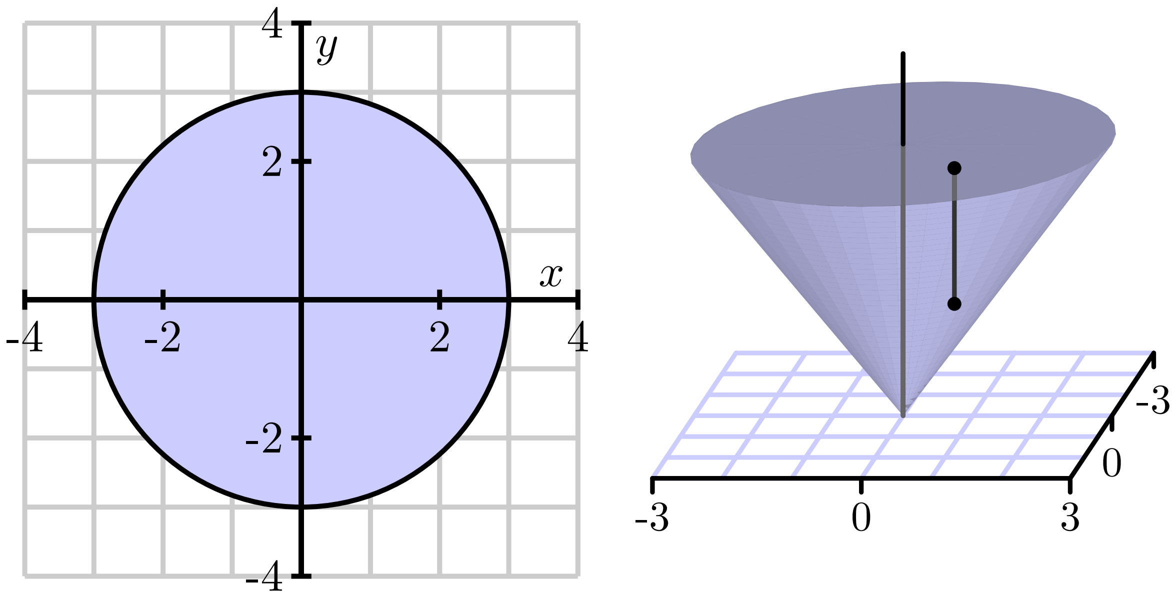


Figure 11.7.4 Left: The cone. Right: Its projection.

i. If we think about slicing up the solid, we can consider slicing the domain of the solid's projection onto the xy-plane (just as we would slice a two-dimensional region in $\mathbb{R}^2$), and then slice in the z-direction as well. The projection of the solid onto the xy-plane is shown at left in Figure 11.7.4. If we decide to first slice the domain of the solid's projection perpendicular to the x-axis, over what range of constant x-values would we have to slice?

ii. If we continue with slicing the domain, what are the limits on y on a typical slice? How do these depend on x? What, therefore, are the limits on the middle integral?

iii. Finally, now that we have thought about slicing up the two-dimensional domain that is the projection of the cone, what are the limits on z in the innermost integral? Note that over any point (x,y) in the plane, a vertical slice in the z direction will involve a range of values from the cone itself to its flat top. In particular, observe that at least one of these limits is not constant but depends on x and y.

iv. In conclusion, write an iterated integral of the form (11.7.1) that represents the mass of the cone S.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.7.3. There are several other ways we could have set up the integral to give the mass of the tetrahedron in Example 11.7.5.

a. How many different iterated integrals could be set up that are equal to the integral in Equation (11.7.2)?

b. Set up an iterated integral, integrating first with respect to z, then x, then y that is equivalent to the integral in Equation (11.7.2). Before you write down the integral, think about Figure 11.7.6, and draw an appropriate two-dimensional image of an important projection.

c. Set up an iterated integral, integrating first with respect to y, then z, then x that is equivalent to the integral in Equation (11.7.2). As in (b), think carefully about the geometry first.

d. Set up an iterated integral, integrating first with respect to x, then y, then z that is equivalent to the integral in Equation (11.7.2).

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.7.4. A solid S is bounded below by the square $z = 0$, $-1 \le x \le 1$, $-1 \le y \le 1$ and above by the surface $z = 2 - x^2 - y^2$. A picture of the solid is shown in Figure 11.7.7.

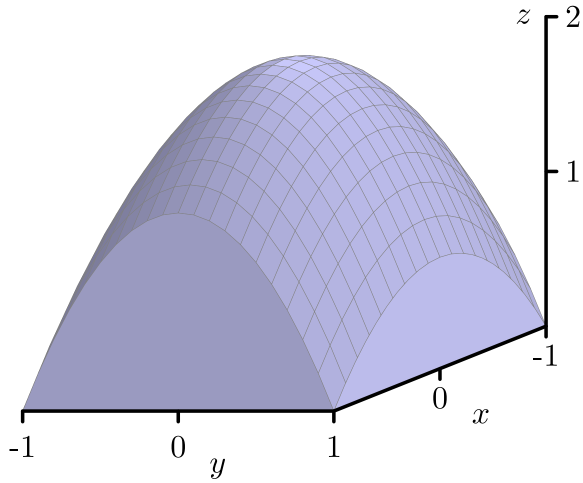


Figure 11.7.7 The solid bounded by the surface $z = 2 - x^2 - y^2$.

a. First, set up an iterated double integral to find the volume of the solid S as a double integral of a solid under a surface. Then set up an iterated triple integral that gives the volume of the solid S. You do not need to evaluate either integral. Compare the two approaches.

b. Set up (but do not evaluate) iterated integral expressions that will tell us the center of mass of S, if the density at point (x, y, z) is $\delta(x, y, z) = x^2 + 1$.

c. Set up (but do not evaluate) an iterated integral to find the average density on S using the density function from part (b).

d. Use technology appropriately to evaluate the iterated integrals you determined in (a), (b), and (c); does the location you determined for the center of mass make sense?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.8 Triple Integrals in Cylindrical and Spherical Coordinates

Preview Activity 11.8.1. In the following questions, we investigate the two new coordinate systems that are the subject of this section: cylindrical and spherical coordinates. Our goal is to consider some examples of how to convert from rectangular coordinates to each of these systems, and vice versa. Triangles and trigonometry prove to be particularly important.

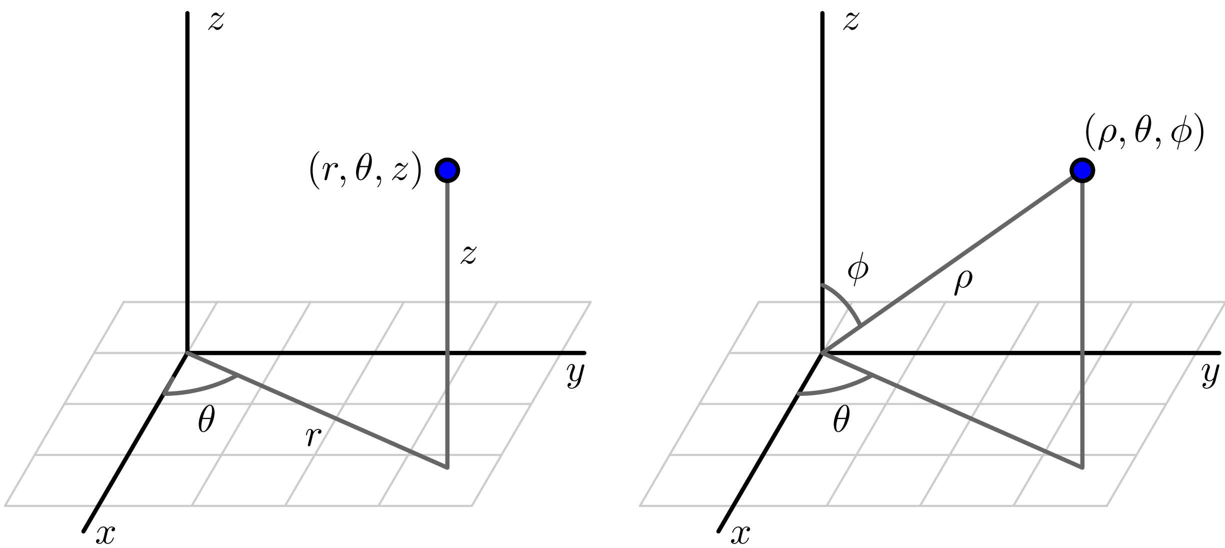


Figure 11.8.1 The cylindrical (left) and spherical (right) coordinates of a point.

a. The cylindrical coordinates of a point in $\mathbb{R}^3$ are given by (r, θ, z) where r and θ are the polar coordinates of the point (x, y) and z is the same z coordinate as in Cartesian coordinates. An illustration is given at left in Figure 11.8.1.

   i. Find cylindrical coordinates for the point whose Cartesian coordinates are $(-1, \sqrt{3}, 3)$. Draw a labeled picture illustrating all of the coordinates.

   ii. Find the Cartesian coordinates of the point whose cylindrical coordinates are $\left(2, \frac{5\pi}{4}, 1\right)$. Draw a labeled picture illustrating all of the coordinates.

b. The spherical coordinates of a point in $\mathbb{R}^3$ are ρ (rho), θ, and ϕ (phi), where ρ is the distance from the point to the origin, θ has the same interpretation it does in polar coordinates, and ϕ is the angle between the positive z axis and the vector from the origin to the point, as illustrated at right in Figure 11.8.1. You should convince yourself that any point in $\mathbb{R}^3$ can be represented in spherical coordinates with $\rho \geq 0$, $0 \leq \theta < 2\pi$, and $0 \leq \phi \leq \pi$.

   For the following questions, consider the point P whose Cartesian coordinates are $(-2, 2, \sqrt{8})$.

   i. What is the distance from P to the origin? Your result is the value of ρ in the spherical coordinates of P.

   ii. Determine the point that is the projection of P onto the xy-plane. Then, use this projection to find the value of θ in the polar coordinates of the projection of P that lies in the plane. Your result is also the value of θ for the spherical coordinates of the point.

   iii. Based on the illustration in Figure 11.8.1, how is the angle ϕ determined by ρ and the z coordinate of P? Use a well-chosen right triangle to find the value of ϕ, which is the final component in the spherical coordinates of P. Draw a carefully labeled picture that clearly illustrates the values of ρ, θ, and ϕ in this example, along with the original rectangular coordinates of P.

   iv. Based on your responses to i., ii., and iii., if we are given the Cartesian coordinates (x, y, z) of a point Q, how are the values of ρ, θ, and ϕ in the spherical coordinates of Q determined by x, y, and z?

Activity 11.8.2. In this activity, we graph some surfaces using cylindrical coordinates. To improve your intuition and test your understanding, you should first think about what each graph should look like before you plot it using appropriate technology.

a. What familiar surface is described by the points in cylindrical coordinates with $r = 2$, $0 \leq \theta \leq 2\pi$, and $0 \leq z \leq 2$? How does this example suggest that we call these coordinates *cylindrical coordinates*? How does your answer change if we restrict θ to $0 \leq \theta \leq \pi$?

b. What familiar surface is described by the points in cylindrical coordinates with $\theta = 2$, $0 \leq r \leq 2$, and $0 \leq z \leq 2$?

c. What familiar surface is described by the points in cylindrical coordinates with $z = 2$, $0 \leq \theta \leq 2\pi$, and $0 \leq r \leq 2$?

d. Plot the graph of the cylindrical equation $z = r$, where $0 \leq \theta \leq 2\pi$ and $0 \leq r \leq 2$. What familiar surface results?

e. Plot the graph of the cylindrical equation $z = \theta$ for $0 \leq \theta \leq 4\pi$. What does this surface look like?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.8.3. A picture of a cylindrical box, $B = \{(r, \theta, z) : r_1 \leq r \leq r_2, \theta_1 \leq \theta \leq \theta_2, z_1 \leq z \leq z_2\}$, is shown in Figure 11.8.2. Let $\Delta r = r_2 - r_1$, $\Delta\theta = \theta_2 - \theta_1$, and $\Delta z = z_2 - z_1$. We want to determine the volume ΔV of B in terms of Δr, $\Delta\theta$, Δz, r, θ, and z.

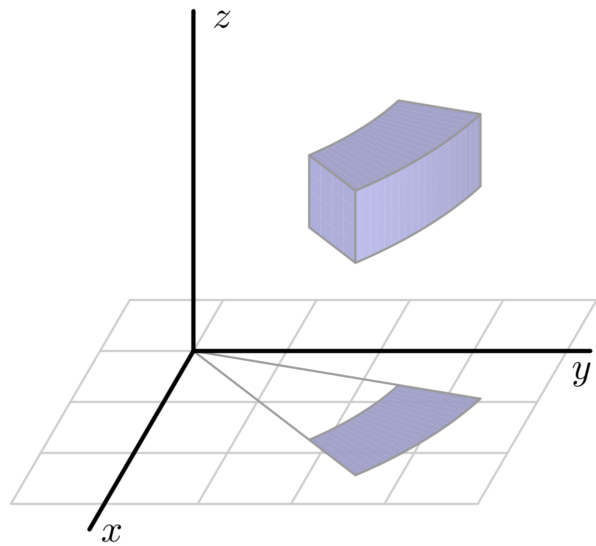


Figure 11.8.2 A cylindrical box.

a. Appropriately label Δr, $\Delta\theta$, and Δz in Figure 11.8.2.

b. Let ΔA be the area of the projection of the box, B, onto the xy-plane, which is shaded blue in Figure 11.8.2. Recall that we previously determined the area ΔA in polar coordinates in terms of r, Δr, and $\Delta\theta$. In light of the fact that we know ΔA and that z is the standard z coordinate from Cartesian coordinates, what is the volume ΔV in cylindrical coordinates?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.8.4. In this activity we work with triple integrals in cylindrical coordinates.

a. Let S be the solid bounded above by the graph of $z = x^2 + y^2$ and below by $z = 0$ on the unit disk in the xy-plane.

   i. The projection of the solid S onto the xy-plane is a disk. Describe this disk using polar coordinates.

   ii. Now describe the surfaces bounding the solid S using cylindrical coordinates.

   iii. Determine an iterated triple integral expression in cylindrical coordinates that gives the volume of S. You do not need to evaluate this integral.

b. Suppose the density of the cone defined by $r = 1 - z$, with $z \geq 0$, is given by $\delta(r, \theta, z) = z$. A picture of the cone is shown at left in Figure 11.8.3, and the projection of the cone onto the xy-plane in given at right in Figure 11.8.3. Set up an iterated integral in cylindrical coordinates that gives the mass of the cone. You do not need to evaluate this integral.

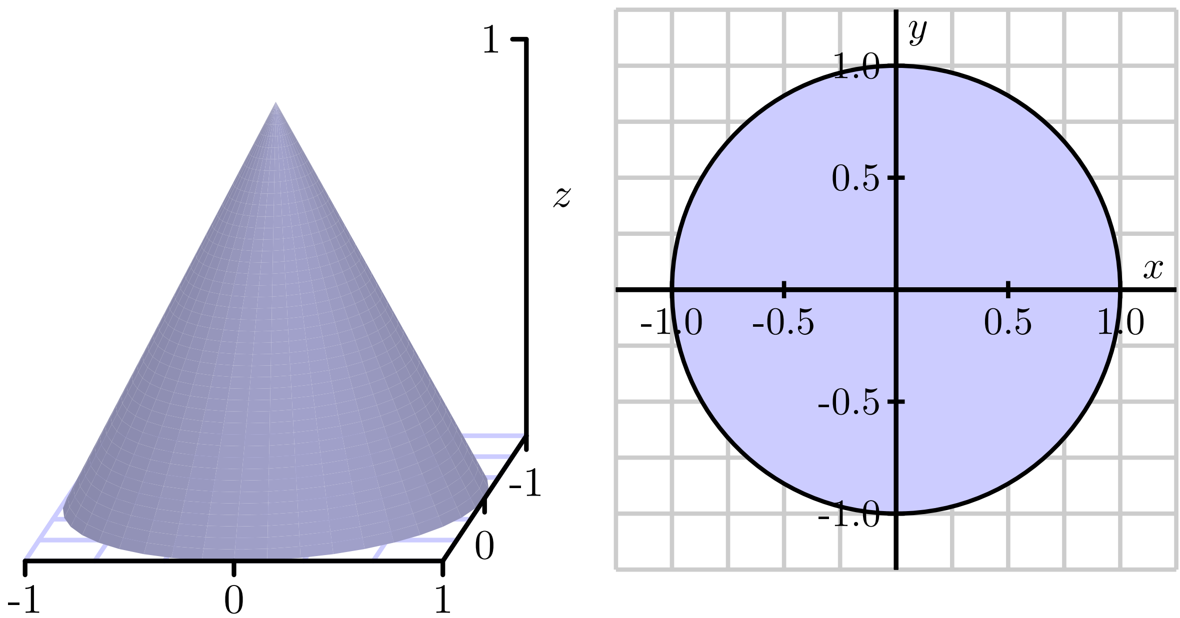


Figure 11.8.3 The cylindrical cone $r = 1 - z$ and its projection onto the xy-plane.

c. Determine an iterated integral expression in cylindrical coordinates whose value is the volume of the solid bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the cone $z = 4 - \sqrt{x^2 + y^2}$. A picture is shown in Figure 11.8.4. You do not need to evaluate this integral.

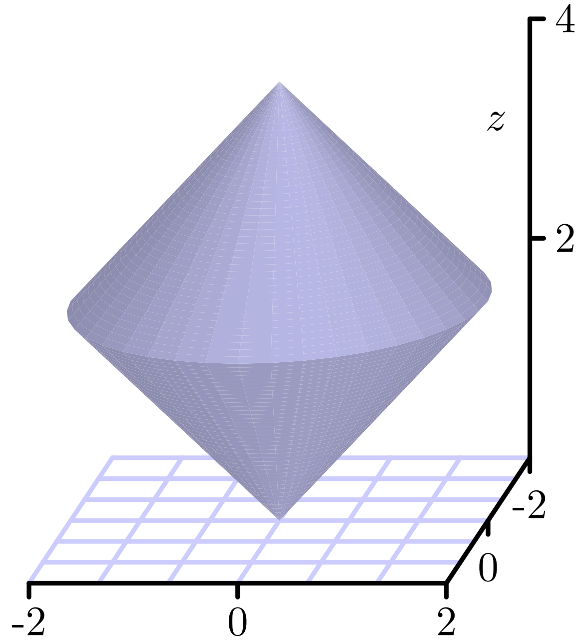


Figure 11.8.4 A solid bounded by the cones $z = \sqrt{x^2 + y^2}$ and $z = 4 - \sqrt{x^2 + y^2}$.

Activity 11.8.5. In this activity, we graph some surfaces using spherical coordinates. To improve your intuition and test your understanding, you should first think about what each graph should look like before you plot it using appropriate technology.

a. What familiar surface is described by the points in spherical coordinates with $\rho = 1$, $0 \leq \theta \leq 2\pi$, and $0 \leq \phi \leq \pi$? How does this particular example demonstrate the reason for the name of this coordinate system? What if we restrict ϕ to $0 \leq \phi \leq \frac{\pi}{2}$?

b. What familiar surface is described by the points in spherical coordinates with $\phi = \frac{\pi}{3}$, $0 \leq \rho \leq 1$, and $0 \leq \theta \leq 2\pi$?

c. What familiar shape is described by the points in spherical coordinates with $\theta = \frac{\pi}{6}$, $0 \leq \rho \leq 1$, and $0 \leq \phi \leq \pi$?

d. Plot the graph of $\rho = \theta$, for $0 \leq \phi \leq \pi$ and $0 \leq \theta \leq 2\pi$. How does the resulting surface appear?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.8.6. To find the volume element dV in spherical coordinates, we need to understand how to determine the volume of a spherical box of the form $\rho_1 \leq \rho \leq \rho_2$ (with $\Delta\rho = \rho_2 - \rho_1$), $\phi_1 \leq \phi \leq \phi_2$ (with $\Delta\phi = \phi_2 - \phi_1$), and $\theta_1 \leq \theta \leq \theta_2$ (with $\Delta\theta = \theta_2 - \theta_1$). An illustration of such a box is given at left in Figure 11.8.6. This spherical box is a bit more complicated than the cylindrical box we encountered earlier. In this situation, it is easier to approximate the volume ΔV than to compute it directly. Here we can approximate the volume ΔV of this spherical box with the volume of a Cartesian box whose sides have the lengths of the sides of this spherical box. In other words,

$$\Delta V \approx |PS|\ |\overset{\frown}{PR}|\ |\overset{\frown}{PQ}|,$$

where $|\overset{\frown}{PR}|$ denotes the length of the circular arc from P to R.

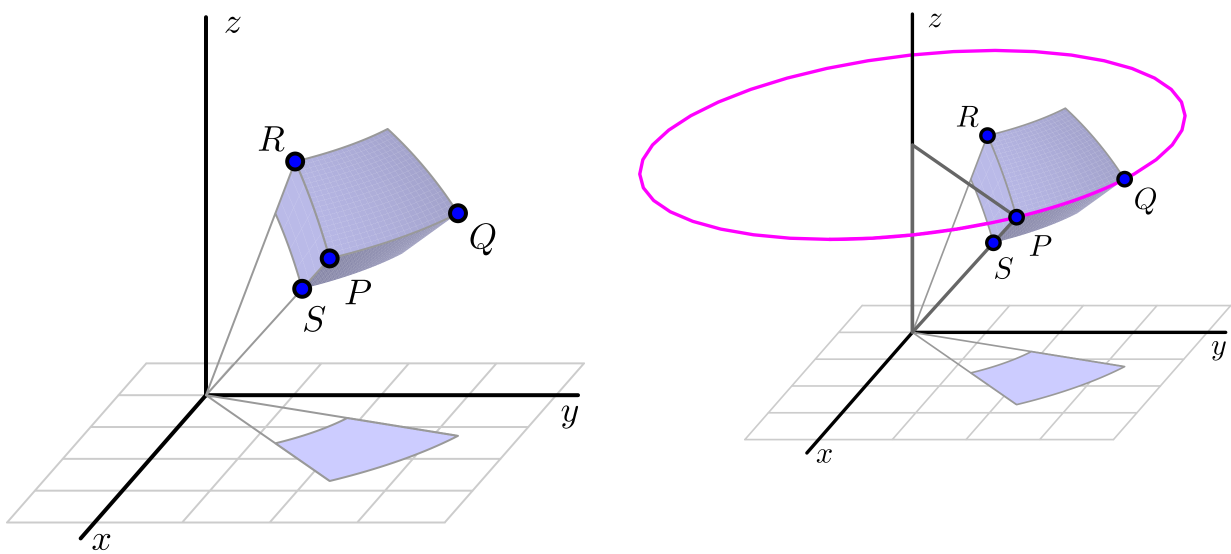


Figure 11.8.6 Left: A spherical box. Right: A spherical volume element.

a. What is the length $|PS|$ in terms of ρ?

b. What is the length of the arc $\overset{\frown}{PR}$? (Hint: The arc $\overset{\frown}{PR}$ is an arc of a circle of radius ρ_2, and arc length along a circle is the product of the angle measure (in radians) and the circle's radius.)

c. What is the length of the arc $\overset{\frown}{PQ}$? (Hint: The arc $\overset{\frown}{PQ}$ lies on a horizontal circle as illustrated at right in Figure 11.8.6. What is the radius of this circle?)

d. Use your work in (a), (b), and (c) to determine an approximation for ΔV in spherical coordinates.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.8.7. We can use spherical coordinates to help us more easily understand some natural geometric objects.

a. Recall that the sphere of radius a has spherical equation $\rho = a$. Set up and evaluate an iterated integral in spherical coordinates to determine the volume of a sphere of radius a.

b. Set up, but do not evaluate, an iterated integral expression in spherical coordinates whose value is the mass of the solid obtained by removing the cone $\phi = \frac{\pi}{4}$ from the sphere $\rho = 2$ if the density δ at the point (x, y, z) is $\delta(x, y, z) = \sqrt{x^2 + y^2 + z^2}$. An illustration of the solid is shown in Figure 11.8.7.

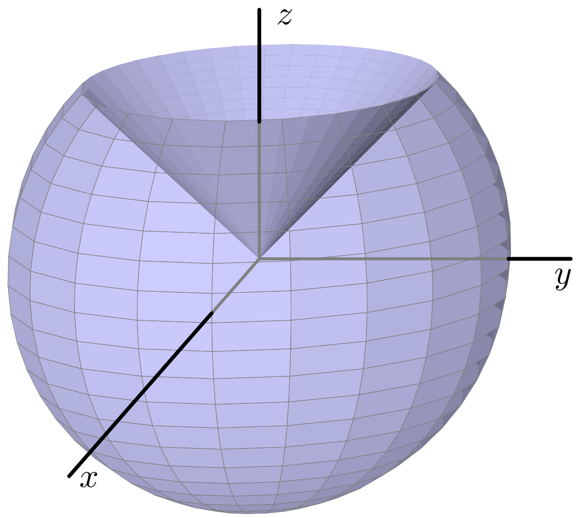


Figure 11.8.7 The solid cut from the sphere $\rho = 2$ by the cone $\phi = \frac{\pi}{4}$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

11.9 Change of Variables

Preview Activity 11.9.1. Consider the double integral

$$I = \iint_D x^2 + y^2 \, dA, \tag{11.9.1}$$

where D is the upper half of the unit disk.

a. i. Write the double integral I given in Equation (11.9.1) as an iterated integral in rectangular coordinates.

   ii. Write the double integral I given in Equation (11.9.1) as an iterated integral in polar coordinates.

b. When we write the double integral (11.9.1) as an iterated integral in polar coordinates we make a change of variables, namely

$$x = r\cos(\theta) \qquad \text{and} \qquad y = r\sin(\theta). \tag{11.9.2}$$

We also then have to change dA to $r\,dr\,d\theta$. This process also identifies a "polar rectangle" $[r_1, r_2] \times [\theta_1, \theta_2]$ with the original Cartesian rectangle, under the transformation[1] in Equation (11.9.2). The vertices of the polar rectangle are transformed into the vertices of a closed and bounded region in rectangular coordinates.

To work with a numerical example, let's now consider the polar rectangle P given by $[1, 2] \times [\frac{\pi}{6}, \frac{\pi}{4}]$, so that $r_1 = 1$, $r_2 = 2$, $\theta_1 = \frac{\pi}{6}$, and $\theta_2 = \frac{\pi}{4}$.

   i. Use the transformation determined by the equations in (11.9.2) to find the rectangular vertices that correspond to the polar vertices in the polar rectangle P. In other words, by substituting appropriate values of r and θ into the two equations in (11.9.2), find the values of the corresponding x and y coordinates for the vertices of the polar rectangle P. Label the point that corresponds to the polar vertex (r_1, θ_1) as (x_1, y_1), the point corresponding to the polar vertex (r_2, θ_1) as (x_2, y_2), the point corresponding to the polar vertex (r_1, θ_2) as (x_3, y_3), and the point corresponding to the polar vertex (r_2, θ_2) as (x_4, y_4).

   ii. Draw a picture of the figure in rectangular coordinates that has the points (x_1, y_1), (x_2, y_2), (x_3, y_3), and (x_4, y_4) as vertices. (Note carefully that because of the trigonometric functions in the transformation, this region will not look like a Cartesian rectangle.) What is the area of this region in rectangular coordinates? How does this area compare to the area of the original polar rectangle?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.9.2. Consider the change of variables

$$x = s + 2t \qquad \text{and} \qquad y = 2s + \sqrt{t}.$$

Let's see what happens to the rectangle $T = [0,1] \times [1,4]$ in the st-plane under this change of variable.

a. Draw a labeled picture of T in the st-plane.

b. Find the image of the st-vertex $(0,1)$ in the xy-plane. Likewise, find the respective images of the other three vertices of the rectangle T: $(0,4)$, $(1,1)$, and $(1,4)$.

c. In the xy-plane, draw a labeled picture of the image, T', of the original st-rectangle T. What appears to be the shape of the image, T'?

d. To transform an integral with a change of variables, we need to determine the area element dA for image of the transformed rectangle. Note that T' is not exactly a parallelogram since the equations that define the transformation are not linear. But we can approximate the area of T' with the area of a parallelogram. How would we find the area of a parallelogram that approximates the area of the xy-figure T'? (Hint: Remember what the cross product of two vectors tells us.)

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.9.3. Find the Jacobian when changing from rectangular to polar coordinates. That is, for the transformation given by $x = r\cos(\theta)$, $y = r\sin(\theta)$, determine a simplified expression for the quantity

$$\frac{\partial x}{\partial r}\frac{\partial y}{\partial \theta} - \frac{\partial x}{\partial \theta}\frac{\partial y}{\partial r}.$$

What do you observe about your result? How is this connected to our earlier work with double integrals in polar coordinates?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.9.4. Consider the problem of finding the area of the region D' defined by the ellipse $x^2 + \frac{y^2}{4} = 1$. Here we will make a change of variables so that the pre-image of the domain is a circle.

- Let $x(s,t) = s$ and $y(s,t) = 2t$. Explain why the pre-image of the original ellipse (which lies in the xy plane) is the circle $s^2 + t^2 = 1$ in the st-plane.
- Recall that the area of the ellipse D' is determined by the double integral $\iint_{D'} 1\,dA$. Explain why

  $$\iint_{D'} 1\,dA = \iint_{D} 2\,ds\,dt$$

  where D is the disk bounded by the circle $s^2 + t^2 = 1$. In particular, explain the source of the "2" in the st integral.
- Without evaluating any of the integrals present, explain why the area of the original elliptical region D' is 2π.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.9.5. Let D' be the region in the xy-plane bounded by the lines $y = 0$, $x = 0$, and $x + y = 1$. We will evaluate the double integral

$$\iint_{D'} \sqrt{x+y}(x-y)^2 \, dA \tag{11.9.4}$$

with a change of variables.

a. Sketch the region D' in the xy-plane.

b. We would like to make a substitution that makes the integrand easier to antidifferentiate. Let $s = x + y$ and $t = x - y$. Explain why this should make antidifferentiation easier by making the corresponding substitutions and writing the new integrand in terms of s and t.

c. Solve the equations $s = x + y$ and $t = x - y$ for x and y. (Doing so determines the standard form of the transformation, since we will have x as a function of s and t, and y as a function of s and t.)

d. To actually execute this change of variables, we need to know the st-region D that corresponds to the xy-region D'.

   i. What st equation corresponds to the xy equation $x + y = 1$?

   ii. What st equation corresponds to the xy equation $x = 0$?

   iii. What st equation corresponds to the xy equation $y = 0$?

   iv. Sketch the st region D that corresponds to the xy domain D'.

e. Make the change of variables indicated by $s = x + y$ and $t = x - y$ in the double integral (11.9.4) and set up an iterated integral in st variables whose value is the original given double integral. Finally, evaluate the iterated integral.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 11.9.6. Consider the solid S' defined by the inequalities $0 \leq x \leq 2$, $\frac{x}{2} \leq y \leq \frac{x}{2}+1$, and $0 \leq z \leq 6$. Consider the transformation defined by $s = \frac{x}{2}$, $t = \frac{x-2y}{2}$, and $u = \frac{z}{3}$. Let $f(x, y, x) = x - 2y + z$.

a. The transformation turns the solid S' in xyz-coordinates into a box S in stu-coordinates. Apply the transformation to the boundries of the solid S' to find stu-coordinatte descriptions of the box S.

b. Find the Jacobian $\frac{\partial(x,y,z)}{\partial(s,t,u)}$.

c. Use the transformation to perform a change of variables and evaluate $\iiint_{S'} f(x, y, z)\, dV$ by evaluating

$$\iiint_S f(x(s,t,u), y(s,t,u), z(s,t,u)) \left| \frac{\partial(x,y,z)}{\partial(s,t,u)} \right| ds\, dt\, du.$$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12 Vector Calculus

12.1 Vector Fields

Preview Activity 12.1.1. It's common when discussing weather to talk about the wind **speed**, but as any student who has gotten this far in the text will know, this nomenclature is imprecise. It's not terribly helpful to tell someone the wind is blowing at 10 $\frac{\text{km}}{\text{h}}$ without telling them the direction in which the wind is blowing. If you're trying to make a decision based on what the wind is doing, you need to know about the direction as well. For instance, if you are taking off in a hot air balloon, the wind direction will determine which direction the chase team should start going to keep track of you. Because of the swirling nature of wind, it makes sense to give the wind **velocity** at each point in a region (two-dimensional or three-dimensional).

(a) Suppose that given a point (x, y) in the plane, you know that the wind velocity at that point is given by the vector $\langle y, x\rangle$. For example, we'd then know that at the point $(1, -1)$, the wind velocity is $\langle -1, 1\rangle$. We will give the wind velocity as a function $\mathbf{F}$, where $\mathbf{F}(x, y) = \langle y, x\rangle$. In the table below, fill in the wind velocity vectors for the given points.

(x, y)	$(2, 1)$	$(0, 0)$	$(-1, 2)$	$(3, -1)$	$(-2, -1)$
$\mathbf{F}(x, y)$					

(b) Suppose that we associate the vector $\mathbf{G}(x, y) = -x\mathbf{j}$ to a point (x, y) in the plane. Complete the table below by giving the vector associated to each of the given points.

(x, y)	$(-2, 0)$	$(-1, 2)$	$(0, -2)$	$(2, 3)$	$(3, 2)$	$(-1, 0)$	$(1, 3)$
$\mathbf{G}(x, y)$							

(c) A table of values of these vector-valued functions is useful to understand the input vs. output nature of a vector field as a function, but perhaps even better is a method of visualizing the vector outputs. A good picture is worth a thousand words (or numbers). Returning to our analogy of the output vector for our vector field being wind velocity, if $\mathbf{F}(2, 1) = \langle 1, 2\rangle$, this means that at the location $(2, 1)$ the wind is moving in the direction given by $\langle 1, 2\rangle$. Thus, we draw the output vector $\langle 1, 2\rangle$ with its initial point at $(2, 1)$.

Using the first set of axes in Figure 12.1.1, plot the vectors $\mathbf{F}(x, y)$ for the five points in the table in part a. The example $\mathbf{F}(1, -1) = \langle -1, 1\rangle$ is drawn for you.

(d) Using the second set of axes in Figure 12.1.1, plot the vectors $\mathbf{G}(x, y)$ for the eight points in the table in part b.

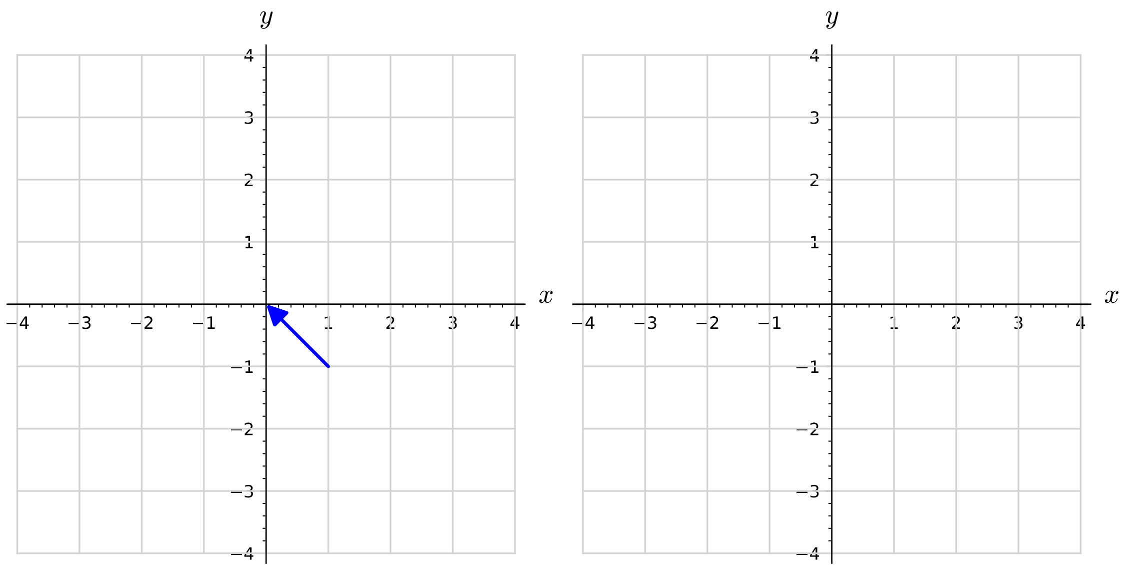


Figure 12.1.1 Axes for plotting some vectors from $\mathbf{F}(x,y)$ and $\mathbf{G}(x,y)$.

Activity 12.1.2. The plot in Figure 12.1.6 illustrates the vector field $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$.

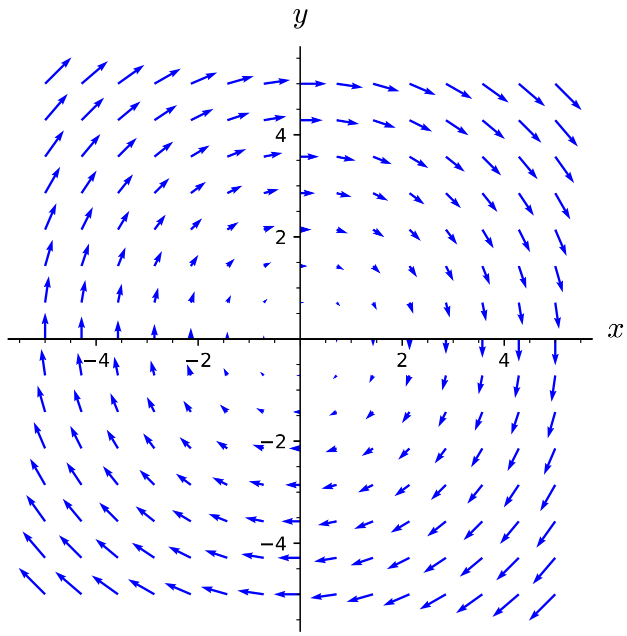


Figure 12.1.6 The vector field $y\mathbf{i} - x\mathbf{j}$

(a) Starting with one of the vectors near the point $(2, 0)$, sketch a curve that follows the direction of the vector field $\mathbf{F}$. To help visualize what you are doing, it may be useful to think of the vector field as the velocity vector field for some flowing water and that you are imagining tracing the path that a tiny particle inserted into the water would follow as the water moves it around.

(b) Repeat the previous step for at least two other starting points not on the curve you previously sketched.

(c) What shape do the curves you sketched in the previous two steps form?

(d) Verify that $\mathbf{F}(x, y)$ is orthogonal to $\langle x, y\rangle$.

(e) Calculate the gradient of the function $f(x, y) = x^2 + y^2$ and write a sentence comparing your result to the vector $x\mathbf{i} + y\mathbf{j}$.

(f) Write a sentence describing the geometric relationship between $\mathbf{F}(x, y)$ and a circle centered at the origin. What is the relationship between $|\mathbf{F}(x, y)|$ and the radius of that circle?

What I Learned

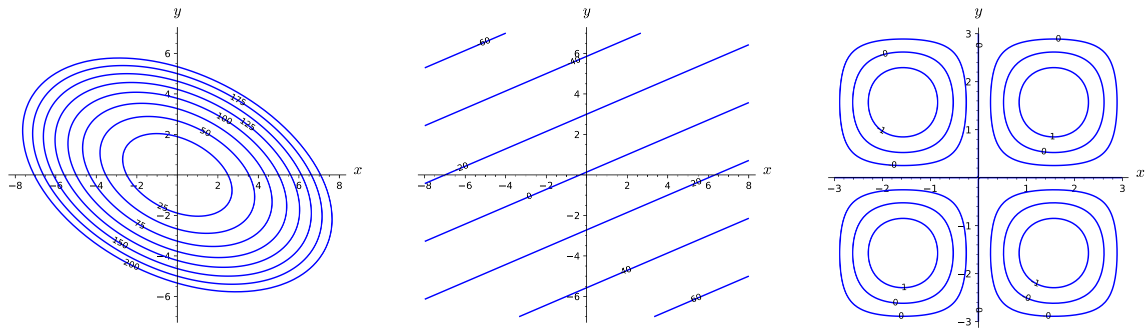
Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.1.3.

(a) In Figure 12.1.7 there are three sets of axes showing level curves for functions f, g, and h, respectively. Sketch at least six vectors in the gradient vector field for each function. In making your sketches, you don't have to worry about getting vector magnitudes precise, but you should ensure that the relative magnitudes (and directions) are correct for each function independently.



(a) f **(b)** g **(c)** h

Figure 12.1.7 Three sets of level curves

(b) Verify that $\mathbf{F}(x, y) = \langle 6xy, 3x^2 + 9\sqrt{y}\rangle$ is a gradient vector field by finding a function f such that $\nabla f(x, y) = \mathbf{F}(x, y)$. For reasons originating in physics, such a function f is called a **potential function** for the vector field $\mathbf{F}$.

(c) Is the function f found in part b unique? That is, can you find another function g such that $\nabla g(x, y) = \mathbf{F}(x, y)$ but $f \neq g$?

(d) Is the vector field $\mathbf{F}(x, y) = 6xy\mathbf{i} + (2x + 9\sqrt{y})\mathbf{j}$ a gradient vector field? Why or why not?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.2 The Idea of a Line Integral

Preview Activity 12.2.1. Recall from Section 9.3 that the work done by a force $\mathbf{F}$ on an object that moves with displacement vector $\mathbf{v}$ is $\mathbf{F} \cdot \mathbf{v}$. In this Preview Activity, we consider the work done by wind on a helicopter at various stages of its journey.

(a) Our intrepid pilot flies for some time and finds that they are 30 km from where they started at a heading of 20° degrees east of due north. During this portion of the trip, the wind is exerting a force of 100 N on the helicopter in the due east direction. Find the work the wind has done on the helicopter during the flight.

(b) Our pilot sees a storm ahead and changes their direction. Some time later, the pilot determines that they are 25 km due north of where they previously checked thier position. The wind is still exerting a force on the helicopter of 100 N in the due east direction. Find the work done by the wind on the helicopter during the second part of the flight.

(c) Find the helicopters's displacement from its original position after the first two parts of its flight and use that to find the work done by the wind on the helicopter during the first two parts of flight.

(d) How does your answer to part 12.2.1.c connect to the answers to part 12.2.1.a and part 12.2.1.b?

(e) In order to get further away from the storm, the pilot turns and flies 45° west of due north for 50 km. The storm the pilot was avoiding has caused the wind to change as well. For this portion of the flight, the wind is exerting a force on the helicopter of 125 N in the south direction. Find the work done by the wind on the helicopter during this part of the flight.

(f) Explain why you cannot take the total displacement of the three parts of the helicopter flight and calculate the total work done by the wind on the helicopter.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.2.2. For each curve below, find a parametrization of the curve. Ensure that each curve's orientation matches the one specified.

(a) The line segment in $\mathbb{R}^3$ from $(0, 1, -2)$ to $(3, -1, 2)$.

(b) The line segment in $\mathbb{R}^3$ from $(3, -1, 2)$ to $(0, 1, -2)$.

(c) The circle of radius 3 (in $\mathbb{R}^2$) centered at the origin, beginning at the point $(0, -3)$ and proceeding clockwise around the circle.

(d) In $\mathbb{R}^2$, the portion of the parabola $y^2 = x$ from the point $(4, 2)$ to the point $(1, -1)$.

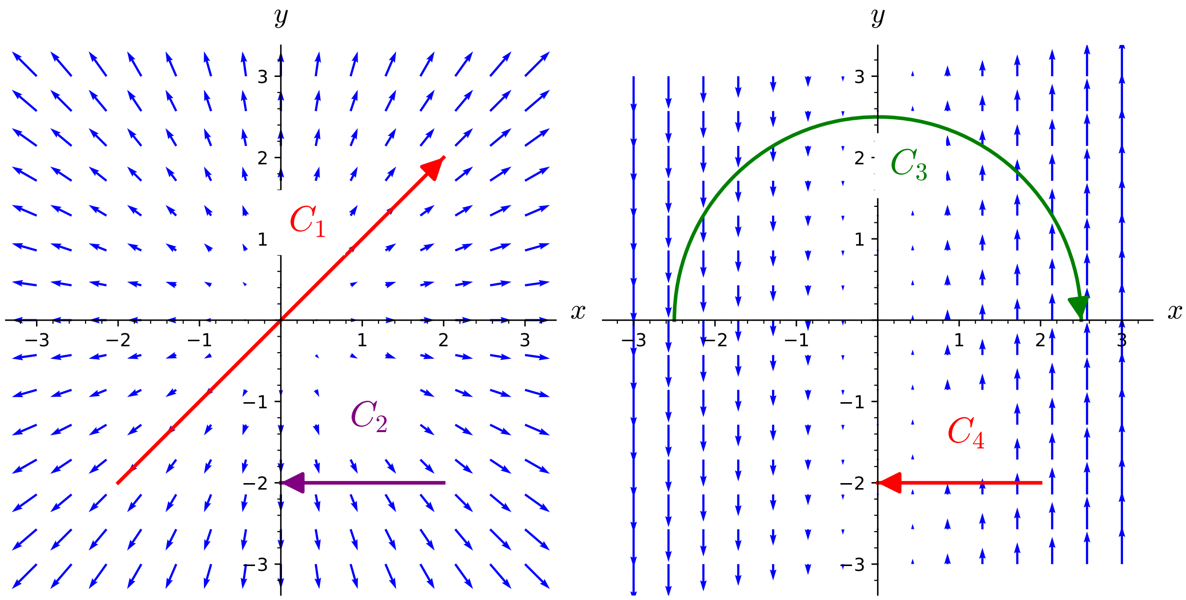
What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.2.3. Shown in Figure 12.2.10 are two vector fields, **F** and **G** and four oriented curves, as labeled in the plots. For each of the line integrals below, determine if its value should be positive, negative, or zero. Do this by thinking about if the vector field is helping or hindering a particle moving along the oriented curve, rather than by doing calculations.



(a) A plot of **F** with paths C_1 and C_2

(b) A plot of **G** with paths C_3 and C_4

Figure 12.2.10 Vector fields and oriented curves

(a) $\displaystyle\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$

(b) $\displaystyle\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$

(c) $\displaystyle\int_{C_3} \mathbf{G} \cdot d\mathbf{r}$

(d) $\displaystyle\int_{C_4} \mathbf{G} \cdot d\mathbf{r}$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.2.4. Figure 12.2.15 shows a vector field $\mathbf{F}$ as well as six oriented curves, as labeled in the plot.

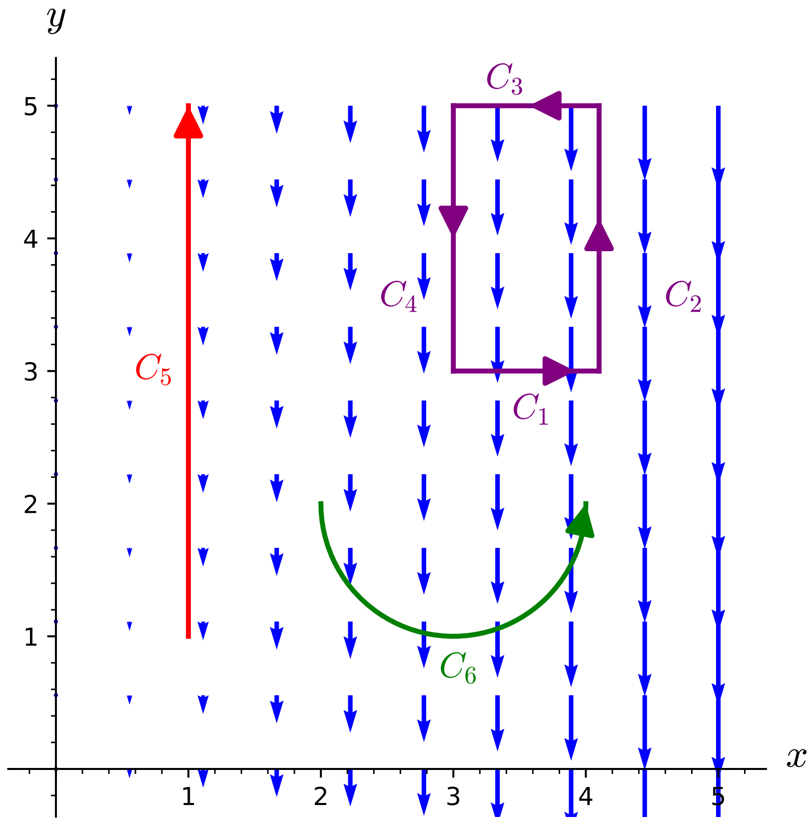


Figure 12.2.15 A vector field $\mathbf{F}$ and six oriented curves.

(a) Is $\int_{C_6} \mathbf{F} \cdot d\mathbf{r}$ positive, negative, or zero? Explain.

(b) Let $C = C_1 + C_2 + C_3 + C_4$. Determine if $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$ is positive, negative, or zero.

(c) Order the line integrals below from smallest to largest.

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} \quad \int_{C_2} \mathbf{F} \cdot d\mathbf{r} \quad \int_{C_3} \mathbf{F} \cdot d\mathbf{r} \quad \int_{C_4} \mathbf{F} \cdot d\mathbf{r} \quad \int_{C_5} \mathbf{F} \cdot d\mathbf{r}$$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.2.5. Determine if the circulation of the vector field around each of the closed curves shown in Figure 12.2.16 is positive, negative, or zero.

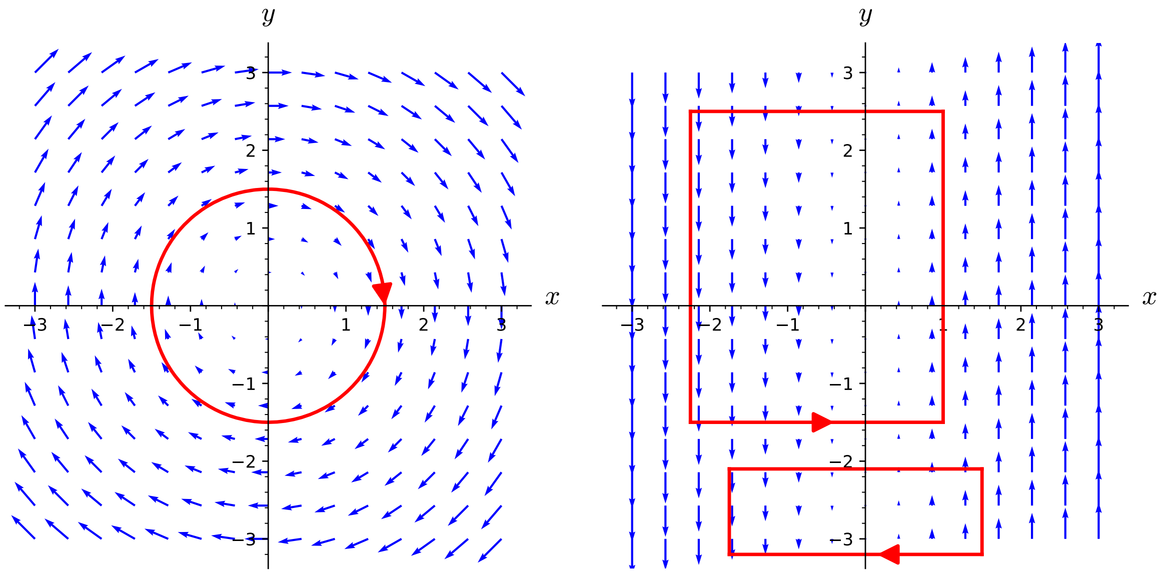


Figure 12.2.16 Vector fields and closed curves

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.3 Using Parametrizations to Calculate Line Integrals

Preview Activity 12.3.1. Let $\mathbf{F} = \langle xy, y^2 \rangle$, let C_1 be the line segment from $(1,1)$ to $(4,1)$, let C_2 be the line segment from $(4,1)$ to $(4,3)$, and let C_3 be the line segment from $(1,1)$ to $(4,3)$. Also let $C = C_1 + C_2$. This vector field and the curves are shown in Figure 12.3.1.

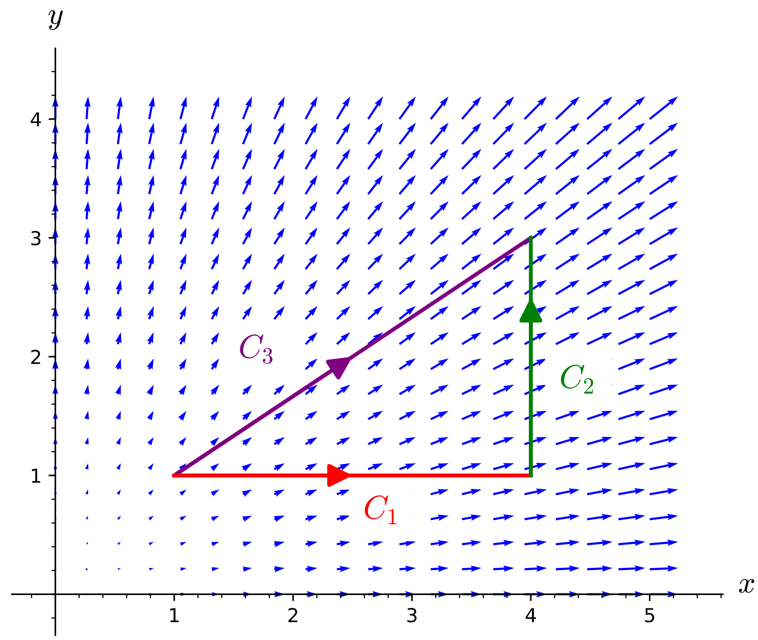


Figure 12.3.1 A vector field $\mathbf{F}$ and three oriented curves

(a) Every point along C_1 has $y = 1$. Therefore, along C_1, the vector field $\mathbf{F}$ can be viewed purely as a function of x. In particular, along C_1, we have $\mathbf{F}(x, 1) = \langle x, 1 \rangle$. Since every point along C_2 has the same x-value, write $\mathbf{F}$ as a function of y only (for the points on C_2).

(b) Recall that $d\mathbf{r} \approx \Delta\mathbf{r}$, and along C_1, we have that $\Delta\mathbf{r} = \Delta x\mathbf{i} \approx dx\mathbf{i}$. Thus, $d\mathbf{r} = \langle dx, 0 \rangle$. We know that along C_1, $\mathbf{F} = \langle x, 1 \rangle$.

- **(i)** Write $\mathbf{F} \cdot d\mathbf{r}$ along C_1 without using a dot product.
- **(ii)** What interval of x-values describes C_1?
- **(iii)** Write $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$ as an integral of the form $\int_a^b f(x)\,dx$ and evaluate the integral.

(c) Use an analogous approach to write $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ as an integral of the form $\int_c^d g(y)\,dy$ and evaluate the integral.

(d) Use the previous parts and a property of line integrals to calculate $\int_C \mathbf{F} \cdot d\mathbf{r}$ without having to evaluate any additional integrals.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.3.2. Is $\mathbf{F} = \langle xy, y^2 \rangle$ a gradient vector field? Why or why not?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.3.3.

(a) Find the work done by the vector field $\mathbf{F}(x, y, z) = 6x^2z\mathbf{i} + 3y^2\mathbf{j} + x\mathbf{k}$ on a particle that moves from the point $(3, 0, 0)$ to the point $(3, 0, 6\pi)$ along the helix given by $\mathbf{r}(t) = \langle 3\cos(t), 3\sin(t), t\rangle$.

(b) Let $\mathbf{F}(x, y) = \langle 0, x\rangle$. Let C be the closed curve consisting of the top half of the circle of radius 2 centered at the origin and the portion of the x-axis from $(2, 0)$ to $(-2, 0)$, oriented clockwise. Find the circulation of $\mathbf{F}$ around C.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.3.4. Let $\mathbf{F}(x, y) = \langle y^2, 2xy + 3\rangle$.

(a) Let C_1 be the portion of the graph of $y = 2x^3 + 3x^2 - 12x - 15$ from $(-2, 5)$ to $(3, 30)$. Calculate $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$.

(b) Let C_2 be the line segment from $(-2, 5)$ to $(3, 30)$. Calculate $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$.

(c) Let C_3 be the circle of radius 3 centered at the origin, oriented counterclockwise. Calculate $\oint_{C_3} \mathbf{F} \cdot d\mathbf{r}$.

(d) To connect the previous parts of this activity, use a graphing utility to plot the curves C_1 and C_2 on the same axes.

- **(i)** What type of curve is $C_1 - C_2$?
- **(ii)** What is the value of $\oint_{C_1 - C_2} \mathbf{F} \cdot d\mathbf{r}$?
- **(iii)** What does your answer to part c allow you to say about the value of the line integral of $\mathbf{F}$ along the top half of C_3 compared to the line integral of $\mathbf{F}$ from $(3, 0)$ to $(-3, 0)$ along the bottom half of the circle of radius 3 centered at the origin?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.3.5.

(a) The typical parametrization of the line segment from $(0,1)$ to $(3,3)$ (the oriented curve C_3 in Example 12.3.9) is $\mathbf{r}(t) = \langle 3t, 1+2t \rangle$ where $0 \le t \le 1$. Use this parametrization to calculate $\int_{C_3} \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = x\mathbf{i}$ and compare your answer to the result of Example 12.3.9.

(b) Calculate $\int_C \langle (3xy + e^z), x^2, (4z + xe^z) \rangle \cdot d\mathbf{r}$ where C is the oriented curve consisting of the line segment from the origin to $(1,1,1)$ followed by the line segment from $(1,1,1)$ to $(0,0,2)$.

(c) Calculate $\int_{C'} \langle 3xy + e^z, x^2, 4z + xe^z \rangle \cdot d\mathbf{r}$ where C_3 is the line segment from $(0,0,0)$ to $(0,0,2)$.

(d) Is the vector field you considered in the previous two parts a gradient vector field? Why or why not? How does this compare to the vector field $\mathbf{F}$ of Activity 12.3.4?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.4 Path-Independent Vector Fields and the Fundamental Theorem of Calculus for Line Integrals

Preview Activity 12.4.1. In Activity 12.3.4, we considered the vector field $\mathbf{F}(x, y) = \langle y^2, 2xy + 3\rangle$ and two different oriented curves from $(-2, 5)$ to $(3, 30)$. We found that the value of the line integral of $\mathbf{F}$ was the same along those two oriented curves.

(a) Verify that $\mathbf{F}(x, y) = \langle y^2, 2xy + 3\rangle$ is a gradient vector field by showing that $\mathbf{F} = \nabla f$ for the function $f(x, y) = xy^2 + 3y$.

(b) Calculate the change in the output of the scalar function f over the curves C_1 and C_2. In other words, what is the difference in the output of f at the start of the curve and the end of the curve? How does this value compare to the value of the line integral $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$ you found in Activity 12.3.4?

(c) Let C_3 be the line segment from $(1, 1)$ to $(3, 4)$. Calculate $\int_{C_3} \mathbf{F} \cdot d\mathbf{r}$ as well as $f(3, 4) - f(1, 1)$. Write a sentence that compares your answer to this part to your result for part 12.4.1.b.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.4.2. Calculate each of the following line integrals.

(a) $\int_C \nabla f \cdot d\mathbf{r}$ if $f(x, y) = 3xy^2 - \sin(x) + e^y$ and C is the top half of the unit circle oriented from $(-1, 0)$ to $(1, 0)$.

(b) $\int_C \nabla g \cdot d\mathbf{r}$ if $g(x, y, z) = xz^2 - 5y^3 \cos(z) + 6$ and C is the portion of the helix $\mathbf{r}(t) = \langle 5\cos(t), 5\sin(t), 3t\rangle$ from $(5, 0, 0)$ to $(0, 5, 9\pi/2)$.

(c) $\int_C \nabla h \cdot d\mathbf{r}$ if $h(x, y, z) = 3y^2 e^{y^3} - 5x\sin(x^3 z) + z^2$ and C is the curve consisting of the line segment from $(0, 0, 0)$ to $(1, 1, 1)$, followed by the line segment from $(1, 1, 1)$ to $(-1, 3, -2)$, followed by the line segment from $(-1, 3, -2)$ to $(0, 0, 10)$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.4.3. Let $\mathbf{G}(x, y, z) = \langle 3e^{y^2} + z\sin(x), 6xye^{y^2} - z, 3z^2 - y - \cos(x)\rangle$ and $\mathbf{H}(x, y, z) = \langle 3x^2y, x^3 + 2yz^3, xz + 3y^2z^2\rangle$.

(a) If $\mathbf{G}$ and $\mathbf{H}$ are to be gradient vector fields, then there are functions g and h for which $\mathbf{G} = \nabla g$ and $\mathbf{H} = \nabla h$. If such functions g and h exist, what would g_y, g_z, h_x, h_y, and h_z be?

(b) Let $g_1(x, y, z) = 3xe^{y^2} + xyz - z\sin(x)$. Calculate $\partial g_1/\partial x$. Could g_1 be a potential function for the vector field $\mathbf{G}$?

(c) Find a function g so that $\partial g/\partial x = 3e^{y^2} + z\sin(x)$. Find a function h so that $\partial h/\partial x = 3x^2y$.

(d) When finding the most general anti-derivative for a function of one variable, we add a constant of integration (usually denoted by C) to capture the fact that any constant will become 0 through differentiation.

(i) When taking the partial derivative with respect to x of a function of x, y, and z, what variables can appear in terms that become 0 in the partial derivative because they are treated as constants?

(ii) What does this tell you should be added to g and h in the previous part to make them the most general possible functions with the desired partial derivatives with respect to x?

(e) Now calculate $\partial g/\partial y$ and $\partial h/\partial y$ based on your choices for part 12.4.3.c. Write a few sentences to explain why this tells you that we must have

$$g(x, y, z) = 3xe^{y^2} - z\cos(x) - yz + m_1(z)$$

and

$$h(x, y, z) = x^3y + y^2z^3 + m_2(z)$$

for some functions m_1 and m_2 depending only on z.

(f) Calculate $\frac{\partial g}{\partial z}$ and $\frac{\partial h}{\partial z}$ for the functions in the part above. Notice that m_1 and m_2 are functions of z alone, so taking a partial derivative with respect to z is the same as taking an ordinary derivative, and thus you may use the notation $m_1'(z)$ and $m_2'(z)$.

(g) Explain why $\mathbf{G}$ is a gradient vector field but $\mathbf{H}$ is not a gradient vector field. Find a potential function for $\mathbf{G}$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.4.4. Calculate each of the following line integrals.

(a) $\int_C \mathbf{F} \cdot d\mathbf{r}$ if $\mathbf{F}(x, y) = \langle 2x, 2y \rangle$ and C is the line segment from $(1, 2)$ to $(-1, 0)$.

(b) $\int_C \mathbf{G} \cdot d\mathbf{r}$ if $\mathbf{G}(x, y) = \langle 4x^3 - 12y\cos(xy), 9y^2 - 12x\cos(xy) \rangle$ and C is the portion of the unit circle from $(0, -1)$ to $(0, 1)$.

(c) $\int_C \mathbf{H} \cdot d\mathbf{r}$ if $\mathbf{H}(x, y, z) = \langle H_1, H_2, H_3 \rangle$ with

$$\begin{aligned}
H_1(x, y, z) &= e^{z^2} + 2xy^3z + \cos(x) - y^3\sin(x) \\
H_2(x, y, z) &= 2ye^{y^2} + 3x^2y^2z + 3y^2z^2 + 3y^2\cos(x) \\
H_3(x, y, z) &= x^2y^3 + 2xze^{z^2} + 2y^3z - 4z^3
\end{aligned}$$

and C is the curve consisting of the line segment from $(1, 1, 1)$ to $(3, 0, 3)$, followed by the line segment from $(3, 0, 3)$ to $(1, 5, -1)$, followed by the line segment from $(1, 5, -1)$ to $(0, 0, 0)$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.4.5. Suppose that $\mathbf{F}$ is a continuous path-independent vector field (in $\mathbb{R}^2$ or $\mathbb{R}^3$) on some region D.

(a) Let P and Q be points in D and let C_1 and C_2 be oriented curves from P to Q. What can you say about $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$ and $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$?

(b) Let $C = C_1 - C_2$. Explain why C is a closed curve.

(c) Calculate $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

(d) Write a sentence that summarizes what we can conclude about line integrals of $\mathbf{F}$ at this point in the activity.

(e) Now let us suppose that $\mathbf{G}$ is a continuous vector field on a region D for which $\oint_C \mathbf{G} \cdot d\mathbf{r} = 0$ for all closed curves C. Pick two points P and Q in D. Let C_1 and C_2 be oriented curves from P to Q. What type of curve is $C = C_1 - C_2$?

(f) What is $\oint_C \mathbf{G} \cdot d\mathbf{r}$? Why?

(g) What does that tell you about the relationship between $\int_{C_1} \mathbf{G} \cdot d\mathbf{r}$ and $\int_{C_2} \mathbf{G} \cdot d\mathbf{r}$?

(h) Explain why this shows that $\mathbf{G}$ is path-independent.

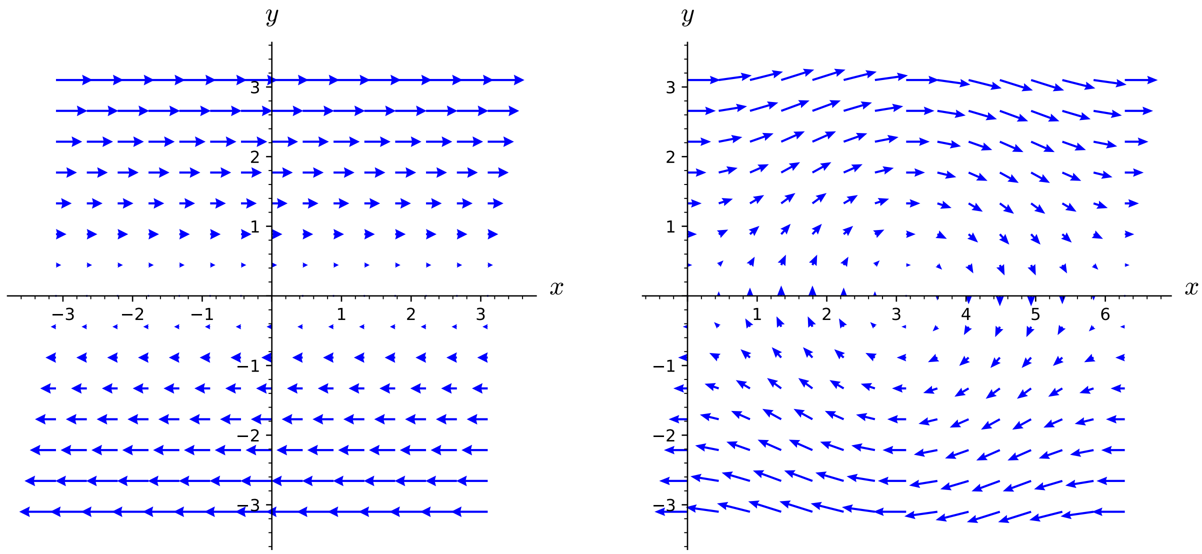
What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.4.6. Explain why neither of the vector fields in Figure 12.4.4 is path-independent.



(a) F **(b) G**

Figure 12.4.4 Two vector fields that are not path-independent.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.4.7. Let $\mathbf{F} = \langle F_1, F_2\rangle$ be a continuous, path-independent vector field on an open, path-connected region D. We will assume that D is in $\mathbb{R}^2$ and $\mathbf{F}$ is a two-dimensional vector field, but the ideas below generalize completely to $\mathbb{R}^3$. We want to define a function f on D by using the vector field $\mathbf{F}$ and line integrals, much like the Second Fundamental Theorem of Calculus allows us to define an antiderivative of a continuous function using a definite integral. To that end, we assign $f(x_0, y_0)$ an arbitrary value. (Setting $f(x_0, y_0) = 0$ is probably convenient, but we won't explicitly tie our hands. Just assume that $f(x_0, y_0)$ is defined to be some number.) Now for any other point (x, y) in D, define

$$f(x, y) = f(x_0, y_0) + \int_C \mathbf{F} \cdot d\mathbf{r},$$

where C is any oriented path from (x_0, y_0) to (x, y). Since D is path-connected, such an oriented path must exist. Since $\mathbf{F}$ is path-independent, f is well-defined. If different paths from (x_0, y_0) to (x, y) gave different values for the line integral, then we would not be sure what $f(x, y)$ really is.
To better understand this mysterious function f we've now defined, let's start looking at its partial derivatives.

(a) Since D is open, there is a disc (perhaps very small) surrounding (x, y) that is contained in D, so fix a point (a, b) in that disc. Since D is path-connected, there is a path C_1 from (x_0, y_0) to (a, b). Let C_y be the line segment from (a, b) to (a, y) and let C_x be the line segment from (a, y) to (x, y). (See Figure 12.4.6.) Rewrite $f(x, y)$ as a sum of $f(x_0, y_0)$ and line integrals along C_1, C_y, and C_x.

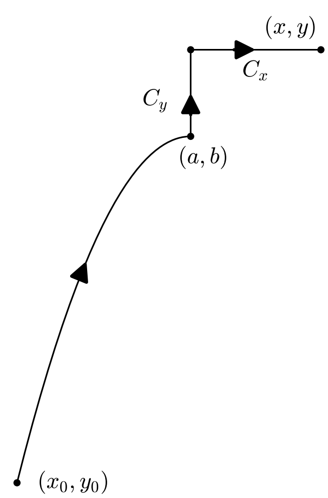


Figure 12.4.6 A piecewise smooth oriented curve from (x_0, y_0) to (x, y).

(b) Notice that we can parametrize C_y by $\langle a, t\rangle$ for $b \le t \le y$. Find a similar parametrization for C_x.

(c) Use the parametrization from above to write $\int_{C_y} \mathbf{F} \cdot d\mathbf{r}$ and $\int_{C_x} \mathbf{F} \cdot d\mathbf{r}$ as single variable integrals in the manner of Section 12.3. Use the fact that $\mathbf{F}(x, y) = \langle F_1(x, y), F_2(x, y)\rangle$ to express your integrals in terms of F_1 and F_2 without any dot products.

(d) Rewrite your expression for $f(x, y)$ using a line integral along C_1 and the single variable integrals above.

(e) Notice that your expression for $f(x, y)$ from the previous part only depends on x as the upper limit of an single variable integral. Use the Second Fundamental Theorem of Calculus to calculate $f_x(x, y)$.

(f) To calculate $f_y(x, y)$, we continue to consider a path C_1 from (x_0, y_0) to (a, b), but now let L_x be the line segment from (a, b) to (x, b) and let L_y be the line segment from (x, b) to (y, b). Modify the process you used to find $f_x(x, y)$ to find $f_y(x, y)$.

(g) What can you conclude about the relationship between ∇f and $\mathbf{F}$? What does this tell you about $\mathbf{F}$ beyond that it is path-independent and continuous?

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.5 Line Integrals of Scalar Functions

Preview Activity 12.5.1. In order to pay for tutition, you take a job driving a mining machine that collects a very valuable mineral called copium. Copium is only produced on the surface and is mined by scooping up the soil at the front of your machine, so the amount of copium ore collected depends on the density of the ore and the distance driven by the mining machine. The plot of land you are mining has been surveyed for the density of copium ore and is presented in the contour plot below.

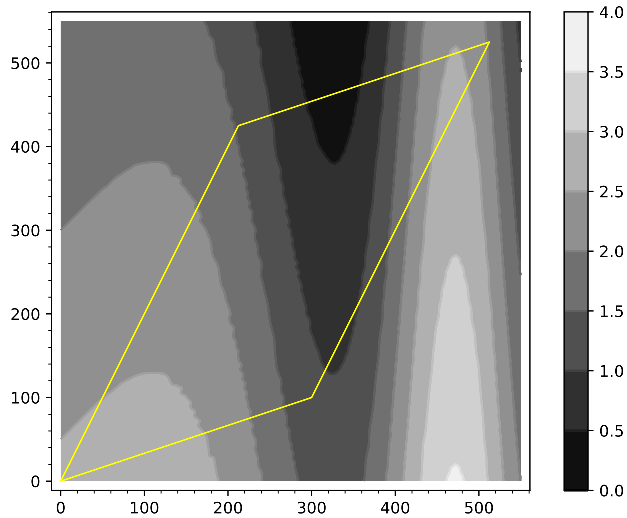


Figure 12.5.1 A plot of land with density of copium deposits

(a) Estimate the amount of copium that would be mined from driving along the left side of plot. You should write a few sentences about how you got your estimate based on the copium density and length of the path. (Did you use more than one piece?)

(b) Estimate the amount of copium that would be mined by driving along the entire outer edge of the plot. You should write a few sentences about how you got your estimate based on the copium density and length of the paths.

(c) Estimate the amount of copium that would be mined from the scraping the curved path shown below. You should use at least 3 segments in your estimate. You should write a few sentences about how you got your estimate based on the copium density and length of the paths.

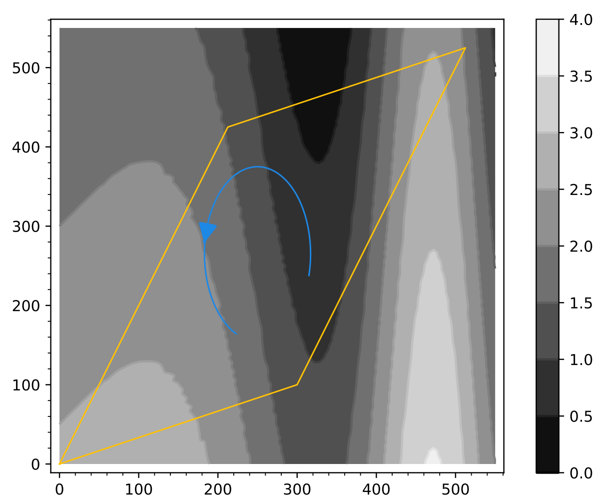


Figure 12.5.2 A plot of land with copium Density and a path plotted in blue

(d) Estimate the amount of copium that would be mined from the scraping the curved path shown below. You should use at least 3 segments in your estimate. You should also explain how and why your answer to this question is different or similar to the previous task.

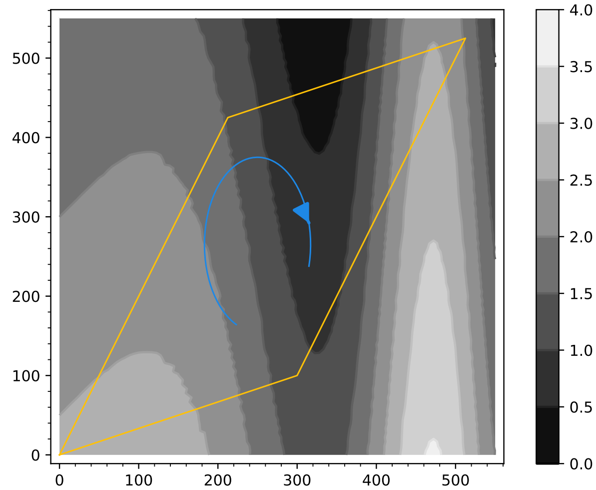


Figure 12.5.3 A plot of land with copium density and a path plotted in blue

Activity 12.5.2. In this activity, we will be making sense of scalar line integrals by examining a few common functions and justifying whether the scalar line integrals given are positive, negative, or zero. Let the functions f_1, f_2, f_3, and f_4 be defined as

- $f_1(x, y, z) = y$
- $f_2(x, y, z) = z$
- $f_3(x, y, z) = x^2$
- $f_4(x, y, z) = x - y$

(a) For each of the paths given below, sketch (in either 2D or 3D) the curve and label at least three points on the curve including the end points (if they exist).

(i) C_1 is the part of the unit circle in the xy-plane centered at the origin that is above the line $y = -x$.

(ii) C_2 is the part of the curve at the intersection of the cylinder given by $x^2 + y^2 = 1$ and the plane $z = x$ such that $y \geq -x$; You may want to consider the circle that is the intersection of $x^2 + y^2 = 1$ and $z = x$, then think about which half of this circle satisfies the inequality $y \geq -x$

(iii) C_3 is the part of the helix given by $\mathbf{r}(t) = \langle \cos(t), \sin(t), \frac{t}{2\pi} \rangle$ with $t \in [0, \pi]$

(b) For each of the functions f_1, f_2, and f_3 defined above, state whether $\int_{C_1} f_i \, ds$ is positive, negative, or zero. Be sure to justify your answer in terms of the function being integrated *and* the particulars of the curve of integration.

(c) For each of the functions f_1, f_2, and f_3, defined above, state whether $\int_{C_2} f_i \, ds$ is positive, negative, or zero. Be sure to justify your answer in terms of the function being integrated *and* the particulars of the curve of integration.

(d) For each of the functions f_1, f_2, and f_3, defined above, state whether $\int_{C_3} f_i \, ds$ is positive, negative, or zero. Be sure to justify your answer in terms of the function being integrated *and* the particulars of the curve of integration.

(e) For the function f_4, defined above, state each of the following integrals is positive, negative, or zero. Be sure to justify your answer in terms of the function being integrated *and* the particulars of the curve of integration. You should consider which parts of the curve being integrated will have positive/negative/zero output for the function f_4.

$$\int_{C_1} f_4 \, ds$$

$$\int_{C_2} f_4 \, ds$$

$$\int_{C_3} f_4 \, ds,$$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.5.3. In this activity, we will examine why we must be careful when using symmetry to make arguments about scalar line integral. Let C_1 and C_2 be the paths shown in Figure 12.5.16. We will consider the function $f(x,y) = x$ for this activity.

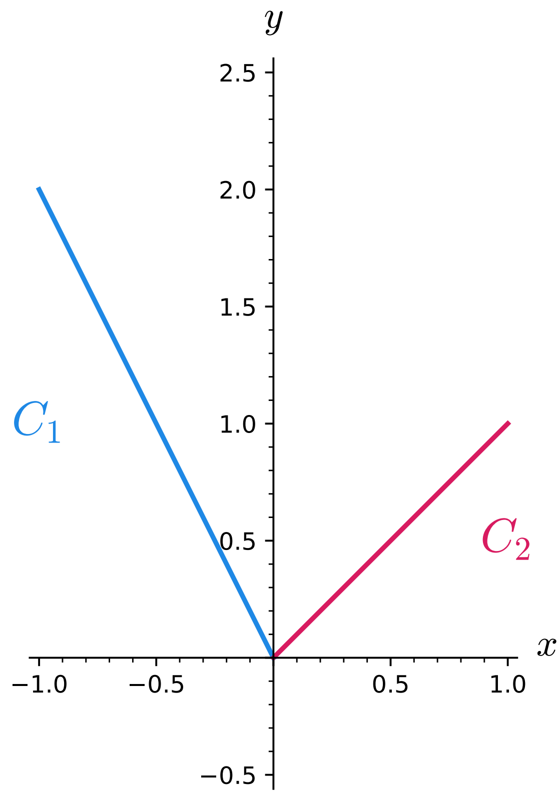


Figure 12.5.16 A plot of paths C_1 and C_2

(a) Parameterize C_1 and C_2 as $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$. (It is fine to have $0 \leq t \leq 1$ for both of your parameterizations.)

(b) Use Theorem 12.5.11 to compute $\int_{C_1} f\, ds$ and $\int_{C_2} f\, ds$.

(c) As with line integrals of vector fields, we have that $\int_{C_1+C_2} f\, ds = \int_{C_1} f\, ds + \int_{C_2} f\, ds$. Use this property to compute $\int_{C_1+C_2} f\, ds$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.5.4 Explaining Properties of Scalar Line Integrals.. In this activity, we will be explaining each of the Properties from Properties of Scalar Line Integrals in the context of our copium mining analogy from Preview Activity 12.5.1. Remember that the curve in our scalar line integral coresponds to the path the mining rig will take and the function in the scalar line integral measures the density of copium at that point on the surface.

(a) Explain in your own words what $\int_C f\,ds$ means in the copium analogy and what exactly would be measured by this scalar line integral.

(b) Explain in your own words what $\int_C (kf)\,ds = k\int_C f\,ds$ means in the copium analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

(c) Explain in your own words what $\int_C (f+g)\,ds = \int_C f\,ds + \int_C g\,ds$ means in the copium analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

(d) Explain in your own words what $\int_{-C} f\,ds = \int_C f\,ds$ means in the copium analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

(e) Explain in your own words what $\int_{C_1+C_2} f\,ds = \int_{C_1} f\,ds + \int_{C_2} f\,ds$ means in the copium analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.6 The Divergence of a Vector Field

Preview Activity 12.6.1. In this preview activity, we will look at several two-dimensional vector fields and try to assess when the vector field has increased or decreased in strength over a given region. We begin with graphs of the three vector fields, **F**, **G**, and **H**. Parts a, b, and c ask you to answer the same three questions about the vector field and square illustrated in each of the figures. Part d asks you to think further about the third vector field.

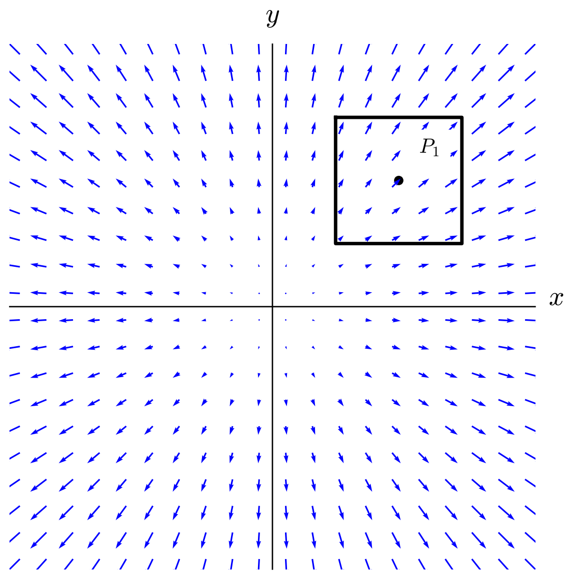


Figure 12.6.2 Vector Field **F**

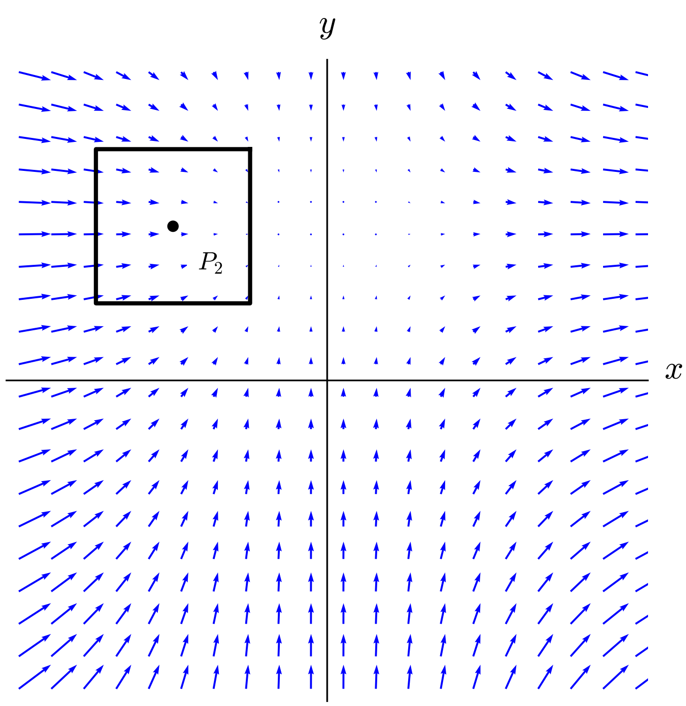


Figure 12.6.3 Vector Field **G**

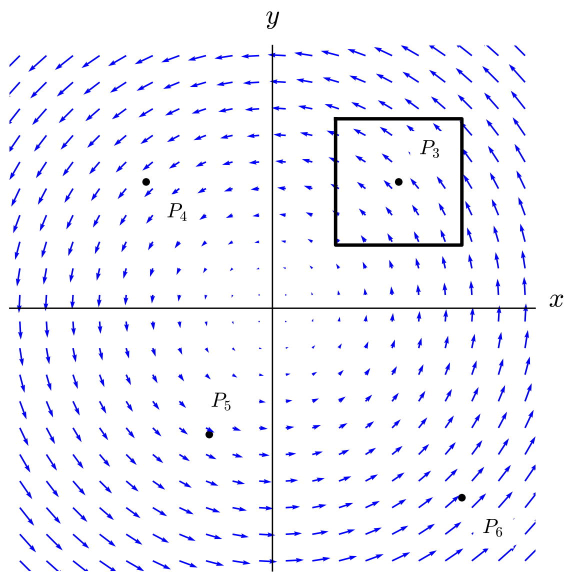


Figure 12.6.4 Vector Field **H**

(a) For each of the vector fields **F**, **G**, and **H** and the square centered on P_1, P_2, and P_3 (respectively), which statement do you think best applies?

i More of the vector field is going into the square than going out.

ii Less of the vector field is going into the square than going out.

iii The same amount of the vector field is going into the square as is going out.

(b) For each of the vector fields **F**, **G**, and **H** (and corresponding square), does your answer to part a suggest that the vector field is being created, destroyed, or is unchanging in strength inside the square? Write a sentence to explain your thinking for each vector field.

(c) Would the answer to parts a or b change if you used a smaller square centered on P_1, P_2, and P_3 for the corresponding vector fields? Write a sentence to explain your thinking for each vector field.

(d) Thinking now only about the vector field **H**, would your answers to parts a, b, or c change if you considered squares around points P_4, P_5, or P_6? Write a couple of sentences to explain your thinking.

Activity 12.6.2 Graphical Representations of Divergence..

(a) For this part of the activity, consider the vector field $\mathbf{F}$ shown in Figure 12.6.12.

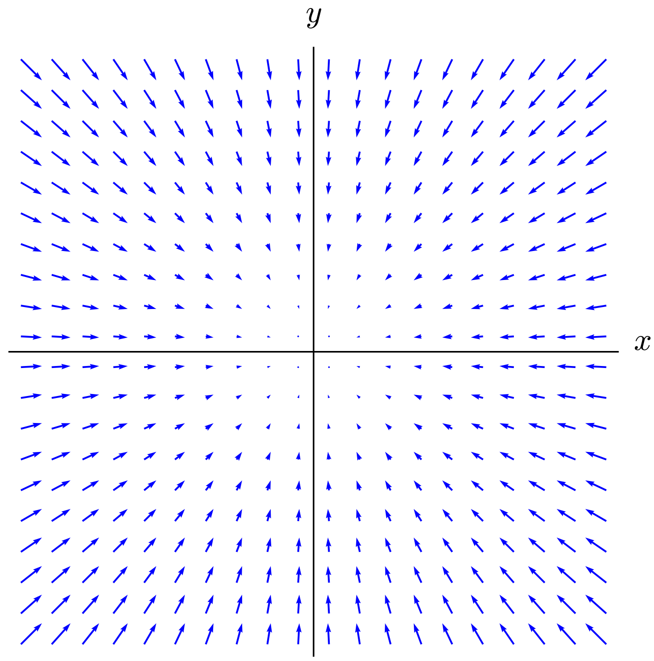


Figure 12.6.12 Vector field $\mathbf{F}$

(i) Draw a circle in the first quadrant of the vector field $\mathbf{F}$ depicted in Figure 12.6.12. Based on the flow of the vector field into or out of the circle, do you think the vector field is increasing in strength, decreasing in strength, or not changing in overall strength in the first quadrant?

(ii) As you move along the flow of the vector field in the first quadrant of Figure 12.6.12, does your vector field increase in magnitude, decrease in magnitude, or have constant magnitude?

(iii) Draw a circle in each of quadrants II, III, and IV. Based on the flow of the vector field into or out of your circles, do you think the vector field is increasing in strength, decreasing in strength, or not changing in overall strength in quadrants III, II, and IV?

(iv) As you move along the flow of the vector field in the third quadrant of Figure 12.6.12, does your vector field increase in magnitude, decrease in magnitude, or have constant magnitude.

(v) Based on your arguments above, describe why the divergence of $\mathbf{F}$ is negative for all points in the xy-plane.

(b) Look at the plot of the vector field $\mathbf{G}$ in Figure 12.6.13 and state whether you think the vector field is increasing in strength, decreasing in strength, or not changing in overall strength in each of the four quadrants. You can make your argument in terms of the change in magnitude along the flow of the vector field or in terms of the net flow into or out of a small region on the plane. You may need to make separate arguments for each of the four quadrants.

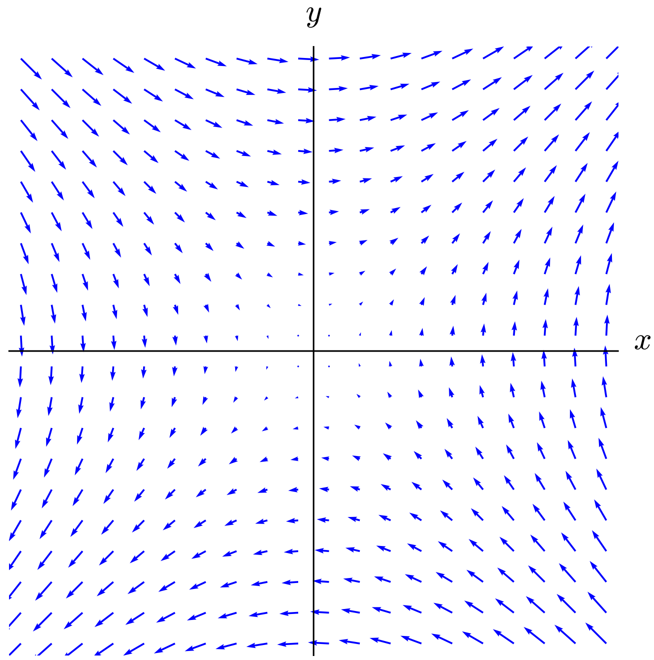


Figure 12.6.13 Vector field **G**

(c) Look at the plot of the vector field **H** in Figure 12.6.14 below and state whether you think the vector field is increasing in strength, decreasing in strength, or not changing in overall strength in each of the four quadrants. You can make your argument in terms of the change in magnitude along the flow of the vector field or in terms of the net flow into or out of a small region on the plane. You may need to make separate arguments for each of the four quadrants.

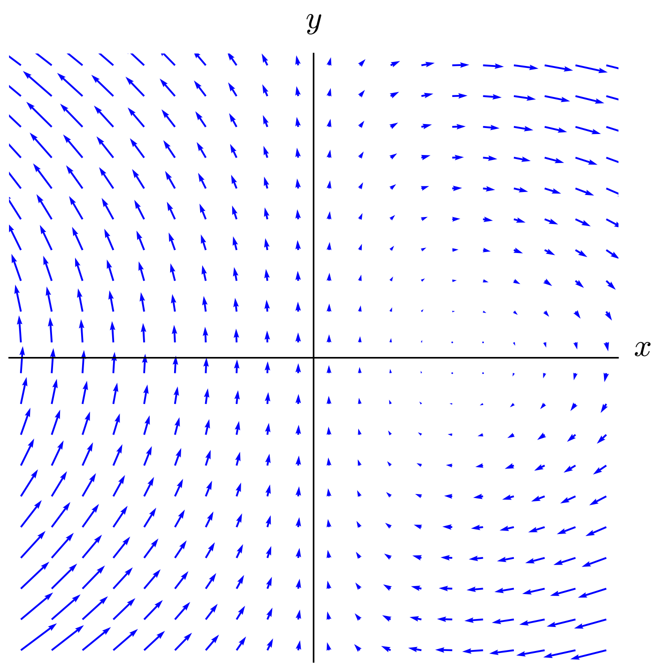


Figure 12.6.14 Vector field **H**

Activity 12.6.3.

(a) Calculate the divergence of the vector fields given below.

- $\mathbf{F}(x, y) = \langle -x, -y \rangle$
- $\mathbf{G}(x, y) = \langle y, x \rangle$
- $\mathbf{H}(x, y) = \langle xy, 1 - x \rangle$

(b) Explain how your answers to the questions in Activity 12.6.2 can be explained by using your results from part a of this activity.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.7 The Curl of a Vector Field

Preview Activity 12.7.1. We would like to understand and measure rotation of a vector field near a particular point. In order to investigate this concept, we will look at some two-dimensional vector fields and think about whether the vector field shown will rotate a small pinwheel or spinner placed at a particular location. The sort of spinner we imagine is illustrated in Figure 12.7.3. It consists of a central axis with a four-bladed paddle placed at one end of the axis. We imagine that the spinner is anchored at a point and the vector field, perhaps thought of as a fluid flow or wind velocity vector field, pushes against the blades of the spinner's paddle. In this activity, we will be trying to assess how the the spinner will rotate around the black axel (as an axis of rotation).

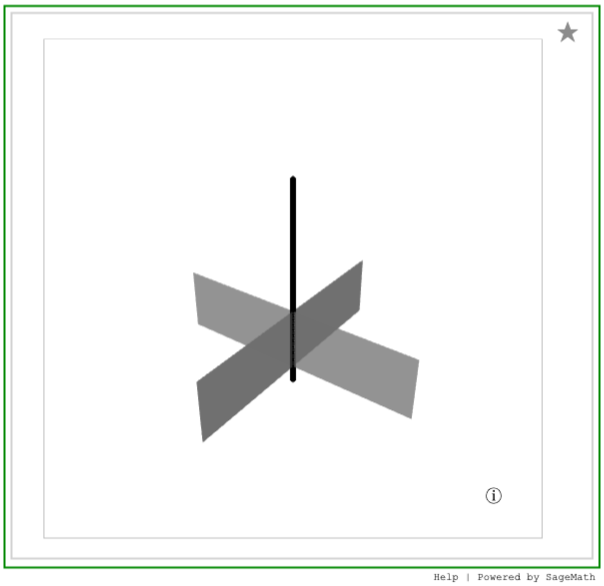


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Figure 12.7.3 A paddle-bladed spinner

To begin our investigation of the rotation of a spinner in a vector field, we will look at the vector field **F** in Figure 12.7.4. As you think about these questions, draw an "X" at each of the points about which you are asked and consider the vector field as being the pattern of a wind blowing across the plane.

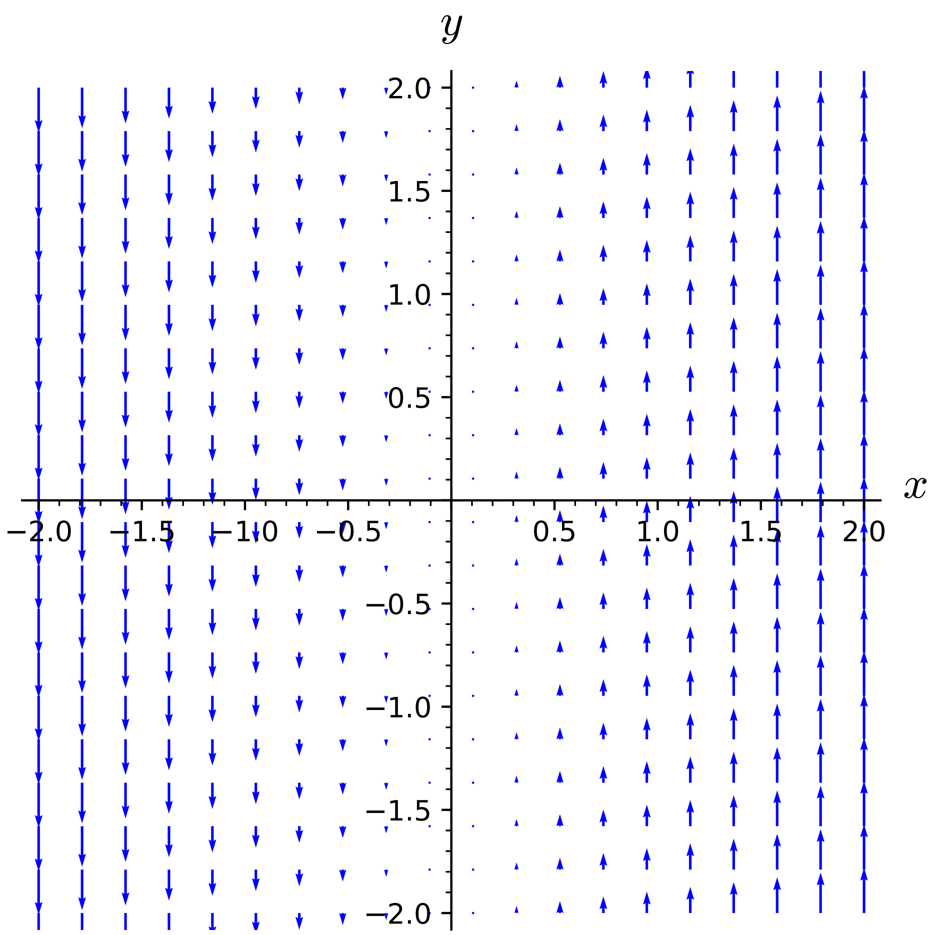

Figure 12.7.4 The vector field **F**

(a) Draw an X at the origin to act as your spinner. Draw a vector on the top right blade of your spinner that represents how the wind will push on that blade. Next, draw a vector on each of the other blades of your spinner that represents how the wind will push on that blade.

(b) Use your vector representations from the previous part to describe if a small spinner placed at the origin would spin clockwise, counterclockwise, or not spin at all.

(c) Now we would like to look at the rotational strength of **F** at the point $(1, 1)$. As before, draw a spinner at this point and draw vectors on each of the blades to represent how the wind will push on that blade. It is important at this stage to draw the relative lengths of the vectors on each blade to scale so you can see which blades have a larger force due to the wind.

(d) Use your vector representations from the previous part to describe if a small spinner placed at at $(1, 1)$ would spin clockwise, counterclockwise, or not spin at all. You should pay attention to which blades will have a stronger force (in comparison to the blade on the other side).

(e) Would a small spinner placed at $(-2, 1)$ spin clockwise, counterclockwise, or not spin at all? Draw a representation of a spinner if necessary to demonstrate your ideas.

(f) Are there any points on the xy-plane where the spinner would not turn counterclockwise?

Activity 12.7.2 Matching Visual Measurements to Algebraic Calculations.. Remember that the circulation density of a vector field $\mathbf{F} = \langle F_1, F_2 \rangle$ at a point (a, b) is calculated as

$$\text{Circulation Density at the point } (a,b) = \frac{\partial F_2}{\partial x}(a,b) - \frac{\partial F_1}{\partial y}(a,b).$$

The circulation density will be positive when a small spinner placed at (a, b) will spin counterclockwise, negative when the spinner will move clockwise, and zero when the spinner does not rotate.
For each of the vector fields given below, answer the following questions:

1. What is the formula for the circulation density of the vector field?
2. For what points will you have a positive circulation density?
3. For what points will you have a negative circulation density?
4. For what points will you have a zero circulation density?

(a) $\langle -y, x \rangle$

(b) $\langle x, y \rangle$

(c) $\langle 1 - x, xy \rangle$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.7.3 Estimating Curl in Three Dimensions..

(a) Consider the vector field **F** plotted in Figure 12.7.14. You can adjust the size of the region around $(1, 1, -2)$ over which the vector field is plotted using the "Zoom" slider. The "Density" slider allows you to adjust the number of vectors plotted. Try to identify any rotation in the three dimensional vector field plot at the point $(1, 1, -2)$. Write a sentence describing how a spinner placed at $(1, 1, -2)$ would rotate, including along which axis it would rotate. Try to state your answer as a vector representing the rotational strength of the vector field at $(1, 1, -2)$.

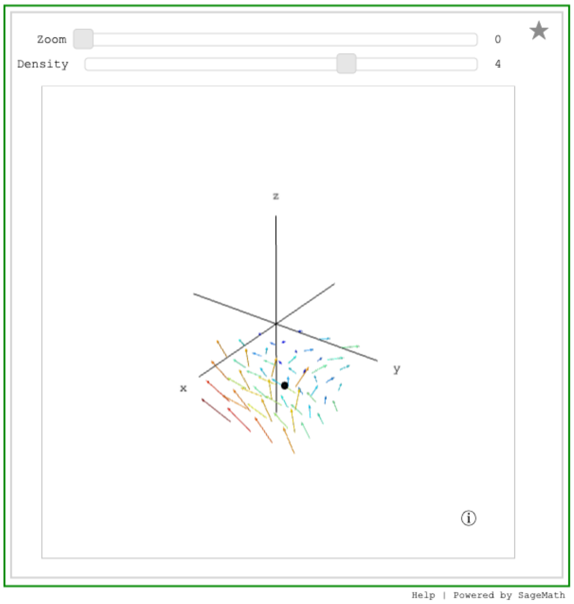

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Figure 12.7.14 A vector field plotted in a region around the point $(1, 1, -2)$

(b) You likely found it difficult to decide how you thought a spinner might rotate in this new, three-dimensional setting. Let's look at the vector field in the plane $z = -2$, as displayed in Figure 12.7.15. Do you think a spinner placed on the red point would rotate clockwise, counterclockwise, or not rotate? If the spinner will rotate, you should think about what the axis of rotation would be and whether the rotation should be positive or negative. Summarize your result as a vector representing the rotational strength of **F** in the plane $z = -2$.

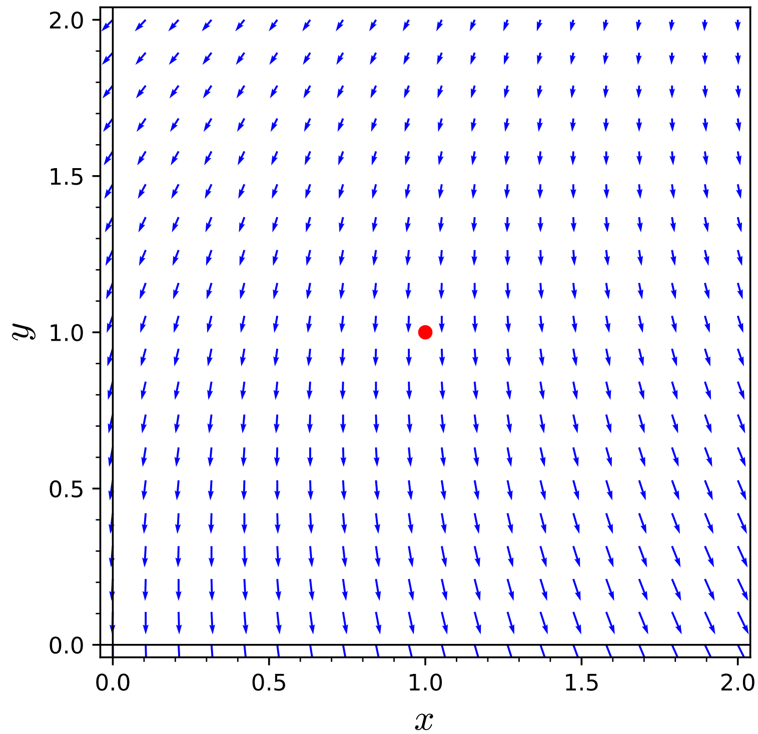

Figure 12.7.15 The trace of $\mathbf{F}$ in the plane $z = -2$

(c) Next we will look at the vector field in the plane $y = 1$, as displayed in Figure 12.7.16. Do you think a spinner placed on the red point would rotate clockwise, counterclockwise, or not rotate? If the spinner will rotate, you should think about what the axis of rotation would be and whether the rotation should be positive or negative. Summarize your result as a vector representing the rotational strength of your vector field in the plane $y = 1$. Do you think the rotation in this figure is stronger or weaker than in Figure 12.7.15?

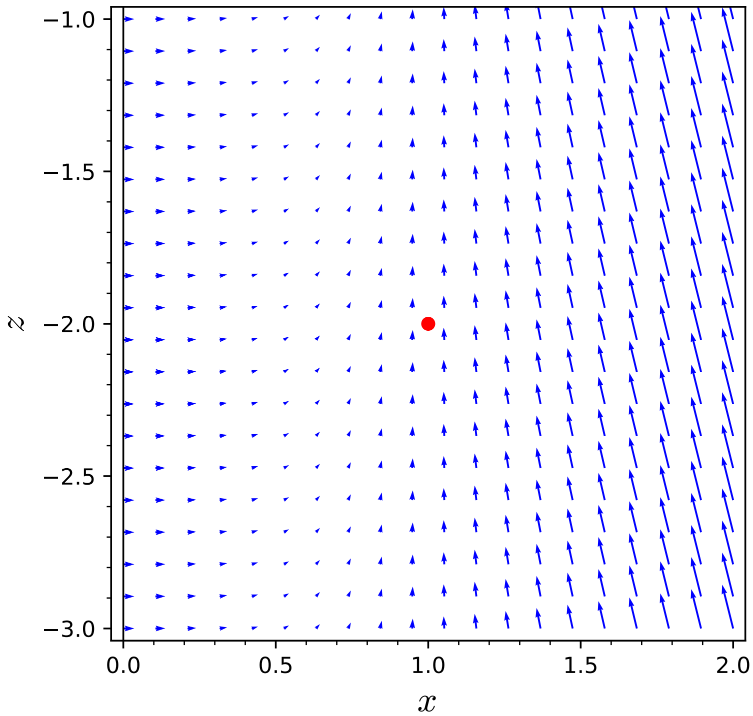

Figure 12.7.16 The trace of **F** in the plane $y = 1$

(d) Finally, consider the trace of **F** in the plane $x = 1$, as displayed in Figure 12.7.17. Do you think a spinner placed on the red point would rotate clockwise, counterclockwise, or not rotate? If the spinner will rotate, you should think about what the axis of rotation would be and whether the rotation should be positive or negative. Summarize your result as a vector representing the rotational strength of your vector field in the plane $x = 1$. How do you think the rotation in this figure compares (i.e., stronger or weaker) to that in Figure 12.7.15 and Figure 12.7.15?

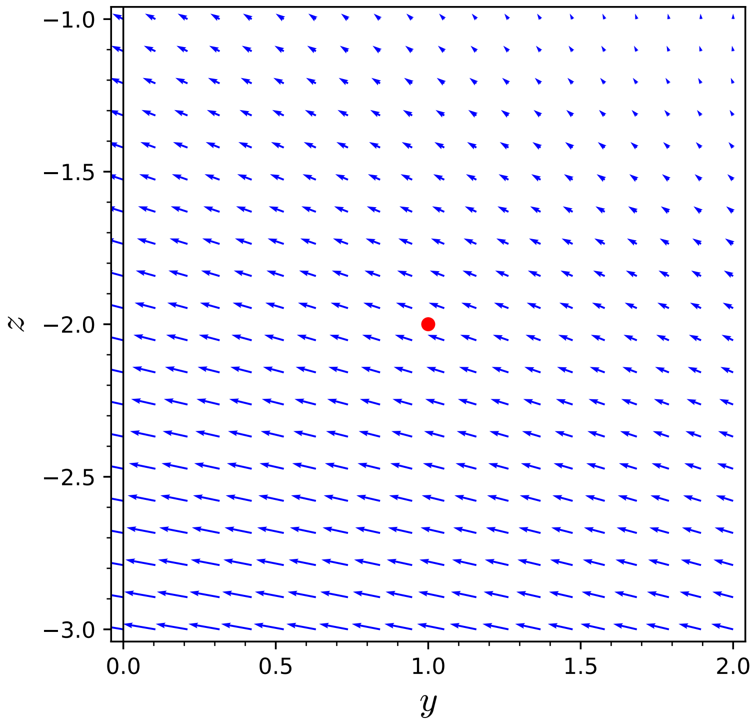

Figure 12.7.17 The trace of **F** in the plane $x = 1$

(e) Summarize your prediction to what you think the three-dimensional rotational strength of the vector field will be at the point $(1, 1, -2)$ in the form of three-dimensional vector.

(f) Compute the curl of $\mathbf{F} = \langle x - y, y + 2z, x^2 \rangle$. Specifically, what is $\text{curl}(\mathbf{F})$ at the point $(1, 1, -2)$?

(g) Compare the result of the curl calculation in part f to your prediction from part e. You likely found it difficult to estimate the magnitude, so your answer there may be incorrect. Hopefully, you did get the signs of the components and their relative strengths (i.e., which is biggest) correct. If you did not, go back and review the previous parts and explain why the calculated components match with the rotational strength for each of the three figures.

Activity 12.7.4. In this activity, we will work on calculating curl algebraically and interpreting it.

(a) Consider the vector field $\mathbf{F} = \langle x, y, z \rangle$. If we plot this vector field in any plane through the origin, we will see the vector field shown in Figure 12.7.19. This two-dimensional vector field has no rotation. The projection of $\text{curl}(\mathbf{F})(0, 0, 0)$ onto any direction therefore must give the zero vector. The only vector that has a zero projection in every direction is the zero vector. Verify this geometric argument for the curl of $\mathbf{F}$ by doing the calculations necessary to show that $\text{curl}(\langle x, y, z \rangle) = \mathbf{0}$.

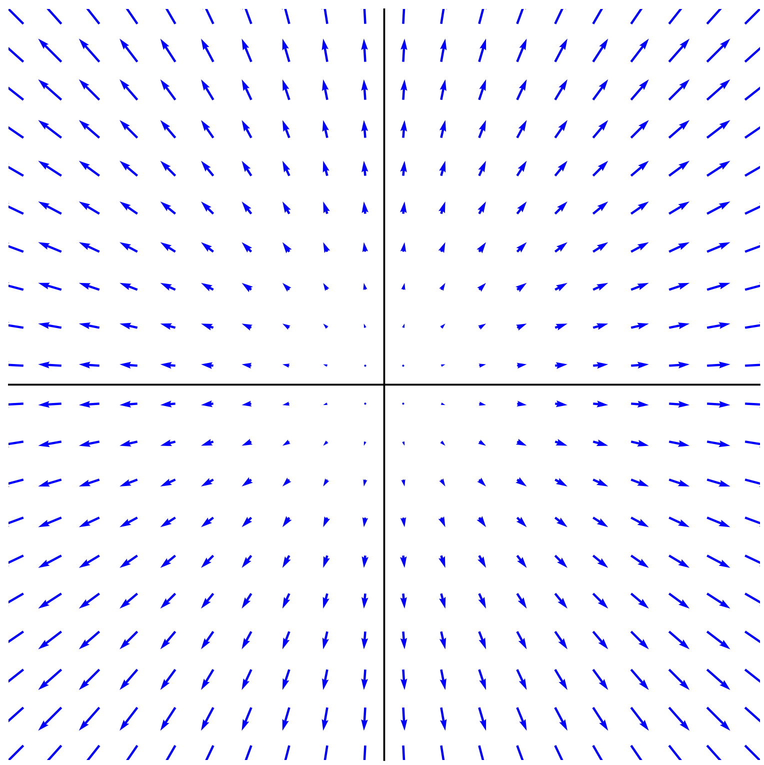

Figure 12.7.19 The vector field $\langle x, y, z \rangle$ on every plane through the origin

(b) Now consider the vector field $\mathbf{F} = \langle -y, x, 0 \rangle$, which is shown in Figure 12.7.20. The figure includes both an interactive three-dimensional plot of $\mathbf{F}$ as well as a two-dimensional plot of $\mathbf{F}$ in the yellow plane, which can be adjusted using the “Angle of Plane” slider.

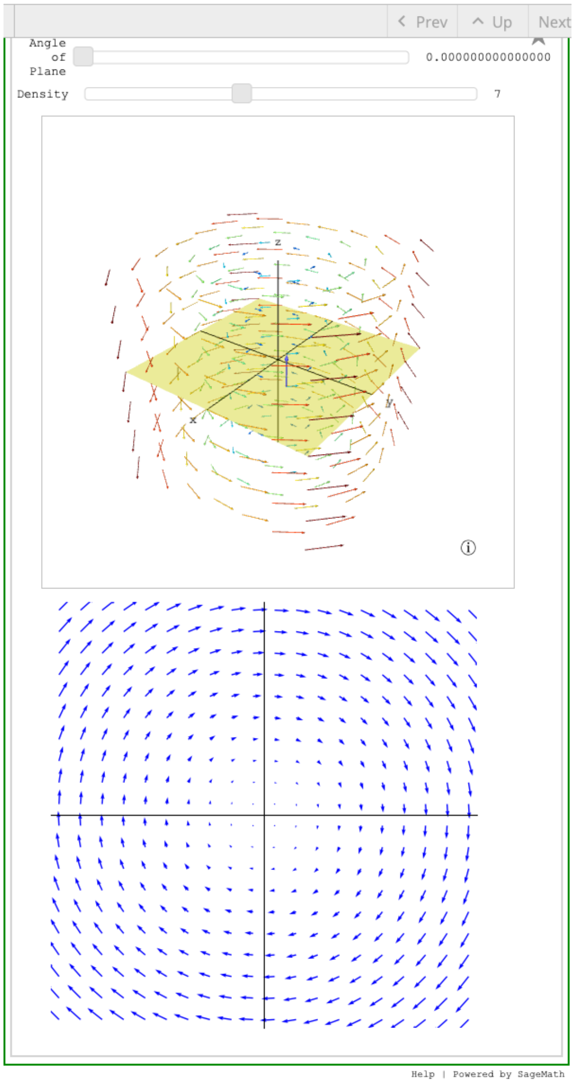

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Figure 12.7.20 Two views of the vector field $\langle -y, x, 0\rangle$

(i) Based on the figure, would you estimate the components of curl($\mathbf{F}$) to be positive, negative, or zero at the origin? Does your answer change if you pick a different point?

(ii) Calculate curl($\mathbf{F}$) algebraically. Does the curl of this vector field vary depending on the point at which the curl is measured?

(iii) Explain why the rotational strength of the vector field on a titled plane through the origin will be given by $2\cos(\theta)$, where θ is the angle between $\mathbf{k}$ and the normal vector of the tilted plane.

(c) In Figure 12.7.21 you can plot a vector field in a region around a point of your choosing in order to look at the rotational properties of the vector field. The check box in Figure 12.7.21 will show the curl vector at the base point specified so you can make sense of your vector field and its curl.

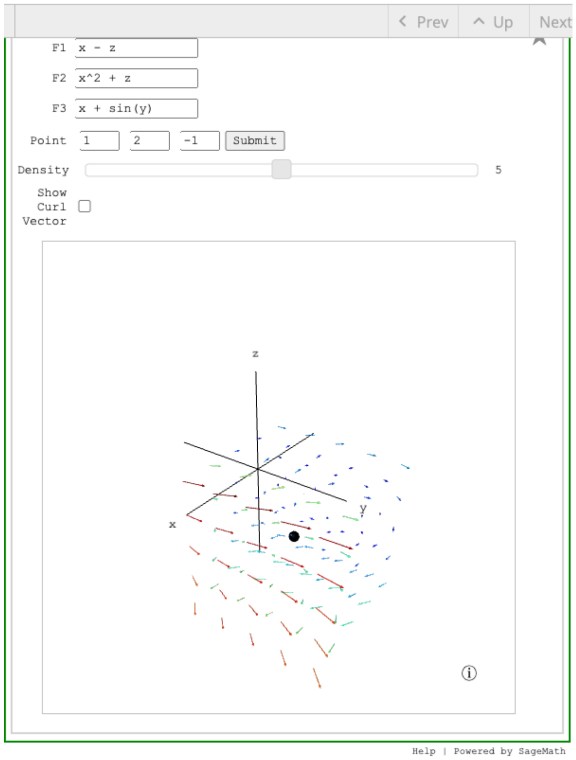

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Figure 12.7.21 A plot of the vector field $\langle F_1, F_2, F_3 \rangle$

Use the figure to estimate the direction of $\text{curl}(\langle x - z, x^2 + z, x + \sin(y) \rangle)$ at the point $(1, 2, -1)$. Confirm your estimate by calculating the curl at this point algebraically. Is there any point at which the direction of greatest rotational strength of this vector field has negative x-component? If there is, find such a point. If not, explain why not.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.8 Green's Theorem

Preview Activity 12.8.1. We will consider the vector field $\mathbf{F} = \langle 2y, 3x^2y \rangle$, which is defined on the entire xy-plane. Suppose that we want to calculate the circulation of $\mathbf{F}$ around the circle C of radius 2, centered at $(0, 0)$, and oriented counterclockwise.

(a) Verify that $\mathbf{F}$ is not path-independent by calculating the circulation of $\mathbf{F}$ around the circle C. The SageMath cell below is set up to assist you with this, but you will need to supply a parametrization of C on line 4.

```
var('r,t,x,y')
F = vector([2*y,3*x^2*y])
#Supply a parametrization of C on the next line inside the [ ]
C = vector([ , ])
integral(F(x=C[0],y=C[1]).dot_product(derivative(C,t)),t,0,2*pi)
```

(b) Recall from Section 12.7 that if $\mathbf{F} = \langle F_1(x, y), F_2(x, y) \rangle$ is a vector field, then the circulation density (in 2D) is given by

$$\text{Circulation Density at the point } (a, b) = \frac{\partial F_2}{\partial x}(a, b) - \frac{\partial F_1}{\partial y}(a, b)$$

What is the circulation density of $\mathbf{F} = \langle 2y, 3x^2y \rangle$?

(c) Sketch the curve C and shade the region it bounds. Describe the region bounded by C in both rectangular and polar coordinates.

(d) Calculate the double integral of $\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}$ over the region inside the circle C. This integral is not the most fun to do by hand, so a SageMath cell has been provided to assist you.

```
integral(integral( FUNCTION HERE , var1, start, stop), var2, start, stop)
```

(e) What do you notice about your results to parts (a) and (d)? Do you think this will happen in general? Write a sentence or two about what you think is happening here.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.8.2. For each of the curves described below, find the circulation of the given vector field around the curve. Do this both by calculating the line integral directly as well as by calculating the double integral from Green's Theorem.

(a) The curve C_1 is the circle of radius 3 centered at the point $(2, 1)$ (oriented counterclockwise) and the vector field is $\mathbf{F} = \langle y^2, 5x + 2xy \rangle$.

(b) The curve C_2 is the triangle with vertices $(0, 0)$, $(3, 0)$, and $(3, 3)$ (oriented counterclockwise) and the vector field is $\mathbf{G} = \langle y^2, 3xy \rangle$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.8.3. Consider the vector field $\mathbf{F} = \frac{-y}{x^2+y^2}\mathbf{i} + \frac{x}{x^2+y^2}\mathbf{j}$. Notice that $\mathbf{F}$ is smooth everywhere in the plane other than at the point $(0,0)$. This vector field is plotted in Figure 12.8.6, but we have not plotted the vectors close to the origin as their magnitudes get so large that they make it hard to interpret the figure. Here Green's Theorem applies to any simple closed curve C that neither passes through $(0,0)$ nor bounds a region containing $(0,0)$.

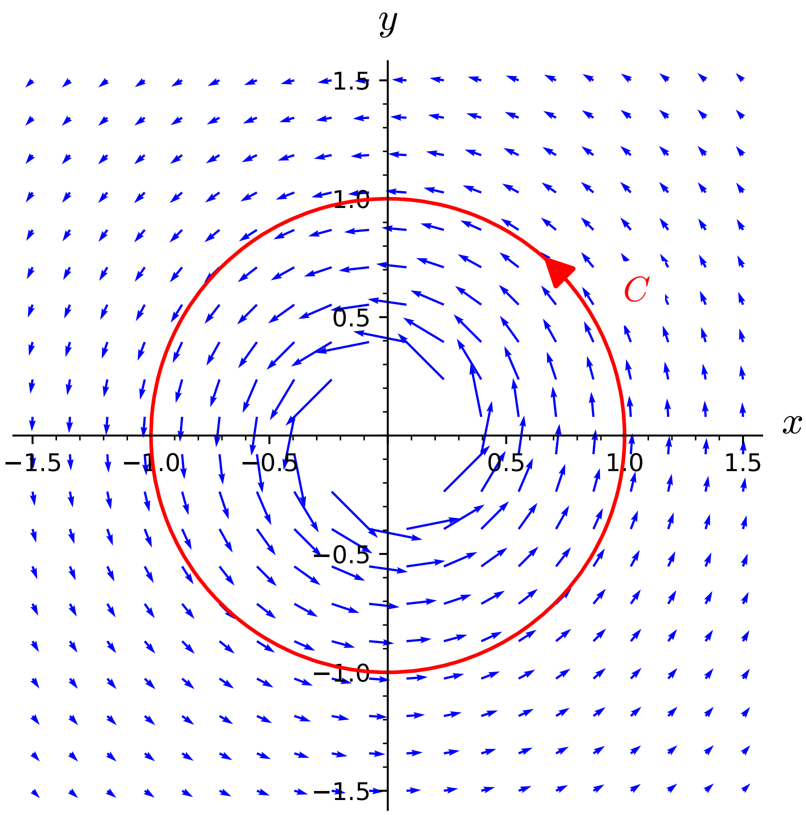

Figure 12.8.6 The vector field $\mathbf{F}$

(a) Find the circulation density of $\mathbf{F}$ (i.e., the integrand of the double integral in Green's Theorem).

(b) Suppose that C is the unit circle centered at the origin. Without doing any calculations, what can you say about $\oint_C \mathbf{F} \cdot d\mathbf{r}$? what does this tell you about if $\mathbf{F}$ is path-independent?

(c) What would you get if you integrated the circulation density of $\mathbf{F}$ over the region bounded by C?

(d) Do the previous two parts contradict Green's Theorem? Explain your reasoning.

(e) Is the vector field $\mathbf{G} = \frac{x}{x^2+y^2}\mathbf{i} + \frac{y}{x^2+y^2}\mathbf{j}$, which is shown in Figure 12.8.7, path-independent? Why or why not?

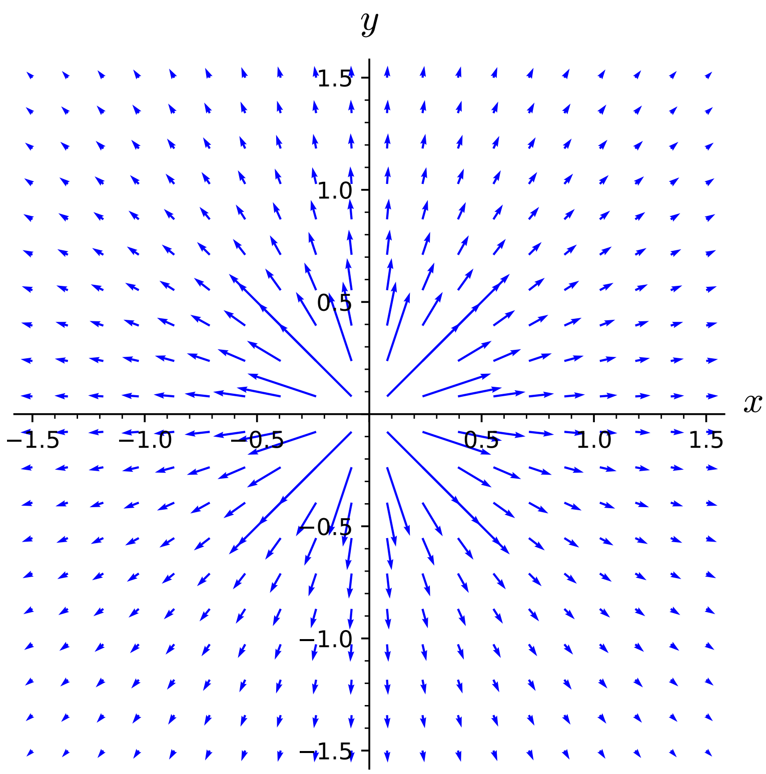

Figure 12.8.7 The vector field **G**

(f) Suppose that C is the unit circle centered at the origin. Find $\oint_C \mathbf{G} \cdot d\mathbf{r}$. Can you do this using Green's Theorem?

12.9 Flux Integrals

Preview Activity 12.9.1. In this preview activity, we will explore the parameterizations of a few familiar surfaces and confirm some of the geometric properties described in the introduction above.

(a) Use the ideas from Section 11.6 to give a parameterization $\mathbf{r}(s,t)$ of the following surface. Be sure to specify the bounds on each of your parameters.

S_1: the right circular cylinder centered on the x-axis of radius 2 when $0 \leq x \leq 5$

(b) Draw a graph of S_1 from the previous part. Label the points that correspond to (s,t) points of $(0,0)$, $(0,1)$, $(1,0)$, and $(2,3)$.

(c) For the parameterization from part a, calculate $\mathbf{r}_s$, $\mathbf{r}_t$, and $\mathbf{r}_s \times \mathbf{r}_t$.

(d) For the parameterization from part a, find the value for $\mathbf{r}_s$,$\mathbf{r}_t$, and $\mathbf{r}_s \times \mathbf{r}_t$ at the (s,t) points of $(0,0)$, $(0,1)$, $(1,0)$, and $(2,3)$.

(e) Draw your vector results from d on your graph and confirm the geometric properties described in the introduction to this section. Namely, $\mathbf{r}_s$ and $\mathbf{r}_t$ should be tangent to the surface, while $\mathbf{r}_s \times \mathbf{r}_t$ should be orthogonal to the surface (in addition to $\mathbf{r}_s$ and $\mathbf{r}_t$).

(f) Repeat steps (a) through (e) for the following two surfaces:

- **(i)** S_2: the sphere centered at the origin of radius 3
- **(ii)** S_3: the first octant portion of the plane $x + 2y + 3z = 6$

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.9.2 Visualizing flux through a surface.. In this activity, you will compare the net flow of different vector fields through our sample surface. In Figure 12.9.5 you can select between five different vector fields. Once you select a vector field, the vector field for a set of points on the surface will be plotted in blue. Each blue vector will also be split into its normal component (in green) and its tangential component (in magenta). The yellow vector defines the direction for positive flow through the surface.
Look at each vector field and order the vector fields from greatest net flow through the surface to least net flow through the surface. Remember that a negative net flow through the surface should be lower in your rankings than any positive net flow.

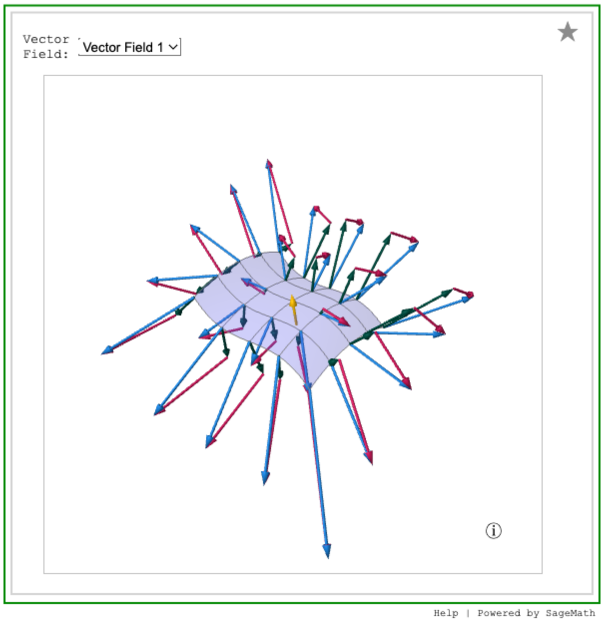

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Figure 12.9.5 Five vector fields flowing through a surface, with decomposition into normal and tangential components

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.9.3 Checking the Visualization for Flux..

(a) Figure 12.9.12 shows a plot of the vector field $\mathbf{F} = \langle y, z, 2 + \sin(x)\rangle$ and a right circular cylinder of radius 2 and height 3 (with open top and bottom). Consider the vector field going into the cylinder (toward the z-axis) as corresponding to positive flux.

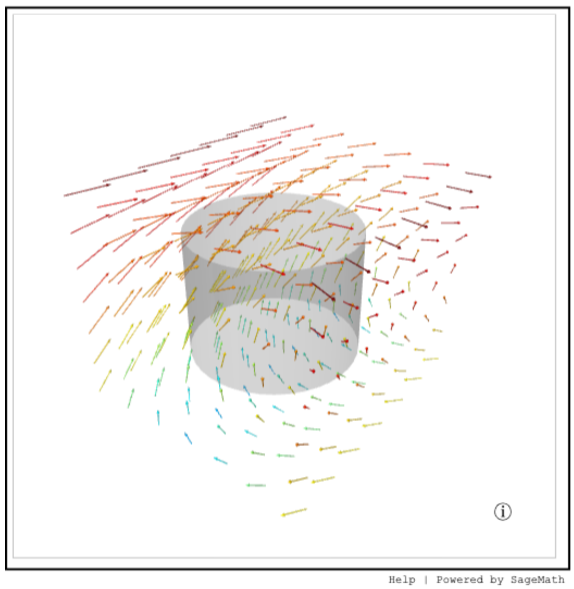

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Figure 12.9.12 The vector field $\mathbf{F} = \langle y, z, 2 + \sin(x)\rangle$ and a right circular cylinder

(i) Reasoning graphically, do you think the flux of $\mathbf{F}$ throught the cylinder will be positive, negative, or zero? Write a few sentences jutifying your answer.

(ii) Parameterize the right circular cylinder of radius 2, centered on the z-axis, for $0 \leq z \leq 3$. Be sure to give bounds on your parameters.

(iii) Based on your parameterization, compute $\mathbf{r}_s$, $\mathbf{r}_t$, and $\mathbf{r}_s \times \mathbf{r}_t$. Confirm that these vectors are either orthogonal or tangent to the right circular cylinder. Is your orthogonal vector pointing in the direction of positive flux or negative flux?

(iv) Use your parametrization to write $\mathbf{F}$ as a function of s and t.

(v) Compute the flux of $\mathbf{F}$ through the parameterized portion of the right circular cylinder.

(vi) Does your computed value for the flux match your prediction from earlier?

(vii) Use Figure 12.9.13 to make an argument about why the flux of $\mathbf{F} = \langle y, z, 2 + \sin(x)\rangle$ through the right circular cylinder is zero.

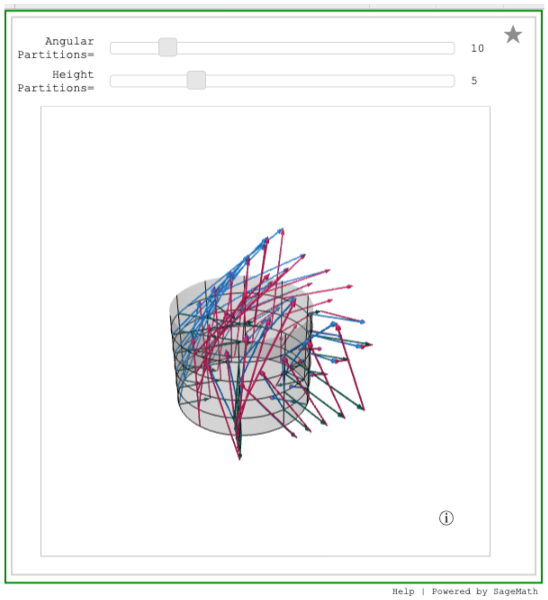

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Figure 12.9.13 The vector field $\mathbf{F} = \langle y, z, 2 + \sin(x) \rangle$ with normal and tangential components plotted on a right circular cylinder

(b) Write a couple of sentences to explalin how the results of the flux calculations would be different if we used the vector field $\mathbf{F} = \left\langle y, z, \cos(xy) + \frac{9}{z^2+6.2} \right\rangle$ and the same right circular cylinder.

(c) write a couple of sentences to explain how the results of the flux calculations would be different if we used the vector field $\mathbf{F} = \langle y, -x, 3 \rangle$ and the same right circular cylinder.

12.10 Surface Integrals of Scalar Valued Functions

Preview Activity 12.10.1. In Preview Activity 12.5.1 we looked at how to understand line integrals of scalar functions through the analogy of running a mining machine along a given path. In short, the amount of copium mined depended on the density of copium at points on the path and the length of the path driven by the mining rig.

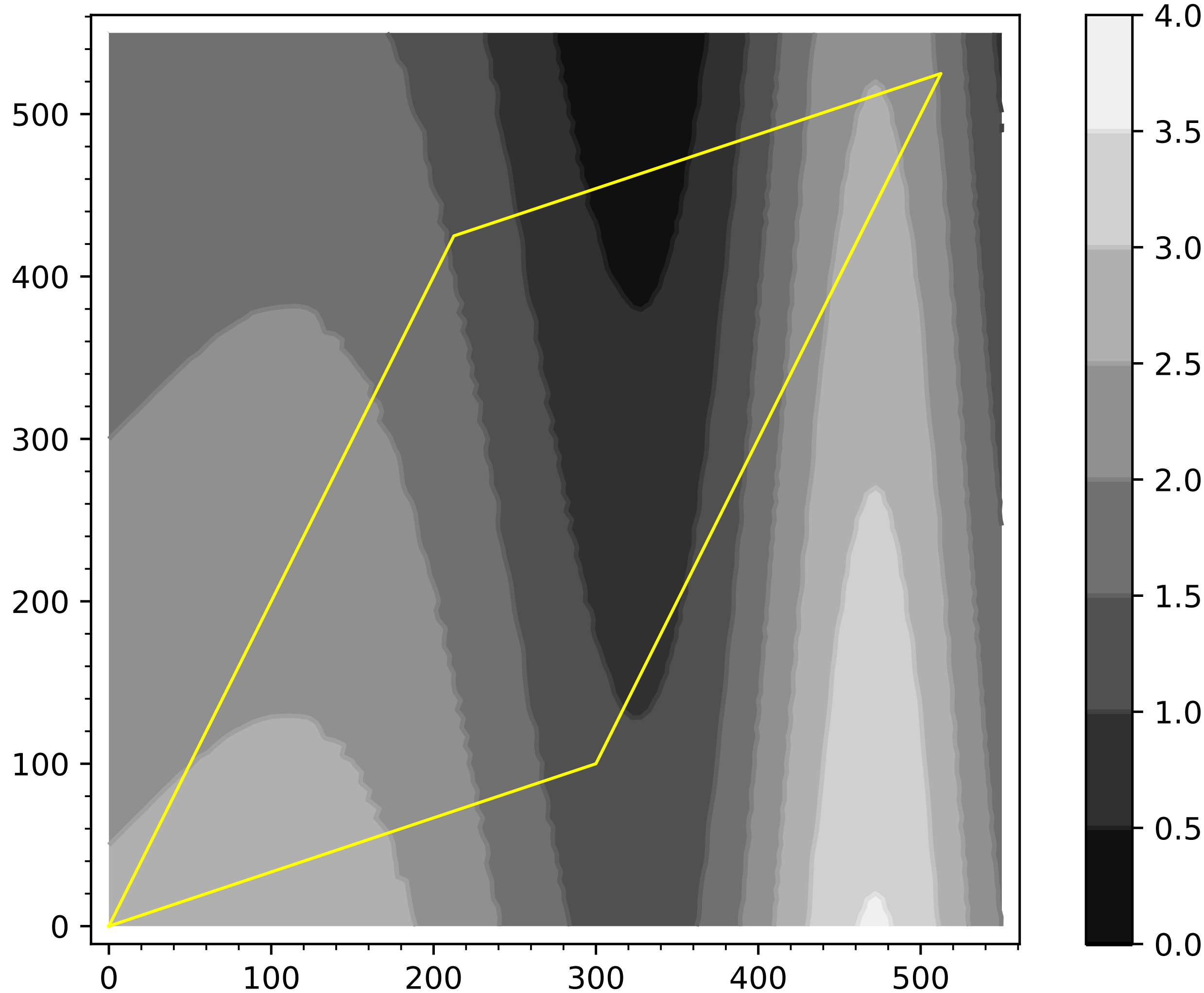

Figure 12.10.1 A plot of land with density of copium deposits and the edges of the plot drawn in yellow

The opening tasks of Preview Activity 12.5.1 had you estimating the amount of copium mined by driving along the edge of a plot of land (drawn in yellow on Figure 12.10.1). This iterpretation meant that the scalar function we were using measured the linear density of copium. Thus, the Riemann sum we computed was the product of linear density of copium with the distance traveled.

(a) In this task, we will interpret Figure 12.10.1 as a contour graph of the density of copium per unit area. This will allow us to compute the total amount of copium in our mining area by setting up a double integral. To approximate the total amount of copium available in our mine, do the following:

(i) Break the mining plot (the area inside the yellow segments of Figure 12.10.1) into three pieces. Estimate the area of the three pieces you are using. Write a few sentences explaining your methods of estimating the areas.

(ii) For each of the three pieces you used in part a.i, estimate the average density of copium on the piece. Write a few sentences explaining your methods of estimating the average density on each piece.

(iii) Give an estimate for the total amount of copium on the mining plot and explain your computation.

(b) What if instead of your mining plot being on a flat piece of land as represented in Figure 12.10.1, your mining plot was on a hill as represented in Figure 12.10.2. If we had the same copium density plot as a function of the (x, y) coordinates, which of the following would you expect to be true?

- the total amount of copium available on Figure 12.10.2 is greater than on Figure 12.10.1
- the total amount of copium available on Figure 12.10.2 is lesser than on Figure 12.10.1
- the total amount of copium available on Figure 12.10.2 is the same as on Figure 12.10.1

Explain your reasoning for your choice.

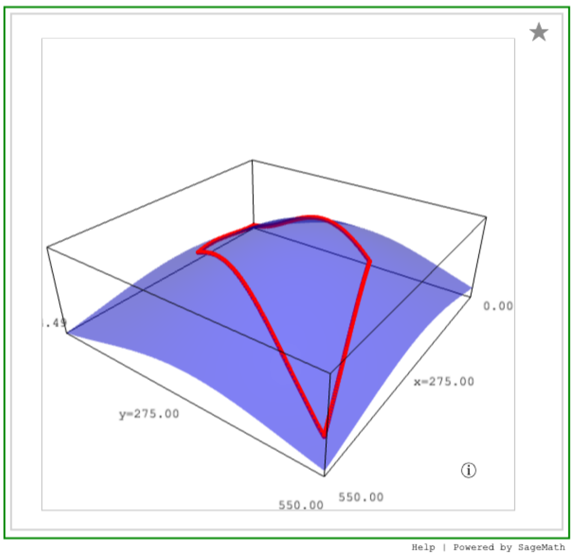

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Figure 12.10.2 A 3D plot of the region including our mining area (shown with boundary in red)

Activity 12.10.2. In this activity, we will try to understand the scalar surface integral by looking at whether the value of the scalar surface integral will be positive, negative, or zero over common surfaces. In each part below, you are given a function and a surface. For each surface, first draw a plot of the surface and make sure you have labeled a proper scale for each coordinate direction. Then reason if the given surface integral is positive, negative, or zero. Be sure to justify your answers in terms of the function being integrated *and* the particulars of the surface of integration.

(a) For S_1 defined as the top half ($z \geq 0$) of the sphere of radius one centered at the origin, consider the surface integral $\iint_{S_1} x\, dS$.

(b) For S_2 defined as the bottom half ($z \leq 0$) of the sphere of radius one centered at the origin, consider the surface integral $\iint_{S_2} z\, dS$.

(c) For S_3 defined as the disc of radius one centered at $(1, 0, 0)$ on the plane $x = 1$, consider the surface integral $\iint_{S_3} x + z\, dS$.

(d) For S_1 as defined above, consider the surface integral $\iint_{S_1} x + z\, dS$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.10.3 Explaining Properties of Scalar Surface Integrals.. In this activity, we will be explaining each of the properties from Properties of Scalar Line Integrals in the context of a new analogy. We have just purchased a plot of land that spans two mountains, Sugar Mountain and Spice Mountain. We will label the plot of land on Sugar Mountain S_1 and the plot of land on Spice Mountain S_2. Unfortunately there are two types of toxic organisms on the surface of your new land, which may explain why you paid so little for the land. Let f be the density of the toxic fungus on your new plot of land and let g be the density of toxic bacteria on the new plot.

(a) Explain in your own words what $\iint_{S_1} f\,dS$ means in the above analogy and what exactly would be measured by this scalar line integral.

(b) Explain in your own words what $\displaystyle\iint_{S_1} (2f)\,dS = 2\iint_{S_1} f\,dS$ means in the new analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

(c) Explain in your own words what $\displaystyle\iint_{S_2} (f+g)\,dS = \iint_{S_2} f\,dS + \iint_{S_2} g\,dS$ means in the new analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

(d) Explain in your own words what $\displaystyle\iint_{-S_2} f\,dS = \iint_{S_2} f\,dS$ means in the new analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

(e) Explain in your own words what $\displaystyle\iint_{S_1+S_2} f\,dS = \iint_{S_1} f\,dS + \iint_{S_2} f\,dS$ means in the new analogy. It may be helpful to describe each side of the equation separately and say why they are equal in the analogy.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.10.4. Compute the value of the scalar surface integral in part c of Subsection 12.10.2. That is, compute $\iint_{S_3} x + z\, dS$ where S_3 is the disc of radius one centered at $(1, 0, 0)$ on the plane $x = 1$. Explain why your answer makes sense geometrically.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.11 Stokes' Theorem

Preview Activity 12.11.1. In this activity, we will look at how we can apply the ideas about circulation along overlapping curves from the beginning of Subsection 12.8.2 to curves in space.

(a) For this part, consider the curves in Figure 12.11.1, where the yellow curve is Y, the blue curve is B, and the magenta curve is M.

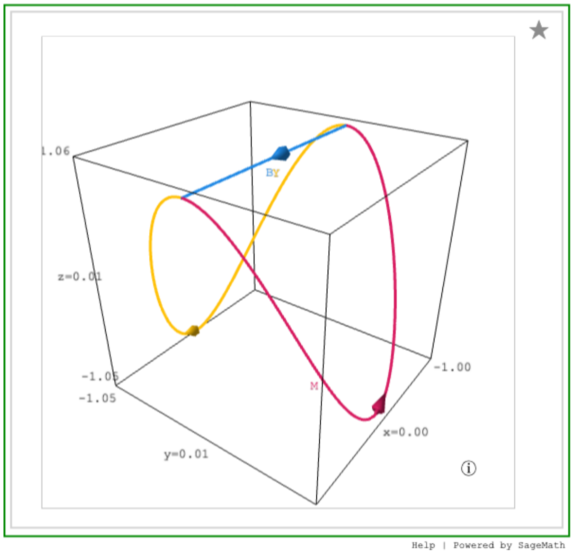

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Figure 12.11.1 Three curves in space

You should also go back and refamiliarize yourself with our notation for combining paths (as used in line integrals) from Convention 12.2.13. In our convention, $Y + M$ would be closed loop, but $Y - M$ would not make sense because the segment $-M$ does not begin where Y begins.

(i) Using the three segments in Figure 12.11.1, write out at least four different closed curves in terms of B, Y, and M. (Remember to consider orientation!)

(ii) Let $C_1 = M + B$ and $C_2 = Y - B$. Describe the curve given by $C_1 + C_2$.

(iii) Write a couple of sentences explaining how the circulation around $C_1 + C_2$ would compare to the circulation around C_1 and the circulation around C_2. Write an equation in terms of $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$, $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$, and $\int_{C_1+C_2} \mathbf{F} \cdot d\mathbf{r}$.

(iv) Explain how your arguments or equations from any of the parts above would or would not change if you considered the curves depicted in Figure 12.11.2.

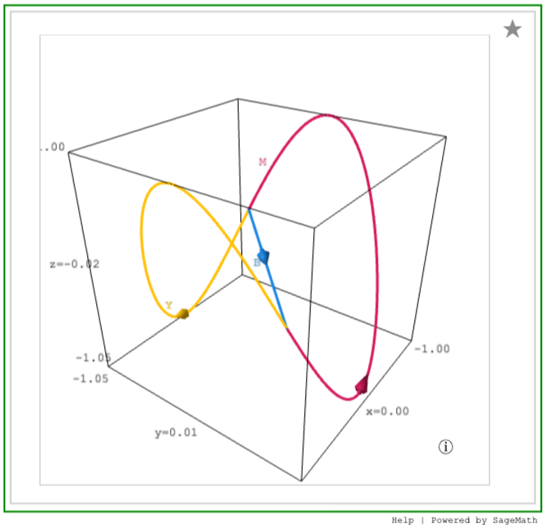

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Figure 12.11.2 Three slightly different curves in space

(b) Let C be the simple closed curve consisting of the yellow and magenta curves in Figure 12.11.1. You can see C plotted in red in Figure 12.11.3. The drop-down allows you to select three different surfaces. You can visually verify that each of the three surfaces contains C. Notice that the scale on the z-axis changes as you select different surfaces.

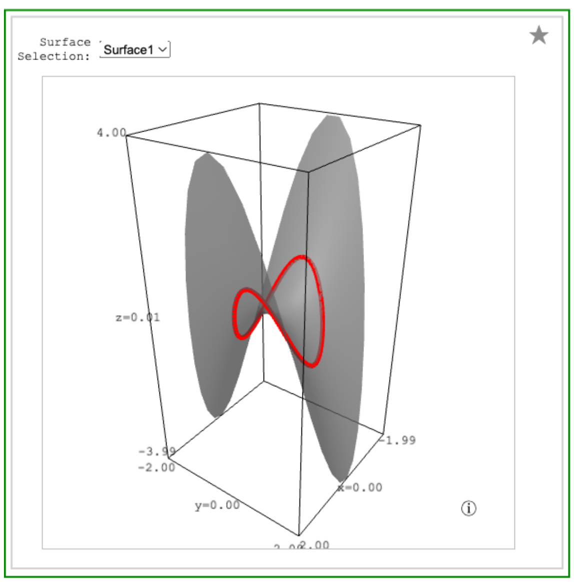

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Figure 12.11.3 Surfaces containing a common simple closed curve

The simple closed curve consisting of the yellow and magenta curves in Figure 12.11.1 can be parameterized by $\langle \cos(t), \sin(t), \cos(2t) \rangle$ with $0 \leq t \leq 2\pi$. Let $C_3 = Y + M$. Use the given parameterization of C_3 to show that C_3 is on each of the following surfaces:

- $x^2 - y^2 = z$
- $z = x^4 - y^4$
- $z = 1 - 2y^2$
- $z = -\cos(\pi\sqrt{x^2 + y^2})(x^2 - y^2)$

Activity 12.11.2. In this activity, we will verify Stokes' Theorem by calculating both a line integral and a flux integral.

(a) Consider the vector field $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ and the circle C_1 parameterized as $\mathbf{r}(t) = \langle \sqrt{2}\cos(t), \sqrt{2}\cos(t), 2\sin(t) \rangle$ for $0 \le t \le 2\pi$.

- **(i)** Calculate $\oint_{C_1} \mathbf{F} \cdot d\mathbf{r}$ directly using the given parametrization.
- **(ii)** Let S_1 be the hemisphere of the sphere of radius 2 centered at the origin with $y \le x$. Calculate the flux of $\text{curl}(\mathbf{F})$ through S_1.
- **(iii)** What could you have observed about $\mathbf{F}$ that would have gotten you the same answer without doing either of the above calculations?

(b) Consider the vector field $\mathbf{G} = x\mathbf{i} + y^2z\mathbf{j} + x^2\mathbf{k}$ and the curve C_2, which is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ with orientation corresponding to the order the points are listed here.

- **(i)** Find the circulation of $\mathbf{G}$ along C_2 by calculating the appropriate line integrals.
- **(ii)** The vertices of C_2 lie in a plane. Let S_2 be the portion of this plane lying in the first octant, i.e., the portion with $x, y, z \ge 0$. Find the flux of $\text{curl}(\mathbf{G})$ through S_2.
- **(iii)** Write a sentence to explain why the sign of your answer to the previous two parts makes sense.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.11.3.

(a) Find the circulation of $\mathbf{F} = \langle 3yz, xz, -xy\rangle$ along the curve C consisting of (given in order of the orientation) the quarter-circle of radius 1 centered at $(0,-2,0)$ in the plane $y=-2$ from $(0,-2,1)$ to $(1,-2,0)$, the line segment from $(1,-2,0)$ to $(1,5,0)$, the quarter-circle of radius 1 centered at $(0,5,0)$ in the plane $y=5$ from $(1,5,0)$ to $(0,5,1)$, and the line segment from $(0,5,1)$ to $(0,-2,1)$.

(b) Find the circulation of $\mathbf{G} = 3z^2\mathbf{i} - (z^2+2x)\mathbf{j} + zy\mathbf{k}$ along the circle in the xy-plane of radius 3 centered at the origin. Assume the counterclockwise orientation of the circle.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.11.4. Because Stokes' Theorem requires us to consider a surface (with normal vector) and the boundary of the surface, this activity will give you a chance to practice identifying the boundary of some surfaces in $\mathbb{R}^3$. For each surface below:

i Describe the boundary in words.

ii Find a parametrization for the boundary.

iii Ensure that a person walking along the boundary in the direction of your parametrization with head pointing in the direction of the surface's normal vector would hold their left hand over the surface.

(a) The surface S_1 is the portion of the sphere $x^2+y^2+z^2=4$ with $z \geq x$. Assume the outward orientation on the sphere.

(b) The surface S_2 is the portion of the sphere $x^2+y^2+z^2=4$ with $z \geq 0$. Assume the outward orientation on the sphere.

(c) The surface S_3 is the portion of the hyperbolic paraboloid $z=x^2-y^2$ with $x^2+y^2 \leq 1$. Assume the "upward" orientation, e.g., the normal vector at $(0,0,0)$ is $\mathbf{k}$.

(d) The surface S_4 is the portion of the cylinder $x^2+y^2=4$ for which $-2 \leq z \leq 2$, assuming the outward orientation.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Activity 12.11.5. In some sense, this activity considers the reverse problem of that considered in Activity 12.11.4. Here, each part of the activity gives you an oriented simple closed curve C in $\mathbb{R}^3$, and your task is to find

- a surface S so that C is the boundary of S and
- a normal vector for the S so that a person walking along C in the direction of the given orientation with head pointing in the direction of your chosen normal vector would have their left hand over S.

You are encouraged to think about multiple possible answers, since as we saw in Preview Activity 12.11.1, there may be more than one reasonable choice of a surface with a particular boundary.

(a) The curve C is the triangle with vertices $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$ with orientation corresponding to the order the points are listed here.

(b) The curve C is the circle parameterized as $\mathbf{r}(t) = \langle \sqrt{2}\cos(t), \sqrt{2}\cos(t), 2\sin(t) \rangle$ for $0 \leq t \leq 2\pi$.

(c) The curve C consists of (given in order of the orientation)

- the quarter-circle $C_1$of radius 2 centered at the origin in the xy-plane from $(2,0,0)$ to $(0,2,0)$,
- the line segment C_2 from $(0,2,0)$ to $(0,2,2)$,
- the quarter-circle C_3 of radius 2 centered at $(0,0,2)$ in the plane $z = 2$ from $(0,2,2)$ to $(2,0,2)$, and
- the line segment C_4 from $(2,0,2)$ to $(2,0,0)$.

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

12.12 The Divergence Theorem

Preview Activity 12.12.1 Locating sources of a poisonous gas.. We will use three different surfaces to examine the flux through closed surfaces. Let S_{top} be the top half of the unit sphere centered at the origin (graphed in magenta in the figures below). Let S_{bottom} be the bottom half of the unit sphere centered at the origin (graphed in yellow). Finally, let S_{mid} be the unit disk centered at the origin in the xy-plane (graphed in blue). With these definitions, S_{top} and S_{bottom} will make a closed surface given by the unit sphere. The surfaces S_{top} and S_{mid} will enclose the top half of the unit ball, while S_{bottom} and S_{mid} will enclose the bottom half of the unit ball.
In this problem we will be using the surfaces defined above and the flux integrals of a poisonous gas through these surfaces to try to determine whether different regions of space are emitting or absorbing the poisonous gas.

(a) In this part, we will consider the unit ball shown in Figure 12.12.4, with boundary and normal vectors as shown in the plot. If the flux integral of a poisonous gas through S_{top} is 15 and the flux integral of the poison gas through S_{bottom} is -3, is the interior of the sphere emitting or absorbing poisonous gas? Explain your reasoning.

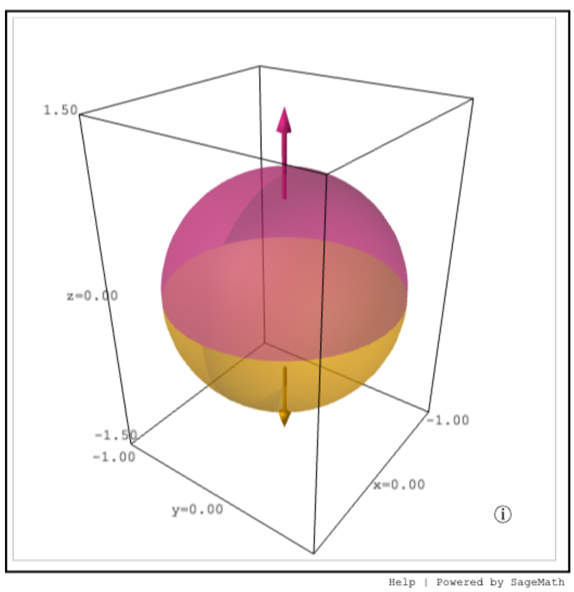

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Figure 12.12.4 Unit ball with boundary given by the combination of S_{top} and S_{bottom}

(b) In this part, we will consider the top half of the unit ball shown in Figure 12.12.5, with boundary and normal vectors as shown in the plot. If the flux integral of a poisonous gas through S_{top} is 15 and the flux integral of the poison gas through S_{mid} is -20, is the top half of the unit ball emitting or absorbing poisonous gas? Explain your reasoning.

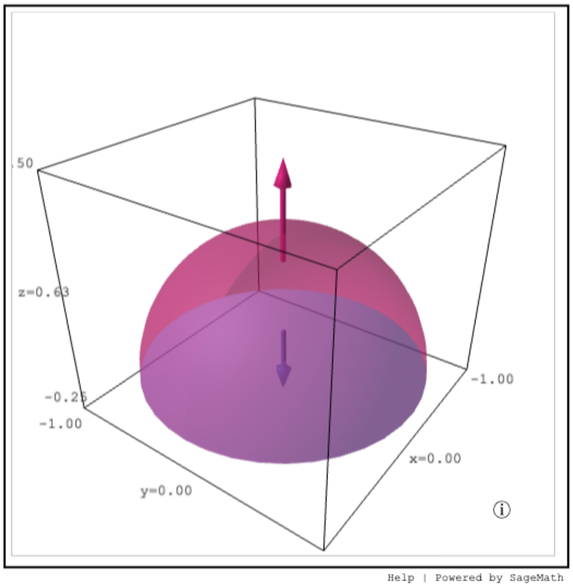

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Figure 12.12.5 Upper half of the unit ball with boundaries given by S_{top} and S_{mid}

(c) In this part, we will consider the bottom half of the unit ball shown in Figure 12.12.6, with boundary and normal vectors as shown in the plot. Based on the information given in the previous two parts, what will the flux integrals of the poison gas be for S_{bottom} and S_{mid} be in this case? Be sure to pay attention to the orientation of what we consider positive flow. Explain your reasoning.

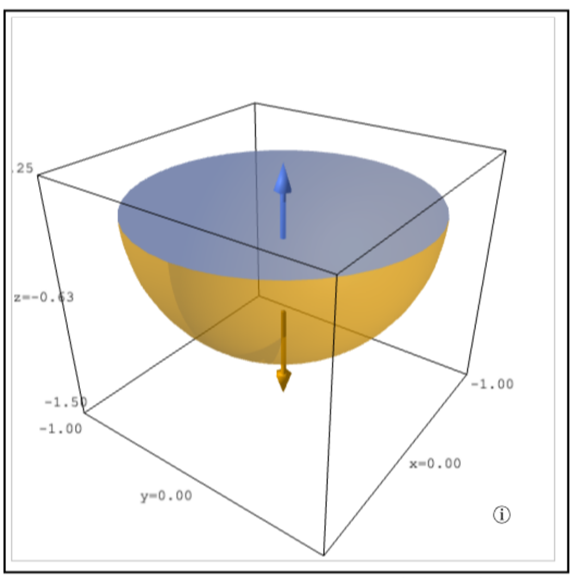

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Figure 12.12.6 Lower half of the unit ball with boundaries given by S_{mid} and S_{bottom}

(d) Using your answer from the previous part, is the bottom half of the unit ball emitting or absorbing poisonous gas? Explain your reasoning.

Activity 12.12.2. In this activity, we will look at calculating both sides of a non-trivial example of the Divergence Theorem. We will look at the region inside the right circular cylinder shown in Figure 12.12.10. Let S be the closed surface formed by combining S_{top} (in yellow), S_{sides} (in blue), and S_{bottom} (in magenta). The solid volume Q is the volume bounded by S. The region shown has radius 2, and its height is 1. The vector field we consider in this activity is given by

$$\mathbf{F} = \langle xy - 2z, y^2 - yz, 3x + z^2 \rangle.$$

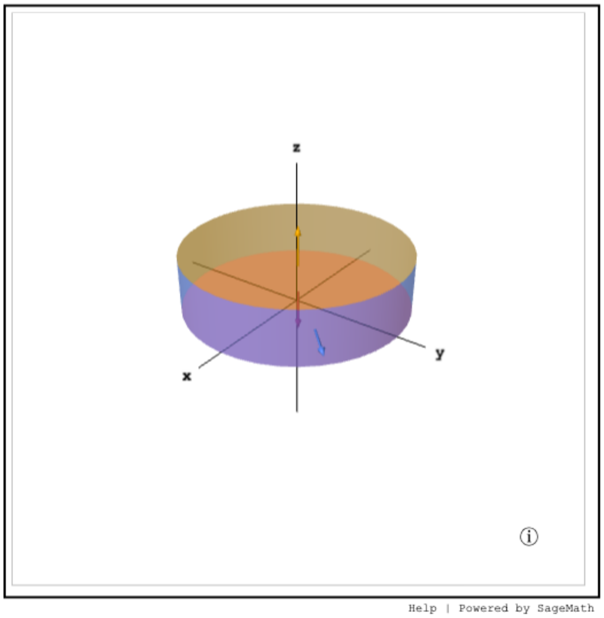

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Figure 12.12.10 A closed cylindrical surface

(a) Figure 12.12.11 shows the vector field $\mathbf{F}$ on a region around S. Without doing any computations, write a couple of sentences to explain if you think the flux of $\mathbf{F}$ through S will be positive, negative, or zero.

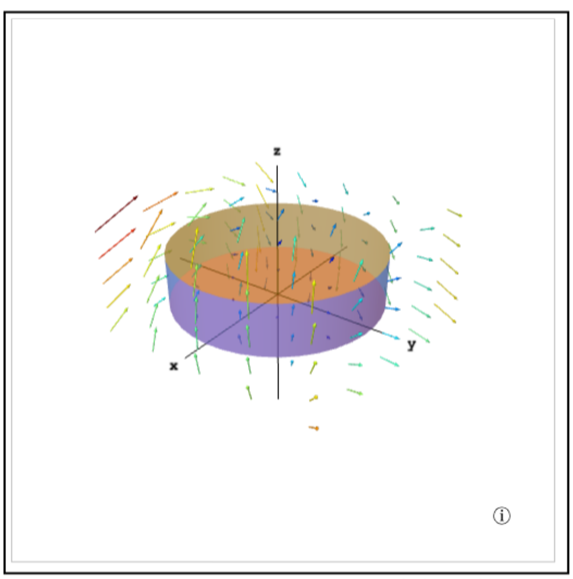

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Figure 12.12.11 The vector field $\mathbf{F} = \langle xy - 2z, y^2 - yz, 3x + z^2 \rangle$ in the region near Q

(b) Parametrize each of the surfaces S_{top}, S_{sides}, and S_{bottom}. Be sure to give bounds for each of your parametrization.

(c) Give inequalities in terms of cylindrical coordinates to describe Q.

(d) Set up and evaluate double integrals to calculate the flux of $\mathbf{F}$ through S_{top}, S_{sides}, and S_{bottom}.

(e) What is the net flux through the closed surface? Be sure to state if the net flux is in or out.

(f) Compute the divergence of $\mathbf{F}$ and use this to explain whether you think $\iiint_Q \operatorname{div}(\mathbf{F})\, dV$ will be positive, negative, or zero.

(g) Set up and compute the triple integral for $\iiint_Q \operatorname{div}(\mathbf{F})\, dV$.

(h) Verify that your answers for part e and part g are the same and thus that the Divergence Theorem holds for this example.

(i) Was the left-hand side or right-hand side of the equation in the Divergence Theorem more tedious to calculate in this example? Do you think this will be true for most other cases where the Divergence Theorem applies?

Activity 12.12.3.

(a) Find the flux of the vector field $\mathbf{F} = \langle 3x^2 + y^5, 5 + e^{z^3}, z\rangle$ through the surface of the solid cube Q in $\mathbb{R}^3$ with $-2 \leq x \leq 2$, $-2 \leq y \leq 2$, and $-2 \leq z \leq 2$.

(b) Find the flux of the vector field $\mathbf{G} = \langle x^3, y^3, z^3\rangle$ through surface consisting of the top half of sphere of radius 3 centered at the origin and the disc of radius 3 in the xy-plane (centered at the origin).

What I Learned

Key takeaways from this activity

Important

Formulas and results to remember

Colophon

This book was authored in PreTeXt.